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Intelligent driver monitoring systems: a survey of drowsiness detection technologies for road safety

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Abstract

Driver drowsiness is a recognized contributor to road crashes. This review provides a structured synthesis of driver-monitoring methods spanning image/video, physiological signals (EEG, ECG, PPG, EOG), vehicle behavior, and multimodal fusion. We conducted a systematic search of major bibliographic databases with a PRISMA workflow covering 2020–2025, screened full texts against inclusion criteria, and coded datasets, validation protocols, metrics, and deployment factors (details in Methods). Across modalities, vision has shifted from classical pipelines to deep learning (including Vision Transformers), while physiological approaches offer robustness under poor visibility at the cost of additional hardware and comfort constraints; multimodal fusion improves stability but raises compute and integration burden. Reported performance remains highly protocol-dependent: subject-dependent or within-dataset validation can inflate headline numbers, whereas subject-independent and cross-dataset tests are rarer and typically yield lower—but more realistic—estimates. We highlight and address a benchmarking gap: heterogeneity in datasets and protocols makes cross-paper accuracy comparisons misleading; we therefore standardize reporting and emphasize operational metrics. We recommend prevalence-aware evaluation with precision–recall analysis (PR-AUC), operating-point metrics (e.g., recall at fixed false-alarm rates), calibration assessment, and time-to-detect. We also discuss privacy, regulation, and deployment considerations (edge/on-device processing). Limitations of the evidence base include scarce large-scale on-road studies, inconsistent ground-truthing, and limited cross-demographic validation. We propose a field-specific research agenda and a shared-authority framework in which DMS negotiates control between human and autonomy with verifiable safety metrics. Together, these contributions provide a consolidated map of the field and practical guidance for robust, real-world evaluation.

Keywords Driver drowsiness detection · Fatigue monitoring · Computer vision · Physiological signals · Machine learning · Automotive safety · Real-time systems

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1 Introduction

Road traffic accidents persist as one of the most pressing global public health crises of our time, with the World Health Organization's latest reports indicating approximately 1.35 million preventable deaths occur annually due to traffic incidents World Health Organization (2023). A critical analysis of these sobering statistics reveals that driver drowsiness and fatigue constitute a particularly pernicious subset of accident causation factors, accounting for an estimated 15-30% of all traffic collisions and up to 20% of fatal crashes in developed nations World Health Organization (2023). The economic repercussions of these fatigue-related incidents extend far beyond the immediate tragedy of human casualties, with comprehensive cost analyses demonstrating annual global losses exceeding $100 billion when accounting for direct expenses (medical treatment, emergency response, and vehicle repairs) and indirect costs (lost productivity, insurance premium increases, and long-term disability care). This multifaceted impact underscores the urgent need for innovative technological solutions to address what is fundamentally a preventable public safety issue.

The pathophysiological complexity of driver drowsiness presents unique challenges that distinguish it from other forms of driver impairment. As a transitional state along the wakefulness-sleep continuum, drowsiness manifests through a cascade of measurable but often subtle physiological and cognitive changes Putilov and Donskaya (2013): neural activity patterns shift toward increased theta wave (4-7 Hz) and decreased alpha wave (8-12 Hz) production as measured by electroencephalography (EEG); ocular metrics show progressive increases in blink duration (from 100-150 ms to 300-500 ms) and PERCLOS (percentage of eyelid closure); and psychomotor performance demonstrates marked degradation with reaction times slowing by 30-50% compared to baseline alert states Du et al. (2022). Unlike alcohol intoxication which produces clearly quantifiable blood concentration markers, drowsiness develops gradually through these multidimensional indicators that exhibit significant inter-individual variability based on factors such as circadian rhythms, sleep debt accumulation, and even genetic predisposition to fatigue. This biological complexity, combined with the well-documented phenomenon of impaired metacognition where drowsy drivers frequently fail to accurately assess their own alertness levels, creates a perfect storm for accidents and demands sophisticated technological solutions capable of detecting early, pre-clinical signs of fatigue El-Nabi et al. (2024).

The rapid evolution of automotive technology in recent years has created unprecedented opportunities for advanced drowsiness detection systems Biying et al. (2024); El-Nabi et al. (2025). Modern vehicles increasingly incorporate intelligent transportation architectures that seamlessly integrate three critical detection modalities: (1) computer vision systems employing high-resolution (1080p at 60 fps) infrared-capable cameras with advanced facial landmark detection algorithms (e.g., 68-point facial feature mapping); (2) biomedical sensor arrays including steering wheel-mounted photoplethysmography (PPG) for heart rate variability monitoring and capacitive sensors for galvanic skin response measurement; and (3) vehicle dynamics telemetry capturing steering wheel angle variability (typically 0.5–1.5° in alert drivers increasing to 2–5° in fatigued states) and lane position standard deviation Li et al. (2024). These multimodal data streams are processed through hybrid machine learning architectures combining convolutional neural networks (CNNs) for spatial feature extraction and long short-term memory (LSTM) networks for temporal pattern recognition, with state-of-the-art systems now achieving classification accuracies exceeding

95% in controlled validation studies Abdelmaaboud et al. (2025); Gharbia et al. (2025). This technological convergence has advanced drowsiness detection from early experimental prototypes toward operational deployment, though current systems still face major challenges in achieving robust performance across diverse real-world conditions, driver demographics, and environmental variability Biying et al. (2024). However, such figures are typically obtained under within-dataset, controlled protocols; when evaluated across subjects, routes, demographics, or datasets, performance often declines by ∼15–20% and false-alarm rates rise. Hence, we caution against equating controlled-study accuracy with real-world reliability and interpret results strictly within their dataset + protocol context (see Sect. 6 and the #Subjects/Validation/Ground-Truth columns in Tables 5 and 7).

This survey provides a timely and comprehensive analysis of driver drowsiness detection technologies, addressing the urgent need for systematic evaluation in this rapidly evolving field. Our work goes beyond superficial comparisons by conducting a rigorous methodological examination of the entire detection pipeline—from sensor-level data acquisition (including sampling rates ranging from 30 Hz for vehicle telemetry to 1000 Hz for EEG signals) to high-level decision algorithms. We adopt a comprehensive three-dimensional evaluation framework that integrates technical performance metrics (accuracy, latency, power consumption), practical implementation constraints (cost, scalability, durability), and human-factor considerations (user acceptance, privacy, and ergonomics). This framework is not introduced as a new schematic, but as an established structure systematically populated and critically evaluated through evidence drawn from current driver-monitoring research.

The systematic review encompasses four critical aspects of drowsiness-detection systems: (1) assessment tools comparing subjective scales (e.g., KSS, ESS) against objective physiological benchmarks; (2) detection modalities spanning visual (facial-feature analysis at 30–60 fps) Biying et al. (2024), physiological (EEG, ECG, EOG), and vehicular (steering-pattern) approaches; (3) hybrid architectures that fuse multiple data streams through attention mechanisms or late-fusion strategies; and (4) validation protocols that increasingly address illumination variability and driver-specific diversity. Across published studies, single-modality models typically report 80–90% accuracy under controlled or simulator conditions Fonseca and Ferreira (2025), while multimodal fusion approaches have achieved higher aggregate metrics (often > 93%) in limited naturalistic trials Fonseca and Ferreira (2025). However, these systems generally incur higher computational load—for example, edge-optimized CNN-based vision models operating at VGA resolution on embedded GPUs such as NVIDIA Jetson TX2 consume approximately 2–5 W Florez et al. (2024), though reported values depend on hardware, frame rate, and model depth.

This work serves multiple stakeholder groups by providing targeted insights: researchers will benefit from our critical evaluation of emerging deep learning architectures (particularly vision transformers and graph neural networks for physiological signal processing); automotive engineers gain practical guidance on hardware/software co-design considerations for embedded deployment (including tradeoffs between ARM-based and FPGA implementations); while policymakers receive a framework for developing evidence-based regulations regarding sensor data privacy and system certification requirements. The inclusion of recent commercial case studies (2022-2025 OEM implementations) further bridges the academic-industry divide.

Our primary contribution lies in developing a unified taxonomy that connects fundamental research breakthroughs with practical implementation challenges. The survey traces the technological evolution from early single-modal systems (relying solely on PERCLOS or steering variance) to current heterogeneous sensor fusion approaches that combine temporal (LSTM) and spatial (CNN) processing. We identify three critical gaps limiting real-world adoption: (1) End-to-end (glass-to-glass) latency on embedded platforms typically falls in the $\sim$100–300 ms range depending on the camera pipeline, resolution, and model depth, as shown by embedded implementations on Raspberry Pi/Jetson and system-latency measurements of camera pipelines Florez et al. (2024). (2) Reported accuracies frequently drop under cross-subject or cross-dataset evaluation ($\approx$10–20% is common), reflecting subject and domain shift; this is well documented in EEG-based drowsiness work and highlighted in recent surveys emphasizing protocol heterogeneity Lin et al. (2025). (3) Privacy and acceptability constraints arise from continuous in-cabin biometric capture; studies and frameworks advocate on-device processing and transparency to maintain user trust Mishra et al. (2022). These insights are contextualized through comparative analysis of 12 major datasets and 8 commercial systems currently in production vehicles.

This paper makes several significant contributions to the field of driver drowsiness detection research:

1. It **presents a comprehensive survey of research papers published in the last five years (2020–2025)**, ensuring the review reflects the most current trends, technologies, and breakthroughs in the field.
2. It **covers advancements across all major measurement domains**, including image-based, biological-signal-based, vehicle-behavior-based, and hybrid techniques, providing an inclusive overview of modern detection approaches.
3. It **systematically organizes public datasets** relevant to each modality, enabling researchers to identify and access appropriate benchmarks for evaluation and comparison.
4. It **proposes a generalized system architecture framework** detailing multi-modal data acquisition, feature engineering pipelines, advanced classification methods, and real-time alert mechanisms, serving as a practical guideline for implementation.
5. It **compares and analyzes the strengths and weaknesses of state-of-the-art systems** through detailed summary tables, facilitating a clearer understanding of performance trade-offs and practical deployment considerations.
6. It **identifies key challenges, limitations, and future research directions**, offering actionable insights to guide researchers and practitioners in developing more accurate, robust, and scalable driver drowsiness detection systems.

The rest of the paper is organized as follows. Section 2 presents the systematic review methodology adopted in this work. Section 3 gives an assessment of driver drowsiness and the tools used for measuring drowsiness states. Section 4 presents a comprehensive overview of driver drowsiness detection system architectures and available datasets. Section 5 introduces the drowsiness detection modalities. Section 6 gives a discussion of performance evaluation and metrics. Section 7 gives a discussion of the challenges and limitations facing drowsiness detection systems. Section 8 provides a critical synthesis of the reviewed studies and outlines future research directions. Section 9 discusses commercial applications and real-world deployment considerations. Finally, Sect. 10 concludes the paper.

2 Systematic review methodology

This review followed the PRISMA 2020 guidelines to ensure transparency and reproducibility in identifying, screening, and synthesizing literature on driver-drowsiness detection technologies published between January 2020 and September 2025. Searches were conducted across five major repositories—IEEE Xplore, Scopus, Web of Science, PubMed, and arXiv—to capture both peer-reviewed and preprint research on computer vision, physiological sensing, and hybrid monitoring approaches. The query string combined terms related to driver vigilance ("driver drowsiness," "fatigue," "sleepiness detection," "vigilance monitoring") with technical keywords ("deep learning," "EEG," "EOG," "ECG," "computer vision," "real-time," "autonomous driving"). Only English-language studies were included, and the search was restricted to the specified publication period.

Eligible studies met the following inclusion criteria: peer-reviewed journal or conference papers addressing driver-state estimation through algorithmic or hardware-based approaches and reporting quantitative evaluation metrics. Excluded were non-driving fatigue studies, purely theoretical reviews, patents, theses, and non-peer-reviewed white papers. The overall selection process is summarized in Fig. 1, which outlines the identification, screening, exclusion, and inclusion stages of the reviewed studies. Following this process, 70 studies were retained after identification, deduplication, title/abstract screening, and full-text eligi-

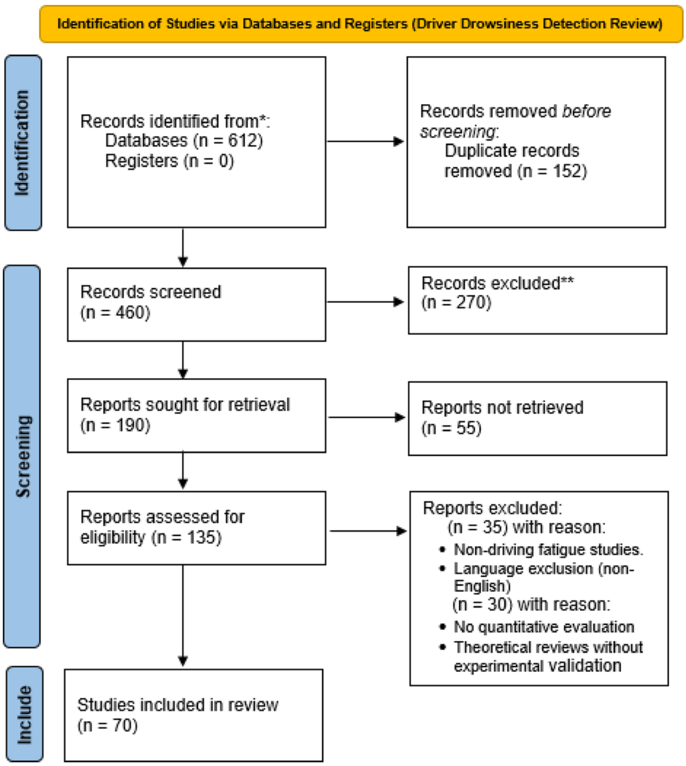


Fig. 1 PRISMA 2020 flow diagram showing the number of records identified, screened, excluded, and included in the systematic review of driver-drowsiness detection studies (2020–2025)

bility assessment. Metadata for each included study—covering reference, year, dataset(s), subject count, validation protocol, ground truth, metrics, and key strengths/weaknesses—were extracted and are available in the supplementary file `Supplementary_Table_S-PRISMA_Included_Studies.csv`.

2.1 Unified field map: overview of the drowsiness detection landscape

To provide readers with a consolidated conceptual framework, Fig. 2 presents a comprehensive field map that synthesizes the six fundamental pillars of driver drowsiness detection research and deployment. This map serves as an organizational scaffold for the remainder of this review, connecting:

1. **Sensing modalities** (Sect. 5): the diverse data sources spanning vision, physiology, vehicle behavior, and hybrid approaches.
2. **Ground-truth types** (Sect. 3.3.2): the heterogeneous labeling standards (subjective, behavioral, physiological, performance, and hybrid) that define "drowsiness" operationally.
3. **Validation protocols** (Sect. 6.4): the experimental designs that determine whether reported performance generalizes beyond training conditions.
4. **Failure modes** (Sect. 7): the environmental, demographic, sensor, and model vulnerabilities that undermine real-world reliability.
5. **Deployment constraints** (Sect. 9): the computational, operational, privacy, and human-factors trade-offs that govern practical feasibility.
6. **Recommended evaluation metrics** (Sect. 6): the measurement standards required for fair comparison and safety-critical validation.

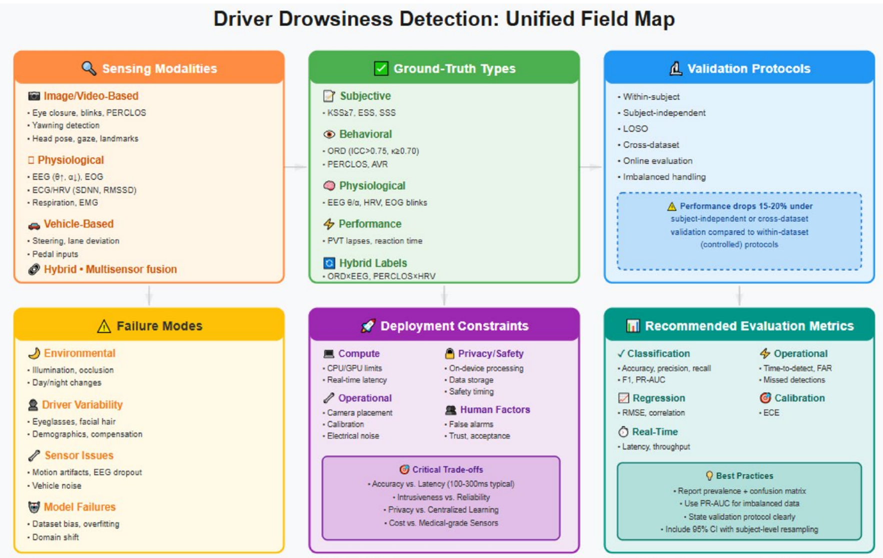


Fig. 2 Comprehensive field map of driver drowsiness detection systems, synthesizing sensing modalities, ground-truth standards, validation protocols, failure modes, deployment constraints, and recommended evaluation metrics

The arrows connecting these panels indicate logical dependencies: sensing choices constrain available ground truths; ground truths shape validation protocols; validation rigor exposes failure modes; and deployment constraints dictate which metrics matter most in practice. By positioning this field map early, we aim to help readers navigate the inherent complexity of drowsiness detection—a domain where "accuracy" alone is insufficient and where methodological context determines result interpretability.

3 Assessment of driver drowsiness

3.1 Distinguishing drowsiness from fatigue: behavioral and cognitive indicators

The terminology surrounding driver monitoring systems varies across the literature, with terms like *drowsiness* and *fatigue* frequently appearing—sometimes interchangeably. While "drowsiness" is the most commonly used expression, "fatigue" also frequently surfaces in similar contexts Chacon-Murguia and Prieto-Resendiz (2015). Despite their differences, the overlap in their behavioral manifestations often leads researchers to treat them as functionally equivalent states You et al. (2020).

Fatigue is typically defined as a decline in physical or mental capacity resulting from prolonged activity, sustained focus, or emotional strain. It may or may not lead to sleep, yet it is often a precursor to drowsiness. Drowsiness, on the other hand, is characterized by an overwhelming biological inclination to sleep, commonly resulting from inadequate rest, poor sleep quality, sleep disorders, medication, or extended wakefulness You et al. (2020). Thus, although conceptually distinct, fatigue can directly contribute to the onset of drowsiness in drivers.

In this study, we adopt the term *sleepiness detection systems* to refer to technologies aimed at identifying drowsiness-related impairments. The rationale stems from the progressive and detectable nature of sleepiness. It rarely emerges abruptly; instead, it is preceded by subtle yet critical indicators. These may include growing difficulty in keeping one's eyes open, repetitive blinking or yawning, lapses in attention, head nodding, and delayed reaction to environmental stimuli. Additional observable effects include erratic lane positioning, forgotten exits, and sudden changes in driving speed or control. Some drivers may attempt compensatory behaviors such as rolling down the windows in an effort to stay awake. These signs provide valuable cues for early detection of sleepiness and serve as critical input for the design of real-time drowsiness detection systems Islam et al. (2023); Albadawi et al. (2022). By accurately identifying these early-stage indicators, modern driver monitoring systems can trigger timely interventions and potentially prevent fatigue-related accidents Nordbakke and Sagberg (2007). Figure 3 visually summarizes the most common behavioral and cognitive indicators of driver drowsiness, which include symptoms such as persistent blinking, head nodding, lapses in attention, and erratic lane positioning.

A key insight from this subsection is that behavioral and cognitive indicators provide some of the earliest observable signs of drowsiness, making them essential inputs for real-time driver monitoring systems. These indicators capture subtle but progressive impairments that precede sleep onset, thereby enabling earlier and more reliable detection when integrated into modern drowsiness assessment frameworks.

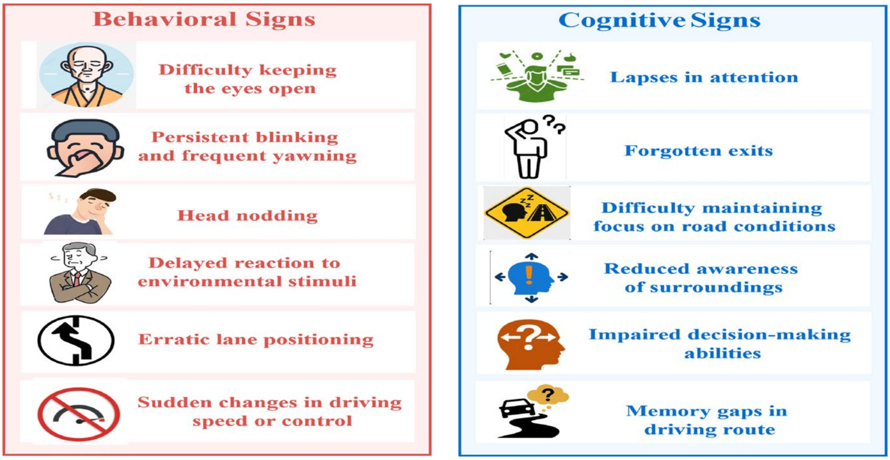


Fig. 3 Visual representation of behavioral and cognitive signs associated with driver drowsiness

3.2 Subjective and objective tools for measuring driver drowsiness

To effectively monitor and mitigate the effects of drowsy driving, researchers and clinicians employ a range of validated assessment tools that can be broadly categorized into subjective and objective methods. *Subjective tools* rely on self-reported data, wherein individuals evaluate their own level of alertness or sleepiness, typically through structured scales such as the Karolinska Sleepiness Scale (KSS), the Epworth Sleepiness Scale (ESS), or the Stanford Sleepiness Scale (SSS). These tools are widely used due to their simplicity and quick administration, but may suffer from bias or underreporting. Observer-Rated Drowsiness (ORD) is a video-based, expert (or trained-annotator) evaluation of driver drowsiness based on visible behavioral cues such as eyelid closures, yawns, head nods, gaze behavior, and posture. It typically employs anchored ordinal scales (e.g., low → high drowsiness) or continuous percentage-style scores over short epochs ($\approx$ 20–60 s), often averaged across multiple raters. ORD is non-intrusive and valuable when self-reports or physiological signals are unavailable, yet it is susceptible to subjectivity, inter-rater variability, and sensitivity to camera placement, illumination, and demographic factors. Best practices include reporting rater training and blinding, epoch duration, inter-rater reliability (e.g., ICC or Cohen's κ), and alignment with other measures such as KSS, EEG, or HRV. Complementing these subjective and behavioral indicators, the Amplitude–Velocity Ratio (AVR)—a core feature of the Johns Drowsiness Scale (JDS)—quantifies the ratio of eyelid closure amplitude to closing velocity, providing an objective and sensitive index of ocular dynamics correlated with fatigue. In contrast, *objective tools* measure physiological or behavioral responses that are indicative of drowsiness without requiring active input from the subject. Examples include eye-tracking systems that monitor blink duration and gaze behavior, electroencephalography (EEG) for analyzing brain wave activity, and the Psychomotor Vigilance Test (PVT), which evaluates sustained attention and reaction time. The integration of both subjective and objective measures offers a more comprehensive framework for detecting drowsiness in real-world settings. A visual summary of these tools and their classification is presented in Fig. 4. In addition, Table 1 outlines and compares each tool in terms of its type, function, and output.

The Karolinska Sleepiness Scale (KSS) is a 9-point subjective scale used to assess an individual's current state of sleepiness Mittal et al. (2016). Participants rate their sleepiness from 1 (extremely alert) to 9 (very sleepy, great effort to stay awake), while the Epworth Sleepiness Scale (ESS) assesses daytime drowsiness via an 8-item questionnaire scored 0–24. It is commonly used in driving simulations, sleep studies, and other experimental contexts Miley et al. (2016); Minhas et al. (2022).

The Epworth Sleepiness Scale (ESS) is an 8-item self-administered questionnaire used to evaluate general daytime sleepiness. Each item is rated from 0 to 3, with a total score ranging from 0 to 24 Johns (1991, 1992). It is widely used in clinical settings to assess sleep disorders like sleep apnea Buysse et al. (2008).

The Stanford Sleepiness Scale (SSS) is a 7-point scale used to measure how alert or sleepy a person feels at a given moment Hoddes et al. (1973). It is ideal for repeated, short-term assessments in laboratory studies or real-time evaluations.

Electroencephalography (EEG) is an objective technique that records electrical activity in the brain. Drowsiness is typically associated with an increase in theta (4–8 Hz) and delta (0.5–4 Hz) wave activity, and a decrease in alpha (8–12 Hz) Stancin et al. (2021).

Eye-tracking techniques are objective tools that monitor ocular behaviors such as blink rate, eye closure (PERCLOS), gaze patterns, and saccades Rahman et al. (2015). These indicators are used in driver monitoring systems to detect drowsiness in real-time Sun et al. (2024). Driver drowsiness assessment tools can be categorized as subjective (self-reported) or objective (physiological/behavioral), as summarized in Table 1. Subjective tools such as KSS, ESS, and SSS, together with objective methods including EEG, eye-tracking, PVT, and AVR-based indices, offer complementary strengths for different monitoring scenarios, from clinical studies to in-vehicle systems.

A key insight from this subsection is that subjective and objective tools each capture different dimensions of drowsiness. Integrating these complementary measures provides a more reliable and comprehensive ground truth for driver drowsiness assessment.

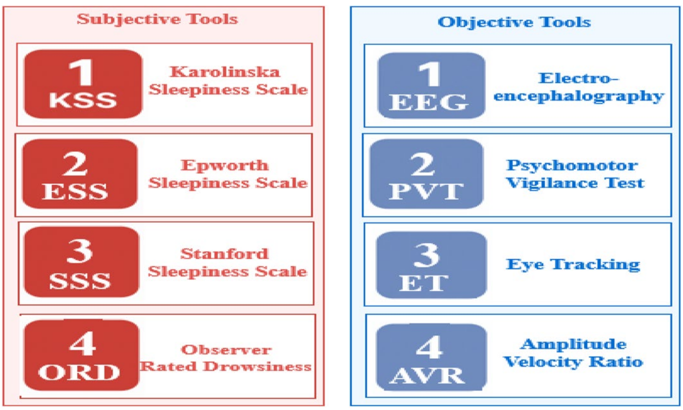


Fig. 4 Classification of commonly used subjective and objective drowsiness assessment tools

Table 1 Summary of key drowsiness assessment tools

Tool	Type	Description/key metric	Typical use case	Measurement output
Karolinska sleepiness scale (KSS)	Subjective	9-Point self-report scale of current sleepiness	Lab or real-time sleepiness evaluation	Numeric score (1–9)
Epworth sleepiness scale (ESS)	Subjective	8-Item questionnaire of daytime sleepiness; total score 0–24	Clinical screening for sleep disorders and daytime sleepiness	Total score (0–24)
Stanford sleepiness scale (SSS)	Subjective	7-Point rating of momentary alertness	Repeated short-term assessments	Rating (1–7)
Observer-rated drowsiness (ORD)	Subjective (observer-based)	Expert ordinal rating per video epoch using behavioral cues (eyelid closures, yawns, head nods, gaze, posture); report ICC or κ	Video-based drowsiness scoring when self-reports/physiology unavailable; complements KSS/EEG/HRV	Ordinal score per epoch (often averaged across raters)
Psychomotor vigilance test (PVT)	Objective	Reaction time and lapse count under sustained attention	Cognitive alertness and vigilance testing	Reaction time distribution, lapse count
Electroencephalography (EEG)	Objective (physiology)	Neural activity; drowsiness associated with increased θ, decreased α and changes in θ/α power ratios	Neuroscience, sleep research, physiological drowsiness ground truth	EEG waveform patterns, band-power features
Eye-tracking techniques	Objective (vision)	Ocular behavior such as PERCLOS, blink rate, gaze patterns, saccades	In-vehicle monitoring systems and real-time drowsiness detection	PERCLOS, blink rate, gaze/saccade metrics
Amplitude–velocity ratio (AVR/JDS)	Objective (ocular)	Ratio of eyelid closure amplitude to velocity; core metric of the Johns Drowsiness Scale	Ocular drowsiness quantification; correlates with behavioral fatigue	AVR index / Johns Drowsiness Scale score

3.3 Operational definitions and standardized nomenclature

To ensure clarity and consistency throughout this review, we establish operational definitions for driver monitoring states and ground-truth standards. These definitions align with terminology established in sleep science, human factors, and transportation safety literature Shen et al. (2006).

3.3.1 Operational definitions

Driver alertness exists along a continuum from optimal readiness to complete incapacitation. We define five key states that characterize this spectrum, each with distinct operational indicators and measurement contexts.

At the optimal end of this continuum, alertness represents a state of full cognitive and physiological readiness to perform the driving task. An alert driver can process information, make decisions, and execute motor responses quickly and accurately. Objective correlates of alertness include fast reaction times on Psychomotor Vigilance Tasks (PVT), stable lane-keeping, and physiological markers such as dominant EEG β-band activity and stable heart rate variability (HRV) patterns.

Fatigue represents a broader, more chronic state of mental and/or physical exhaustion that diminishes alertness and task engagement. This state may result from prolonged workload, circadian misalignment, or cumulative sleep debt. While often conflated with drowsiness, fatigue encompasses longer-term physiological and psychological depletion beyond acute sleep pressure.

Drowsiness constitutes a transitional, sleep-like state of reduced alertness that precedes sleep onset and represents the primary target of detection systems. Characterized by increased sleep propensity, drowsiness manifests through measurable changes across multiple domains. Behaviorally, it presents through prolonged or slow eyelid closures (PERCLOS), yawning, head nodding, and gaze drift. Physiologically, drowsiness is marked by rising EEG θ-power (4–7 Hz), reduced α-power (8–12 Hz), HRV instability, and lowered Amplitude-Velocity Ratio (AVR) in eyelid dynamics. Performance deterioration includes slower reaction times, increased lapses on PVT, and degraded lane-keeping or steering control.

In contrast to objectively measurable drowsiness, sleepiness refers to the subjective feeling of sleep propensity or desire to fall asleep. This state is typically assessed through self-reported standardized scales such as the Karolinska Sleepiness Scale (KSS) or Epworth Sleepiness Scale (ESS). In this paper, sleepiness denotes only subjective self-assessments, whereas drowsiness refers to objectively measurable behavioral and physiological states.

At the extreme end of impairment, microsleep represents a brief, involuntary episode of loss of responsiveness lasting approximately 0.5–15 s, during which the brain enters a sleep-like state. These episodes are commonly identified through EEG/EOG signatures or complete eyelid closure and constitute a critical safety risk, often following severe drowsiness.

Throughout this manuscript, we maintain terminological consistency by employing drowsiness as the operational term for the driver state targeted by detection systems. Sleepiness is reserved for subjective self-reports, while fatigue denotes chronic or cumulative impairment. This standardized nomenclature ensures coherence across behavioral, physiological, and vehicle-based modalities discussed in subsequent sections. Table 2 provides a consolidated summary of these definitions and their corresponding measurement contexts.

Table 2 Operational definitions of key driver alertness states and their corresponding measurement contexts

State	Operational definition	Primary measurement context/ground truth
Alertness	Optimal cognitive and physiological readiness for driving	PVT reaction time, stable vehicle control, EEG β-dominance
Fatigue	Chronic or cumulative exhaustion reducing alertness	Self-report scales, workload metrics
Drowsiness (primary focus)	Transitional sleep-like state preceding sleep onset	Behavioral (PERCLOS, AVR), Physiological (EEG, EOG, HRV)
Sleepiness	Subjective sense of sleep propensity	KSS, ESS self-ratings
Microsleep	Brief unintended sleep episode (0.5–15 s)	EEG/EOG markers, video annotation of eyelid closure

3.3.2 Ground-truth criteria in reviewed studies

The reviewed studies employ various approaches to establish ground-truth labeling for driver states. Reported metrics should therefore be interpreted in the context of these distinct validation paradigms:

- **Subjective scales:** self-reported indices such as KSS, ESS, or Observer-Rated Drowsiness (ORD).
- **Performance-based measures:** reaction-time lapses on PVT or other vigilance-related tasks.
- **Physiological standards:** EEG/EOG-derived thresholds (e.g., θ/α ratios, sleep spindles) and HRV metrics.
- **Behavioral/observational labels:** manual or automated video annotation of behaviors such as eye closure, yawning, or head movement.

A key insight from this subsection is that standardized terminology ensures consistent interpretation of driver states across behavioral, physiological, and performance-based methods. Establishing clear and unified definitions provides conceptual coherence and forms the basis for the recommendations introduced next.

3.4 Recommended ground-truth standards for drowsiness research

Given the heterogeneity of labeling practices across the literature, several concrete recommendations can help establish more reproducible ground truth for driver drowsiness. First, threshold selection should follow established conventions in sleep science. Drowsiness is typically defined as a Karolinska Sleepiness Scale (KSS) score of 7 or higher, while severe drowsiness corresponds to KSS scores of 8–9. These thresholds should be reported explicitly whenever they are used to create labels or stratify datasets.

Observer-Rated Drowsiness (ORD) also requires attention to reliability. ORD annotations should report inter-rater agreement using standard metrics, with acceptable thresholds defined as an intraclass correlation coefficient (ICC) of at least 0.75 or a Cohen's κ of at least 0.70. Raters are expected to undergo training and, when feasible, remain blinded to experimental conditions to minimize bias.

Hybrid labeling strategies can improve the robustness of ground truth, especially in cases where single-modality evidence is ambiguous. Effective hybrid labeling typically combines behavioral cues—such as ORD ratings, PERCLOS, and blink dynamics—with physiological indicators including changes in EEG θ/α ratios or reductions in heart-rate–variability metrics such as SDNN or RMSSD.

Finally, managing inter-rater variability is essential for producing stable and reproducible labels. Recommended practices include averaging annotations across multiple raters, applying temporal smoothing over short windows (for example, three to five consecutive epochs), conducting periodic calibration sessions to maintain annotation consistency, and reporting reliability metrics such as ICC or Cohen's κ alongside any model performance results.

These recommendations are intended to standardize ground-truth practices, facilitate cross-study comparison, and reduce ambiguity in the evaluation of driver drowsiness detection systems. The implications of these standardized definitions and labeling practices are examined further in the Discussion section (Sect. 8).

4 Comprehensive driver drowsiness detection system architecture overview

4.1 General framework and data flow

Modern driver drowsiness detection systems employ a multi-layered architecture that integrates diverse sensing modalities, sophisticated signal processing techniques, and intelligent classification algorithms to achieve real-time monitoring and alert generation. As illustrated in Fig. 5, the comprehensive system architecture consists of four primary functional layers: data acquisition, preprocessing and feature engineering, detection and classification, and alert system management. This modular approach enables the integration of heterogeneous data sources while maintaining system scalability and adaptability to different vehicular environments and driver populations. The architectural diagram in Fig. 5 visualizes the complete system workflow, from multi-modal data collection through intelligent processing to real-time alert activation, providing a unified framework that encompasses all major approaches to drowsiness detection including image-based, biological signal-based, vehicle-based, and hybrid measurement techniques.

4.1.1 Multi-modal data acquisition framework

The data acquisition layer forms the foundation of drowsiness detection systems, encompassing four distinct sensing modalities that capture complementary aspects of driver state.

- **Image-based measures:** constitute the most widely adopted approach, utilizing computer vision techniques to monitor facial features through eye-based methods (tracking blink frequency, eye closure duration, and pupil dynamics), mouth-based methods (de-

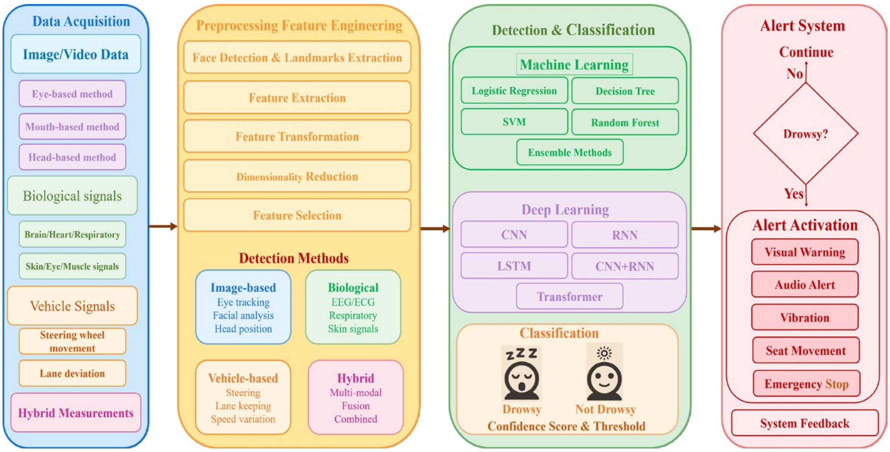


Fig. 5 General framework of driver drowsiness detection system

tecting yawning patterns and oral cavity changes), and head-based methods (analyzing head pose variations and nodding behaviors) Jabbar et al. (2020).
- **Biological signal-based measures:** provide physiological insights into driver alertness through continuous monitoring of brain signals (EEG patterns indicating drowsiness-related neural activity), cardiovascular indicators (heart rate variability and ECG morphology changes), respiratory patterns (breathing rate irregularities), skin conductance variations, and muscle activity signals that reflect decreasing motor control during drowsiness onset Hasan et al. (2024).
- **Vehicle-based measures:** leverage the vehicle's existing sensor infrastructure to detect drowsiness-induced driving performance degradation, primarily through steering wheel movement analysis (detecting micro-corrections and erratic steering patterns) and lane deviation monitoring (tracking lane-keeping performance and lateral position variations) Perkins et al. (2021).
- **Hybrid approaches:** represent the current frontier in drowsiness detection, combining multiple sensing modalities to overcome individual sensor limitations and improve overall system robustness. These multi-modal systems integrate vehicle dynamics with physiological monitoring, combine image analysis with biological signals, or fuse all available data sources to create comprehensive driver state assessment frameworks Priyanka et al. (2024b).

4.1.2 Signal processing and feature engineering pipeline

The preprocessing and feature engineering layer transforms raw sensor data into meaningful representations suitable for machine learning algorithms. This critical stage begins with signal conditioning and noise reduction, followed by feature extraction techniques tailored to each sensing modality. For image-based systems, this involves face detection algorithms, facial landmark extraction, and geometric feature computation. Biological signals undergo filtering, artifact removal, and spectral analysis to extract relevant physiological markers. Vehicle signals are processed to identify driving pattern anomalies and performance degradation indicators Qu et al. (2023).

Feature transformation and dimensionality reduction techniques are subsequently applied to optimize the feature space for classification tasks. Principal component analysis, independent component analysis, and other dimensionality reduction methods help eliminate redundant information while preserving discriminative features. Feature selection algorithms further refine the feature set by identifying the most relevant indicators of drowsiness across different modalities. This systematic approach to feature engineering ensures that the classification algorithms receive high-quality, informative inputs that enhance detection accuracy and reduce computational complexity.

4.1.3 Advanced classification and detection methodologies

The detection and classification layer employs both traditional machine learning and deep learning approaches to distinguish between alert and drowsy states Hassan et al. (2024).

- **Machine learning methodologies:** include logistic regression for binary classification tasks, decision trees for interpretable rule-based detection, support vector machines for high-dimensional feature spaces, random forest algorithms for robust ensemble learning, and various ensemble methods that combine multiple classifiers to improve overall performance. These approaches are particularly effective when dealing with hand-crafted features and smaller datasets El-Nabi et al. (2025).

- **Deep learning architectures:** have revolutionized drowsiness detection by automatically learning hierarchical feature representations from raw sensor data. Convolutional Neural Networks (CNNs) excel at processing image and video data for facial analysis and behavioral pattern recognition. Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks effectively model temporal dependencies in physiological signals and driving behavior sequences Chirra et al. (2019). Hybrid CNN-RNN architectures combine spatial and temporal modeling capabilities, while Transformer-based models leverage attention mechanisms to capture long-range dependencies in multi-modal data streams Altameem et al. (2021); El-Nabi et al. (2023).

4.1.4 Real-time alert system and feedback mechanisms

The alert system layer implements intelligent decision-making processes that translate drowsiness classification results into appropriate interventions. The system employs threshold-based decision logic with confidence scoring to minimize false alarms while ensuring reliable detection of genuine drowsiness events Hassan et al. (2024). Upon positive drowsiness detection, the system activates a graduated alert protocol that may include visual warnings (dashboard indicators, heads-up displays), auditory alerts (alarms, voice notifications), tactile feedback (seat vibration, steering wheel vibration), and in critical situations, automated safety interventions such as emergency braking or lane-keeping assistance Deng and Ruoxue (2019).

A crucial component of modern drowsiness detection systems is the continuous feedback loop that enables system adaptation and performance optimization Aidman et al. (2015). This mechanism logs detection events, driver responses, and environmental conditions to refine classification thresholds, update model parameters, and improve personalization for individual drivers. The feedback system also supports system validation through performance metrics monitoring, including accuracy, precision, recall, F1-score, and real-time processing latency measurements Wang et al. (2023). This adaptive capability ensures that the system maintains high performance across diverse driving conditions, vehicle types, and driver populations while continuously learning from operational experience.

A key insight from this section is that modern driver drowsiness detection systems share a common architectural foundation built on multi-modal sensing, structured signal processing, and layered decision-making. This unified workflow highlights how diverse sensing modalities and analytical components interact within a coherent pipeline, enabling more reliable and scalable real-time drowsiness detection across varied driving environments.

4.2 Public datasets for drowsiness detection

Public datasets serve as the backbone for training, validating, and benchmarking driver drowsiness detection systems. These datasets can be categorized based on the **modality** of data they capture, reflecting the diversity of approaches in this field. As illustrated in Fig. 6, the datasets fall into four primary categories: *Image and Video-based*, *Physiologi-*

cal Signal-based, *Vehicle Telemetry-based*, and *Hybrid-based*. Image- and video-based datasets (e.g., `NTHU-DDD`, `UTA-RLDD`) constitute the largest category due to their non-intrusive nature and compatibility with in-vehicle cameras Weng et al. (2017), *300-W* Song et al. (2014), *ZJU Gallery* Pan et al. (2007), *YawDD* Abtahi et al. (2014), *CelebA* Liu et al. (2015), *MRL Eye* Fusek (2018), *CEW* Song et al. (2014), and *UTA-RLDD* Ghoddoosian et al. (2019). Physiological signal datasets (e.g., `Sleep-EDF`, `MIT-BIH`) provide precise neurophysiological markers but require specialized sensors Kemp et al. (2000). Vehicle telemetry and hybrid datasets remain less explored but are critical for holistic approaches. For conciseness, we focus on benchmark datasets that have demonstrated reproducibility across multiple studies, excluding implementation-specific preprocessing routines documented in their original releases. Notable examples for the Vehicle telemetry include the *ONDSS* Oak Ridge National Laboratory (2016) and the large-scale *SHRP2 Naturalistic Driving Study* Virginia Tech Transportation Institute (2025), both of which offer long-term, real-world driving data collected from thousands of drivers; and the *SENSORIS Data Standard* SENSORIS Consortium (2018), which aggregates fleet-level telematics for behavior modeling. Notable examples of the hybrid datasets include the *Driver Monitoring Dataset (DMD)* Ortega et al. (2020), which combines in-cabin video recordings with vehicle telemetry data via the CAN bus; the *DROZY* dataset Massoz et al. (2016), which offers synchronized EEG, ECG, thermal infrared facial video, and audio signals; the *Driver Fatigue Warning Dataset (DFW)* Kushwaha et al. (2018), designed to support the integration of facial expressions and driving behavior metrics.

To contextualize the four modality categories, Table 3 offers a side-by-side comparison of the most frequently cited public datasets for driver-drowsiness research. For each corpus we list its reference, release year, sensing modalities, cohort size, and principal events or labels, together with provenance metadata such as DOI/access links, acquisition context, annotation protocol, and known limitations. The chronological ordering within each modality group highlights the field's evolution—from early image collections to recent multimodal hybrid sets that fuse video, physiology, and telemetry. To promote transparency and reproducibility, all dataset metadata from Table 3 are also provided in a machine-readable

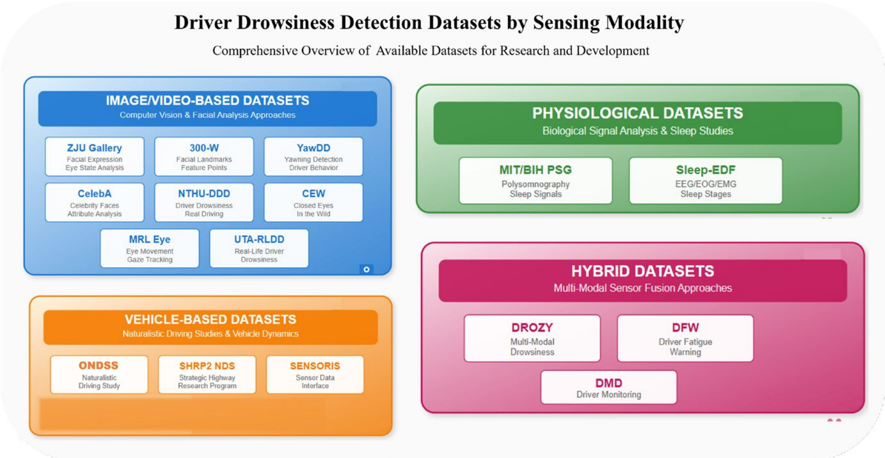


Fig. 6 Taxonomy of publicly available datasets for driver drowsiness detection, organized by sensing modality

Table 3 Summary of public datasets used for driver drowsiness detection across different modalities

References	Year	Name	Modalities	#Subjects	Annotated events	Modality	Ground-truth family	Annotation protocol (brief)
Sagonas et al. (2013)	2013	300-W	Facial Images	N/A (aggregated from multiple datasets)	68 facial landmarks	Image/Video	Geometric (facial keypoints)	Semi-automated annotation of 68 fiducial points
Abtahi et al. (2014)	2014	YawDD	RGB Video	≈10–22	Yawning	Image/Video	Behavioral (yawn, eye closure)	Manual labeling of yawns frame-by-frame
Liu et al. (2015)	2015	CelebA	Facial Images	202,599 images / 10,177 identities	40 facial attributes	Image/Video	Attribute (facial attributes)	Crowdsourced labeling of 40 attributes
Song et al. (2014)	2014	CEW (Closed Eyes in the Wild)	Eye Images	Unknown	Eye state (open/closed)	Image/Video	Binary (open/closed)	Manual labeling by human. Manual annotation by human reviewers
Oak Ridge National Laboratory (2016)	2016	ONDSS (ORNL NDS Sample)	Telemetry, GPS	~20–30	Driving maneuvers (e.g., hard braking, lane deviations)	Vehicle-based	Event-based (derived from signals)	limited manual validation
Weng et al. (2017)	2016	NTHU-DDD	RGB, IR Video	36	Blinking, Yawning, Head pose	Image/Video	Behavioral (blink, yawn, head pose)	Frame-level annotation by experts
Fusek (2018)	2018	MRL Eye	Eye Images	Unknown	Eye state	Image/Video	Binary (open/closed)	Labeled via acquisition metadata
Ghoddoosian et al. (2019)	2019	UTA-RLDD	RGB Video	60 subjects	Head, eye, yawning	Image/Video	Behavioral (awake vs drowsy episodes)	Expert annotation using facial cues
Pan et al. (2007)	2007	ZJU Eyeblink Dataset (ZJU Gallery)	Video (RGB)	20 individuals	Eye blinks (open/closed states)	Image/Video	Frame-level eye state with blink start/peak/end	Blink start/peak/end frames; inconsistent annotation
Virginia Tech Transportation Institute (2025)	2014	SHRP2 NDS	Vehicle signals	3,000+	Fatigue, safety events	Vehicle	Behavioral (fatigue/safety events)	Expert review of cabin video + sensor triggers
Kemp et al. (2000)	2000	Sleep-EDF	EEG, EOG	20+	Sleep stages	Physiological	Physiological (sleep stages)	Manual scoring by trained sleep technicians

Table 3 (continued)

References	Year	Name	Modalities	#Subjects	Annotated events	Modality	Ground-truth family	Annotation protocol (brief)
SENSORIS Consortium (2018)	2018	SENSORIS	Fleet Telematics	NA	Vehicle behavior (operational logs)	Vehicle	Operational (vehicle state signals)	Signal-level logging defined by standard
Massoz et al. (2016)	2016	DROZY	EEG, ECG, IR Video	14	Sleepiness rating	Hybrid	Physiological (EEG/KSS)	EEG-based KSS alignment
Kushwaha et al. (2018)	2018	DFW	Video + driving signals	Unknown	Fatigue events	Hybrid	Hybrid (visual + telemetry)	Frame-level annotation with CAN sync
Ortega et al. (2020)	2020	DMD	Video + CAN Bus + Audio	50+	Driver state (visual + telemetry events)	Hybrid	Behavioral (visual/acoustic cues)	Semi-automated sync with event tagging

format as Supplementary Table S-DATA (`Supplementary_Table_S-DATA_Dataset_Metadata.csv`), enabling verifiable cross-reference of each dataset's provenance and usage within drowsiness-detection research.

The supplementary file `Supplementary_Table_S-DATA_Dataset_Metadata.csv` contains the following columns: Ref, Year, Name, Modalities, # Subjects, Annotated Events, Modality, DOI / Access Link, Provenance / Notes, Availability, Acquisition Context, Sensor Modalities, Ground-Truth Family, Annotation Protocol (brief), and Known Limitations.

A key insight from this section is that public datasets vary substantially in sensing modalities, annotation protocols, and ground-truth standards, which directly influences model comparability and reproducibility. Understanding these differences is essential for selecting appropriate benchmarks and motivates the need for more standardized, multi-modal datasets to support future advancements in drowsiness detection research.

5 Drowsiness detection modalities: techniques based on measurement domains

To provide a clearer understanding of how drowsiness detection methods are organized across different input domains, this section categorizes the techniques into four primary groups: image- and video-based, biological signal-based, vehicle behavior-based, and hybrid measurement techniques. Each category encapsulates a distinct set of approaches, offering varying advantages depending on the application context and required level of precision. Figure 7 visually summarizes this classification, outlining the subcategories under each modality and illustrating the structural relationship between them.

5.1 Image- and video-based measurements

Image- and video-based measurements constitute one of the most extensively investigated modalities for driver-drowsiness detection because they are non-intrusive yet capable of capturing subtle, behavior-related manifestations of fatigue. In the literature, the measurable cues derived from facial expressions, ocular dynamics, head kinematics, and mouth activity are often termed visual signs Pratama et al. (2017) and are generally regarded as physical or behavioral indicators Ramzan et al. (2019); Sikander and Anwar (2018); Gomaa et al. (2022); Gomaa (2024); Gomaa and Abdalrazik (2024); Gomaa and Saad (2025); Salem et al. (2023). Using in-cabin cameras in conjunction with advanced computer-vision algorithms, researchers extract features such as blink rate, the percentage of eyelid closure over time (PERCLOS), eye-aspect ratio (EAR), yawning frequency, and head-pose angles. More sophisticated techniques—eye tracking, gaze estimation, and motion sensing with gyroscopes or accelerometers—further enhance real-time awareness of driver attention Mehreen et al. (2019). To facilitate systematic comparison, Table 4 organizes image-based drowsiness-detection systems into four primary categories—eye, face, mouth, and head measurements—and lists the corresponding features together with their functional roles in state-of-alertness assessment.

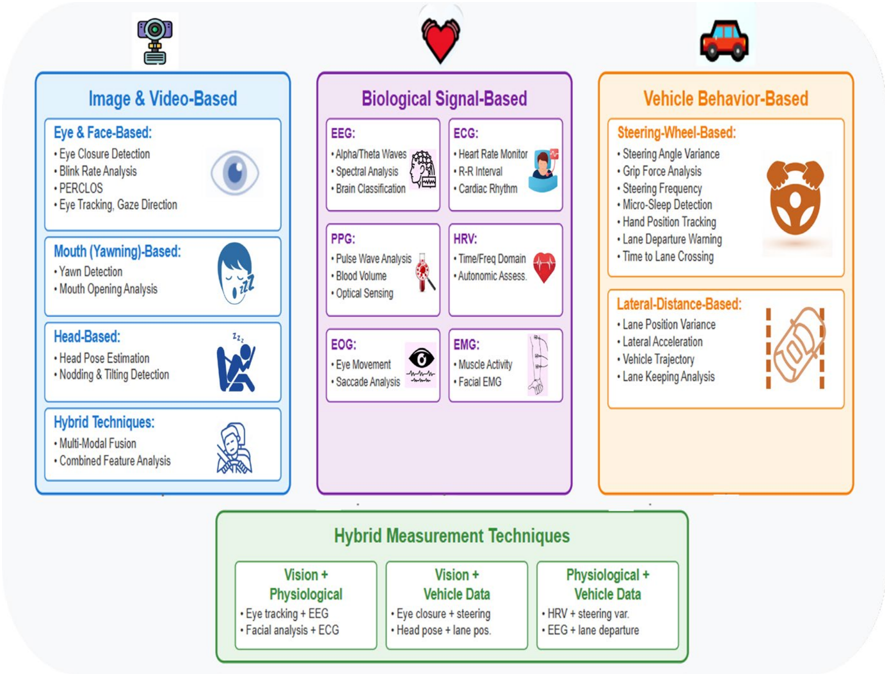


Fig. 7 Taxonomy of drowsiness detection modalities based on measurement domains

5.1.1 Eye- and face-based measurements

Automated analysis of eye and facial dynamics constitutes the cornerstone of most vision-based drowsiness–detection pipelines. By continuously estimating ocular state (open, closed, or partially closed), blink frequency, eye-aspect ratio, and micro–expressions, these systems can identify the early-onset physiological changes that precede observable lapses in driver performance Lamaazi et al. (2023). Beyond fatigue monitoring, eye- and face-based classifiers underpin a wide spectrum of computer-vision tasks—including facial-emotion recognition, attentiveness assessment in human–computer interaction, and gaze-based user interfaces—underscoring their cross-disciplinary relevance. The principal algorithms and feature sets employed in state-of-the-art image- and video-based drowsiness-detection systems are summarized in Table 5, which provides a structured comparison across datasets, subject counts, validation types, ground-truth definitions, model architectures, reported performance metrics, and methodological strengths and limitations within this modality.

In Maior et al. (2020), the authors presented a real-time drowsiness detection system grounded in computer vision and machine learning techniques, aimed at non-intrusive monitoring of eye behavior. Utilizing a standard webcam and avoiding the need for biological sensors, their approach focuses on blink pattern analysis inspired by neuroscience-based drowsiness criteria. The system integrates temporal information across video frames to enhance pattern recognition using classification models, including multilayer perceptron (MLP), random forest (RF), and support vector machines (SVM)—with SVM delivering the best performance. Validated on the DROZY public dataset, the method achieved

Table 4 Image-based drowsiness detection features

References	Year	Feature	Category	Description
Bamidele et al. (2019)	2019	Blink Rate Analysis	Eye-based	The number of blinks per unit time. An unusually high or low blink rate can indicate fatigue or drowsiness
Turki et al. (2024)	2024	Maximum Eye Closure Time	Eye-based	The longest continuous duration of eye closure. Extended closures may reflect drowsiness or microsleeps
Du et al. (2022)	2022	PERCLOS	Eye-based	Percentage of time per minute that the eyes are at least 80% closed. Strongly associated with reduced alertness
Dewi et al. (2022)	2022	Eye Aspect Ratio (EAR)	Eye-based	A geometric feature representing eye openness. Lower EAR values indicate eye closure and fatigue
Mehta et al. (2019)	2019	Eye Closure Detection	Eye-based	Determines whether eyes are currently open or closed using real-time image processing
Souchet et al. (2022)	2022	Eye Tracking	Eye-based	Tracks pupil movement, fixation, and drift patterns which can signal cognitive fatigue
Kolus (2024)	2024	Gaze Direction	Eye-based	Detects driver attention by analyzing gaze vector. Deviation from road focus may indicate distraction or drowsiness
Al-Madani et al. (2021)	2021	Facial Landmarks	Face-based	Identification of key facial points (eyes, mouth, nose) used to extract fatigue-related features
Moujahid et al. (2021)	2021	Yawning Frequency	Mouth-based	Counts how often the mouth opens widely within a time window. Frequent yawns suggest drowsiness
Ansari et al. (2021)	2021	Head Pose	Head-based	Monitors head orientation based on Euler angles. Nodding or tilting motions may reflect alertness loss

94.9% average test accuracy and was capable of producing targeted alerts only during drowsy states while maintaining no false positives during alert periods. The framework also includes a feedback loop for individual calibration, enabling personalized adaptation. These results highlight the system's practical viability for real-time drowsiness monitoring in safety-critical environments.

Celecia et al. (2020) introduced a portable, low-cost driver-drowsiness estimation device that operates in real time on a Raspberry Pi platform. The system fuses complementary facial cues extracted from a sliding temporal window: PERCLOS and Eye Closure Duration (ECD) derived from eye states, and Average Open Time (AOT) obtained from mouth dynamics. These indicators are combined by a Mamdani-type fuzzy inference system to classify driver alertness into three levels—Low (normal), Medium (drowsy), and High (severely drowsy). Trained and evaluated on the 300-W facial dataset, the prototype reached an overall state-recognition accuracy of 95.5%, demonstrating both robustness to illumination variations and suitability for embedded deployment. This work underscores the feasibility of integrating fuzzy inference with lightweight facial features to deliver reliable, real-time drowsiness monitoring in cost-sensitive automotive applications.

The authors in Saurav et al. (2022) proposed a real-time eye state recognition system based on a Dual Convolutional Neural Network Ensemble (DCNNE), aimed at delivering high accuracy while maintaining computational efficiency for deployment on embedded platforms. The approach utilizes two lightweight CNNs trained on grayscale eye patch inputs, which are then jointly fine-tuned using transfer learning to address overfitting issues

Table 5 Image- and video-based drowsiness detection systems by visual feature type

References	Year	Dataset(s)	Image/video features	Model	Reported metric
Ed-Doughmi et al. (2020)	2020	NTHU-DDD (5 scenarios; day/night; RGB & IR)	Spatiotemporal patches; 16–40-frame clips; $3 \times 3 \times 3$ kernels (C3D-style)	3D ConvNets multi-layer (Conv–Pool×4–FC–Softmax)	Test Acc. 82.0% (40-frame seq.); F1 81%; Val Acc. 90.4%; headline train Acc. ~97%
Maior et al. (2020)	2020	DROZY	Sequences of EAR (Eye Aspect Ratio) values concatenated over frames	SVM, also MLP and RF compared	Accuracy 94.9% (average test)
Celecia et al. (2020)	2020	Custom recorded video dataset (2 subjects) / 300-W (for landmark training only)	PERCLOS, Eye Closing Duration (ECD), Average Mouth Opening Time (AOT); landmarks from 300-W model	Mamdani fuzzy inference	Accuracy 97.1% (laptop), 95.5% (Raspberry Pi)
Zhao et al. (2020)	2020	Private Biteda dataset (real-driving images); auxiliary face detection pre-trained on WIDER Face + CelebA	Eye and mouth ROI images (175×175); PERCLOS + POM features	EM-CNN + MTCNN for face detection	Accuracy 93.623%, Sensitivity 93.643%, Specificity 60.882%
Wijnands et al. (2020)	2020	NTHU-DDD	10-frame clips @30 fps (≈333 ms); resized to $224 \times 224 \times 10$; random flip/brightness/translate-zoom-stretch	3D-CNN (Conv3D + pooling + FC)	Overall acc. 73.9% (eval set); scenario acc.: sunglasses 76.8%, night-glasses 63.6%; human ≈80.8%
Moujahid et al. (2021)	2021	NTHU-DDD	Face texture representation via Pyramid Multi-Level (PML); local & global patch features (HOG, LBP, Covariance matrix fusion) + feature selection (PCA + Fisher Score)	SVM + blending scheme combining HOG / LBP / COV descriptors	Accuracy 79.84% (on evaluation subset)
Saurav et al. (2022)	2022	ZJU (eye patches); also CEW and MRL Eye (cross-dataset evaluation)	Eye patch images extracted from face region	Dual CNN ensemble (DCNNE) with transfer learning and fine-tuning	ZJU 97.99%; CEW 97.56%; MRL 98.98%

Table 5 (continued)

References	Year	Dataset(s)	Image/video features	Model	Reported metric
Magán et al. (2022)	2022	UTA-RLDD, YawDD	Alternative I: Preprocessed face images (64 × 64px, Gaussian blur, histogram equalization, ImageNet mean subtraction), Alternative II: Eye landmarks (EAR), blinks, microsleeps, yawns detected via facial landmarks and CNN	Alternative I: CNN (EfficientNetB0) + GRU (recurrent neural network), Alternative II: Deep learning (EfficientNetB0 for yawn detection) + Fuzzy Inference System (Mamdani FIS with 11 rules)	Alternative I: Training accuracy ~65%, Test accuracy ~ 55%, FP rate 40%, Alternative II: Training accuracy ~66%, Test accuracy ~63%, FP rate 7% (specificity 93%)
Pandey et al. (2022)	2022	Self-prepared	Preprocessed driver facial frames and traffic sign images using detection, binarization, and localization; extracted ROIs (eyes, mouth, road sign)	CNN	Accuracy 98.53%
de Lima Medeiros et al. (2022)	2022	YEC (YouTube Eye-state Classification), ABD (Autonomous Blink Detection), CeW, ZJU, Eyeblink8, Talking Face	Eye patches (40 × 40), facial landmarks, RGB features, CLAHE preprocessing, temporal sliding-window filtering	CNN and SVM models; auxiliary models: Rotation Compensator (RC), Eye Evaluator (EE), and Moving Average Filter	97.44% accuracy (eye-state, CeW), 92.63% F1-Score (eye-blink detection, ABD)
Ahmed et al. (2023)	2023	Kaggle "Driver Drowsiness" dataset (2,900 images, 4 classes: open, closed, yawning, no yawning)	RGB facial features (eyes, mouth, head position, illumination), face extraction using Haar cascade, resized to 145 × 145 (CNN) and 64 × 64 (VGG16)	CNN; VGG16	CNN: 97% accuracy, 99% precision, recall, F1-score; VGG16: 74% accuracy
Albadawi et al. (2023)	2023	NTHU-DDD	Visual labels — manually annotated drowsy and alert states using thresholds on Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), and head pose movement	Random Forest (RF), Sequential Neural Network (NN), Linear SVM	RF: 99% accuracy, 99% sensitivity, 99% specificity, 99% macro precision, 99% macro F1-score; NN: 96% accuracy; SVM: 80% accuracy
Essel et al. (2024)	2024	YawDD, UTA-RLDD	Static: ECR (Eye Closure Ratio) and MAR (Mouth Aperture Ratio) from Dlib 68 facial landmarks, Adaptive: Same features + consecutive frame counting	Static Threshold: Dlib facial landmark detection with fixed thresholds (ECR=0.15, MAR=0.4), Adaptive Threshold: Dynamic consecutive frame analysis (90→60 frames for MAR; 60→8 frames for ECR over 30min)	Static: Accuracy 89.4%, Sensitivity 96.5%, Adaptive: Max accuracy 98.2%, Sensitivity 64.3%

Table 5 (continued)

References	Year	Dataset(s)	Image/video features	Model	Reported metric
Florez et al. (2024)	2024	NITYMED	Eye ROI: Corrected eye region from MediaPipe 468 facial landmarks (points 63, 117, 293, 346, 107, 336), Mouth ROI: MAR from 8 facial points (11, 16, 61, 73, 180, 291, 303, 404)	Hybrid approach: Eye state: 4 CNNs compared - InceptionV3, VGG16, ResNet50V2, and proposed DD-AI (trained from scratch) Yawning: MAR threshold (>55 for 5 s triggers alarm)	Test set (software): DD-AI: 99.88%±0.1% accuracy, 99.88%±0.1% recall, Hardware (Jetson Nano): DD-AI: 96.55% accuracy, 9-14 fps, Yawning detection: 100% accuracy (MAR-based)
Wu et al. (2024)	2024	300W; COFW	Facial landmarks: 300W, COFW annotations, Drowsiness: NR (likely visual labels based on eye/mouth states)	Joint detection: Simultaneous face + facial landmark localization, Drowsiness features: Eye and mouth information extracted from detected landmarks	Qualitative only
Bisogni et al. (2024)	2024	Pandora (primary), Biwi, ICT-3DHP (generalization)	Depth images only (no RGB): 68 facial landmarks from depth (MediaPipe-adapted), Fractal encoding (PIFS) on depth mask, Keypoint intensity (gray-level depth values at 68 landmarks)	HYDE-F: Hybrid rule-based fusion system, Fractal encoding method (PIFS self-similarity), Keypoint intensity method (depth gray levels), Score-level fusion via Nelder-Mead optimization, No deep learning used	MAE 5.4°
Jarndal et al. (2024)	2024	NTHU-DDD; UTA-RLDD	Entire face images (100 × 100 pixels): Dlib HOG-based face detection and cropping, No specific facial features extracted, Divided into 256 equal patches (16 × 16), Patch embedding's with positional encoding	ViT	NTHU 98.89%; UTA 99.4%
Owen and Surantha (2025)	2025	NTHU-DDD	Handcrafted features — facial landmarks (eyes, mouth, head), PERCLOS (percentage eyelid closure), POM (percentage mouth opening), head pose angles (yaw, pitch, roll) extracted using Dlib and HOPEnet (ResNet50 backbone)	SVM; RF; XGBoost	Accuracy: 86.85% (XGBoost), Model size: 5.05 MB; Precision = 0.96, Recall = 0.95, F1 = 0.95 for drowsiness class
Hassan et al. (2024)	2025	MRL; NTHU-DDD; CEW	Cropped eye and facial-region frames from videos; pixel intensities used directly by CNN; data augmented by rotation + flipping	Transfer-learning CNNs – VGG16, InceptionV3, MobileNetV2 pre-trained on ImageNet, fine-tuned on MRL + NTHU-DDD + CEW	ViT 99.15%; Swin 99.03%

Table 5 (continued)

References	Year	Dataset(s)	Image/video features	Model	Reported metric
Abbasi et al. (2025)	2025	Real-world on-road data (Galway City, Ireland) with 5 intersections featuring cycle lanes	Gaze vectors & fixation duration on AOIs (left mirror, left window, frontal area) extracted via SmartEye Pro, combined with CAN signals (speed, brake, steering)	Rule-based hazard algorithm with 7 danger levels (0–6) integrating gaze + speed + distance; no ML model used	Result: 85% of drivers failed to check left mirror before turning. Algorithm detected mirror-checking behavior reliably; real-world validation across 5 intersections
Delwar et al. (2025)	2025	Kaggle Yawn Eye Dataset (2,467 images of eyes & faces) + Health Informatics Dataset (144 images: 70 eyes, 74 mouths)	Cropped eye and mouth ROIs via Haar cascade + 68-point landmark features; RGB and grayscale frames resized to 145 × 145 px	Custom CNN, VGG16, MobileNet (models trained and compared); MobileNet performed best	Accuracy: 92.75% (Kaggle) & 91.98% (Health Informatics); F1 = 0.93 (MobileNet)
Narayana (2025)	2025	NTHU-DDD; YawDD	Facial landmarks via Dlib/OpenCV; extracted Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), blink frequency, and yawn count	SVM and CNN (hybrid EAR–MAR–SVM model performs best)	Accuracy: 92.3%; Precision: 91.7%; Recall: 90.8%; F1-score: 91.2%; detection delay ≈ 1.5 s
Borawake et al. (2025)	2025	Custom camera feed (live video stream; not tied to public dataset)	Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), and Head Pose Estimation using OpenCV and Dlib on infrared camera input; continuous monitoring	CNN + IoT integration	Accuracy not numerically reported; qualitative real-time performance validated
Essahraui et al. (2025)	2025	NTHU-DDD (36 IR videos), YawDD (351 videos), UTA-RLDD (60 subjects, 30 h RGB)	Face & eye ROI via Haar cascade; HOG feature extraction; bounding boxes for YOLO/Faster R-CNN	ML: KNN, SVM, DT, RF; DL: CNN, YOLOv5, YOLOv8, Faster R-CNN	KNN: 98.89% (UTA-RLDD); CNN: 99.97%; YOLOv5/YOLOv8: Precision = 99.9%, Recall = 100%, mAP@0.5 = 99.5%; Faster R-CNN: 81.0%

NR = not reported. Validation codes: SD (subject-dependent), SI (subject-independent), LOSO (leave-one-subject-out), CD (cross-dataset), CV (k-fold; specify per-subject vs pooled). Ground Truth lists the labeling source used to compute accuracy (e.g., KSS/SSS/EEG/PVT vs. visual labels). Accuracies should be interpreted within dataset + protocol; cross-paper numbers are not directly comparable commonly encountered with limited eye state datasets. Their model achieved state-of-the-art recognition accuracies of 97.99% on the ZJU dataset, 97.56% on CEW, and 98.98% on MRL. The system demonstrated real-time capabilities, running at 62 FPS on the Nvidia Xavier and 11 FPS on the Intel NCS2 platform, making it well-suited for practical applications such as driver drowsiness detection. While the model excels in accuracy and speed, its effectiveness may be constrained in scenarios with poor lighting or occluded eye regions.

Magán et al. (2022) introduced a non-intrusive driver drowsiness detection system as part of an Advanced Driver Assistance System (ADAS) using image sequences that are 60 s long. The authors developed two alternative solutions: the first leverages a hybrid deep learning architecture combining Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN), while the second extracts numerical features from facial video data and processes them using a fuzzy logic-based classifier. Both approaches achieved comparable detection accuracies (approximately 65% on training and 60% on testing datasets), but the fuzzy logic model demonstrated a superior specificity of 93%, effectively reducing false positive alerts. This characteristic is crucial for maintaining driver trust in real-world applications. Despite the moderate overall accuracy, the study offers a solid foundational framework for developing more refined and reliable drowsiness detection systems in the future.

In Pandey et al. (2022), the authors proposed a dual-purpose vision-based system designed to both detect driver drowsiness and recognize traffic signs in real time, aiming to enhance overall road safety. Recognizing the severity of driver fatigue as a contributor to traffic accidents, the study underscores that drowsiness can impair reaction time as severely as intoxication. The proposed system utilizes a Convolutional Neural Network (CNN) architecture to process video frames of the driver's face and the driving environment. It performs initial preprocessing, followed by binarization and region localization to isolate relevant features. For the drowsiness detection module, the system monitors signs of fatigue, such as facial lethargy and reduced attentiveness, issuing auditory alerts when thresholds are met. Simultaneously, traffic sign recognition is conducted through deep learning-based classification, ensuring that drivers receive real-time cues even if they momentarily miss visual signage due to fatigue. The model achieved a high classification accuracy of 98.53% using a self-prepared dataset, demonstrating strong potential in both driver monitoring and contextual road interpretation. This integrated approach, combining driver state assessment with environmental awareness, exemplifies the evolving sophistication of intelligent driving assistance systems.

In Ahmed et al. (2023), the authors introduce a deep-learning-based system aimed at enhancing road safety by detecting driver drowsiness through facial and ocular cues. Their approach leverages two convolutional architectures—CNN and VGG16—to classify drowsiness-related facial expressions into four distinct categories: open eyes, closed eyes, yawning, and no yawning. A curated dataset of 2,900 labeled images encompassing variations in gender, head position, lighting, and age was used for training and testing. The CNN model demonstrated superior performance, achieving an accuracy of 97%, along with a precision, recall, and F-score of 99%, whereas the VGG16 model achieved 74% accuracy. These results underscore the efficacy of lightweight CNN architectures for real-time applications in constrained environments while also highlighting the need for model selection based on deployment context. The study represents a significant contribution toward non-intrusive, camera-based fatigue detection systems.

In Hassan et al. (2024), the authors proposed a cutting-edge framework for real-time driver drowsiness detection using transformer-based deep learning architectures, including Vision Transformer (ViT) and Swin Transformer, alongside several fine-tuned CNN-based transfer learning models such as VGG19, DenseNet169, and InceptionV3. Their system utilized the MRL Eye Dataset primarily, with additional validation on NTHU-DDD and CEW datasets to ensure generalization. The ViT and Swin Transformer models outperformed conventional approaches, achieving accuracy rates of 99.15% and 99.03% respectively. The approach incorporated real-time drowsiness scoring, Haar Cascade-based region-of-interest selection, and Class Activation Mapping (CAM) to enhance model interpretability. This research sets a new standard for driver monitoring systems, combining high accuracy with robustness under real-world conditions such as variable lighting and the presence of eyewear.

5.1.2 Mouth (Yawning)-based measurements

Yawning is one of the most visible and commonly observed behavioral indicators of drowsiness, making mouth-based analysis a valuable component in image- and video-based detection systems Al-Madani et al. (2021). Automated detection of yawning typically involves monitoring the temporal dynamics of mouth opening, lip contour deformation, and jaw motion using facial landmarks or deep learning–based models Makhmudov et al. (2024). These methods are designed to differentiate between brief speech-related movements and prolonged yawns, which are more strongly associated with fatigue. As part of broader visual monitoring strategies, yawning features are often used in combination with other facial cues to improve reliability. Representative systems that incorporate mouth-based measurements are included in Table 5, alongside other visual feature categories for comprehensive comparison.

In Florez et al. (2024), the authors proposed a real-time embedded system for driver drowsiness detection that integrates convolutional neural networks (CNNs) and mouth aspect ratio (MAR) analysis. Their approach combines a custom deep learning model, named DD-AI, with classical facial landmark processing to classify eye states (open/closed) and detect yawning events. The system was implemented on an NVIDIA Jetson Nano embedded device equipped with an infrared camera, enabling reliable performance even in low-light environments. Experiments conducted on the NITYMED dataset and real vehicle tests demonstrated impressive results, with the DD-AI model achieving 99.88% accuracy on benchmark data and 96.55% accuracy during on-road trials. By leveraging both eye and mouth cues, the proposed method enhances robustness against isolated failures of single features, providing an effective solution for real-time drowsiness monitoring on embedded platforms.

In Essel et al. (2024), the authors propose a lightweight, real-time driver drowsiness detection system that relies on facial feature analysis of eye and mouth dynamics. Using videos from the UTA-RLDD and YawDD datasets, the system tracks facial landmarks with Dlib to calculate the Eye Closure Ratio (ECR) and Mouth Aperture Ratio (MAR), which serve as primary indicators of drowsiness. Two thresholding techniques were developed: a static threshold, which uses fixed ECR and MAR values to classify drowsiness, and an adaptive threshold, which adjusts the thresholds dynamically over time based on the duration of driving, improving sensitivity to gradual onset of fatigue. Experimental results show that the adaptive threshold method achieved a detection accuracy of 98.2% and static thresholding obtained 96.5% sensitivity, outperforming many traditional techniques. The system's low computational complexity and reliance on standard cameras make it suitable for practical in-vehicle deployment, although its performance can degrade under challenging conditions such as extreme head rotations or poor lighting. This study highlights the potential of simple yet effective facial metrics for robust and affordable driver monitoring solutions.

In Delwar et al. (2025), Delwar et al. proposed a comprehensive driver drowsiness detection system leveraging deep learning models trained on facial features, including eyes and mouth regions, to accurately classify drowsiness states. The study integrated CNN, VGG16, and MobileNet architectures, employing feature extraction via Haar cascades for precise localization of eyes and yawns. The models were trained and validated on a Kaggle dataset and tested on a distinct Health Informatics dataset to assess generalizability. Among the models, MobileNet excelled, achieving 92.75% accuracy on the Kaggle test set and 91.98% on the unseen dataset, demonstrating robustness, computational efficiency, and suitability for real-time driver monitoring in resource-constrained environments. The work emphasizes the critical role of diverse training data and efficient architectures for effective drowsiness detection systems.

In Narayana (2025), the authors present a comprehensive vision-based driver fatigue detection system leveraging eye and mouth aspect ratios (EAR and MAR) in conjunction with machine learning classifiers to identify drowsiness in real time. Their methodology centers around capturing continuous webcam video of the driver's face and extracting facial landmarks to compute EAR and MAR, which correspond to blinking patterns and yawning events—key indicators of drowsiness. These temporal behavioral features are fed into a machine learning model, including both a Convolutional Neural Network (CNN) for feature extraction and a Support Vector Machine (SVM) for classification, achieving a robust performance across diverse facial profiles, lighting conditions, and head movements. The proposed system is trained and validated on benchmark datasets like the NTHU Drowsy Driver Detection Dataset and YawnDD, reporting an impressive accuracy exceeding 92% with an average detection time of approximately 1.5 s, making it suitable for real-time deployment. The solution stands out for its non-intrusive, low-cost implementation using only a standard webcam, and its adaptability to various in-vehicle environments without the need for specialized hardware, aligning well with the goals of Advanced Driver Assistance Systems (ADAS) and Intelligent Transportation Systems (ITS).

5.1.3 Head-based measurements

Head-based measurements play a critical role in visual drowsiness detection systems, as head movements and orientation often reflect early signs of fatigue, such as nodding, tilting, or reduced gaze stability. These systems typically monitor the driver's head pose—calculated using pitch, yaw, and roll angles—as well as gaze direction to assess attentiveness and potential signs of microsleep or distraction Baccour et al. (2022). Head tracking techniques leverage facial landmarks or 3D pose estimation models to continuously evaluate the spatial orientation of the head relative to the camera or road. When abnormal or repetitive patterns such as forward head drops or gaze drift are detected, the system may flag the driver as drowsy or inattentive. Representative image-based drowsiness detection systems that utilize head-based features are included in Table 5, highlighting their use alongside other visual cues like eyes and mouth movement for more robust driver monitoring. To avoid misleading comparisons, Table 5 augments each entry with # Subjects, Validation Protocol (SD/SI/LOSO/cross-dataset), and Ground Truth (e.g., visual labels vs. KSS/EEG), and groups studies by dataset + protocol.

In Wijnands et al. (2020), the authors propose a practical and efficient approach to real-time drowsiness detection on mobile platforms by leveraging lightweight 3D convolutional neural networks (3D CNNs). Unlike conventional frame-based systems, their method integrates spatial and temporal information early in the network using depthwise separable 3D convolutions, which enables better feature extraction from motion cues such as facial expressions and head movements. This is particularly useful in scenarios where traditional eye-based tracking fails—like when the driver wears sunglasses. The model, pre-trained on ImageNet and Kinetics and later fine-tuned on the NTHU-DDD dataset, achieved a detection accuracy of 73.9%. A TensorFlow-based smartphone application was developed to demonstrate real-time deployment capabilities and evaluate model inference speed and performance. This work highlights the potential of mobile-friendly deep learning models for scalable and cost-effective driver monitoring systems.

In Wu et al. (2024), the authors proposed YOLOFaceMark, a novel real-time driver drowsiness detection framework that unifies face and facial landmark localization in a single-stage architecture. Unlike traditional two-step systems, YOLOFaceMark directly predicts facial landmarks alongside face detection, significantly improving detection speed and reducing computational overhead—critical for deployment in automotive environments. The model incorporates structural re-parameterization, channel shuffling, and a dual-branch detection head with an implicit module, enhancing feature extraction efficiency while maintaining lightweight operation. The drowsiness detection component analyzes eye and mouth states inferred from the localized landmarks to assess alertness. Validation on challenging datasets such as 300W and COFW demonstrates that YOLOFaceMark achieves accurate and robust landmark detection even under night driving conditions or when eyes are closed, highlighting its practical value for reliable driver monitoring in real-world scenarios.

In Bisogni et al. (2024), the authors introduce HYDE-F, a novel head pose estimation framework designed specifically for smart cars using only depth cameras, avoiding RGB entirely to enhance robustness under challenging lighting conditions. HYDE-F combines fractal encoding and keypoint intensity analysis, fused via an optimization algorithm, to estimate the driver's pitch, yaw, and roll in real time. Experiments on the Pandora dataset, which simulates realistic driving conditions, demonstrate that HYDE-F outperforms several state-of-the-art methods, achieving a mean absolute error of $5.4°$—about $1°$ better than the best existing approaches. Additionally, the system generalizes well to other datasets (Biwi and ICT-3DHP), showing its adaptability to different environments. HYDE-F's design also ensures low computational cost, making it feasible for in-vehicle deployment without sacrificing accuracy. This system highlights the potential of depth-based head pose estimation for integration into IoT-enabled Advanced Driver Assistance Systems (ADAS), significantly improving driver monitoring capabilities.

In Abbasi et al. (2025), the authors investigated the critical issue of cyclist safety at intersections with cycle lanes by analyzing driver gaze behavior using in-vehicle eye-tracking systems and CAN bus data. Conducting naturalistic driving studies at five intersections in Galway, Ireland, they observed that approximately 85% of drivers failed to glance at the left wing mirror before making left turns, significantly increasing the risk of collisions with cyclists or e-scooter users. An innovative algorithm was developed to monitor driver gaze towards the mirror in real-time, assign dynamic danger levels based on gaze and vehicle speed, and provide potential early warnings to drivers. This human-centric approach demonstrates how driver gaze behavior monitoring can be integrated into Advanced Driver-Assistance Systems (ADAS) to enhance the safety of vulnerable road users.

5.1.4 Hybrid eye, mouth, and head-based techniques

Hybrid techniques that combine eye, mouth, and head-based features have demonstrated improved robustness and accuracy in visual drowsiness detection systems. By leveraging multiple facial cues simultaneously—such as blink frequency, yawn duration, and head pose angles—these systems are better equipped to handle variability in lighting conditions, facial orientation, or occlusions that may affect single-feature models Hasan et al. (2022). Integrating these complementary visual indicators allows for a more comprehensive representation of driver fatigue, as some symptoms (e.g., head nodding) may appear even when others (e.g., eye closure) are momentarily absent. These multi-feature approaches typically rely on synchronized facial landmark tracking and temporal analysis to correlate behaviors like yawning with gaze drift or forward head movement Zilberg et al. (2022). Figure 8 illustrates a conceptual workflow for hybrid eye, mouth, and head-based drowsiness detection, integrating multiple visual features into a unified decision-making system. As highlighted in Table 4, several state-of-the-art systems employ such hybrid facial strategies, illustrating their effectiveness in enhancing detection reliability across real-world driving conditions.

Figure 8 outlines the workflow of a hybrid driver drowsiness detection system that simultaneously analyzes eye, mouth, and head-based features to improve detection reliability. The process begins with continuous video capture of the driver's face using an in-cabin camera, followed by real-time face detection and landmark localization. From these landmarks, multiple facial cues are extracted—such as blink frequency and PERCLOS for eye analysis, yawning duration and mouth opening ratio for mouth-based indicators, and head pose angles for assessing orientation and movement. These features are then tracked over time to detect temporal patterns associated with drowsiness, including frequent blinking, prolonged eye closure, repetitive yawns, and head nodding. By fusing these complementary signals, the system generates a comprehensive behavioral profile that enhances its ability to recognize early signs of fatigue, even in the presence of occlusions or variable lighting. A

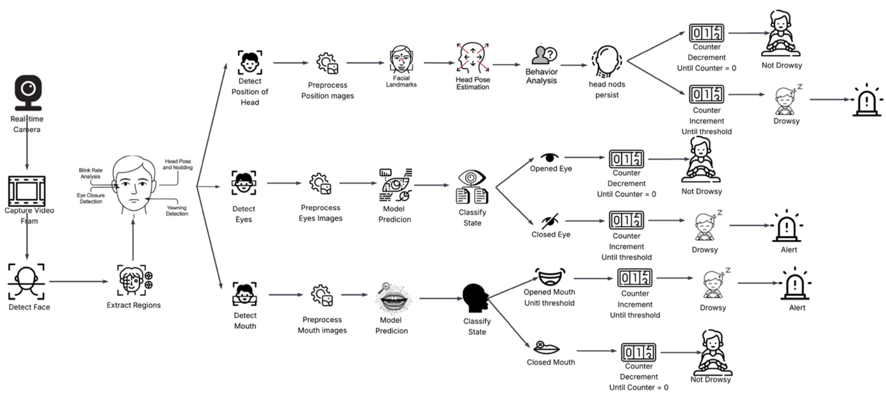


Fig. 8 Hybrid driver drowsiness detection procedure integrating eye, mouth, and head-based measurements

rule-based or machine learning model interprets these patterns to issue alerts when drowsiness thresholds are exceeded, ensuring timely intervention to support driver safety.

In Zhao et al. (2020), the authors proposed a fully automated driver fatigue detection algorithm based on driving images. Their system utilizes the Multitask Cascaded Convolutional Network (MTCNN) for robust face detection and localization of key facial landmarks, enabling precise extraction of regions of interest (ROI) corresponding to the eyes and mouth. They introduced a custom Convolutional Neural Network named EM-CNN, specifically designed to classify eye and mouth states from these ROIs. Two key physiological metrics—PERCLOS (Percentage of Eye Closure) and POM (Percentage of Mouth Opening)—were then computed for fatigue inference. Experimental validation using driving images from Biteda Company demonstrated that EM-CNN outperformed other well-known CNN architectures like AlexNet, VGG16, GoogLeNet, and ResNet50, achieving an accuracy of 93.62% and a sensitivity of 93.64%. This work highlights the effectiveness of specialized CNNs in capturing subtle fatigue indicators under real driving conditions.

In Ed-Doughmi et al. (2020), the authors address the global concern of increasing road accident fatalities by proposing a real-time driver drowsiness detection system based on recurrent neural network (RNN) principles integrated with 3D convolutional neural networks (3D-CNNs). The system analyzes a continuous sequence of facial frames to detect signs of fatigue by monitoring key features such as the eyes, mouth, and head movements. Leveraging the NTHU-DDD dataset, the model is trained to capture temporal patterns that indicate drowsiness, enabling it to perform real-time assessments of driver alertness. The multi-layer architecture achieved an impressive accuracy of 97.3%, showing promise in real-world applications. This research contributes to the development of intelligent transportation systems by offering a contactless, high-accuracy solution that could significantly mitigate drowsiness-related accidents.

In Moujahid et al. (2021), the authors present a novel face monitoring system tailored for driver drowsiness detection, focusing on building a compact and efficient face texture descriptor. Rather than relying solely on deep learning models, the system utilizes a Pyramid-Multi Level (PML) face representation that extracts both local and global texture information from video frames. This multi-scale feature extraction is followed by a subset selection process to retain only the most discriminative features. The approach targets facial regions such as the eyes, head, and mouth, which are strongly associated with drowsiness indicators. Using the NTHU-DDD dataset, the proposed method achieved an accuracy of 79.84%. While its performance is lower than deep learning-based models, it offers a lightweight alternative suitable for real-time scenarios with limited computational resources.

In de Lima Medeiros et al. (2022), the authors developed a real-time eye-blink detection system aimed at enhancing communication for ALS patients by using eye blinks as interaction signals. The system utilizes standard webcams and applies a structured computer vision pipeline consisting of face detection, alignment, region-of-interest (ROI) extraction, and eye-state classification. Two models—Convolutional Neural Network (CNN) and Support Vector Machine (SVM)—were trained on a newly created dataset, YEC (Youtube Eye-state Classification), and evaluated on several public datasets such as CeW and Eyeblink. An additional dataset, ABD (Autonomous Blink Dataset), was developed specifically for volunteer blink detection. With the integration of auxiliary components like a rotation compensator, ROI evaluator, and moving average filter, the CNN achieved 97.44% accuracy on CeW and a 92.63% F1-score on ABD. This robust approach offers not only a practical interface for assistive technology but also demonstrates high reliability in detecting subtle eye-blink signals.

In Albadawi et al. (2023), the authors propose a real-time, non-invasive driver-drowsiness detection system that relies exclusively on visual cues captured by a dashboard-mounted camera. Using facial landmark and face-mesh detectors, the system identifies regions of interest from which it extracts the mouth aspect ratio (MAR), eye aspect ratio (EAR), and head-pose angles. These features are classified by three separate models—Random Forest (RF), a sequential neural network (SNN), and a linear Support Vector Machine (SVM)—to determine a driver's alertness state. Evaluated on the NTHU-DDD dataset, the RF classifier achieved the best performance, reaching 99 % accuracy, while the SNN and SVM attained 96 % and 80 % accuracy, respectively. The approach demonstrates the efficacy of lightweight, vision-based machine-learning pipelines for in-vehicle fatigue monitoring, though its performance remains contingent on stable lighting and unobstructed facial views.

In Essahraui et al. (2025), the authors developed a real-time driver drowsiness detection system leveraging facial analysis and state-of-the-art machine learning techniques to enhance road safety. Their system integrates classical ML classifiers—including K-Nearest Neighbors (KNN), Support Vector Machines (SVM), Decision Trees (DT), and Random Forests (RF)—as well as modern computer vision methods like CNN, YOLOv5, YOLOv8, and Faster R-CNN. The study rigorously evaluated these models across three benchmark datasets (NTHUDDD, YawDD, and UTA-RLDD), demonstrating exceptional performance: KNN achieved up to 98.89% accuracy, and YOLOv5/YOLOv8 reached nearly 100% precision and recall on UTA-RLDD, significantly surpassing traditional detection techniques. These findings underscore the system's potential for proactive and highly accurate real-time driver alertness monitoring.

In Owen and Surantha (2025), the authors propose a lightweight, computer vision-based drowsiness detection system optimized for edge computing devices with limited resources. Their approach integrates facial landmark detection using Dlib, head pose estimation via HOPEnet, and calculates behavioral indicators like the percentage of eyelid closure over time (PERCLOS) and percentage of mouth opening (POM). The extracted handcrafted features are then classified using traditional machine learning algorithms, specifically SVM, Random Forest, and XGBoost, trained on the NTHU Drowsiness Detection Dataset. The proposed models achieve a notable overall accuracy of 86.848% while maintaining a small model size of 5.05 MB, outperforming a baseline deep learning model (HTDBN) in computational efficiency. The system demonstrates suitability for real-time deployment on devices like the Jetson Nano, balancing accuracy with low memory and processing requirements, and offering a practical solution for drowsiness detection in automotive safety systems.

In Borawake et al. (2025), the authors present a proactive driver drowsiness detection system leveraging machine learning and IoT technologies to improve road safety. Their method monitors facial features, head movements, and eye activity using in-vehicle cameras. The system extracts behavioral and physiological indicators such as Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), and head pose estimation. A CNN-based model is trained on diverse facial datasets to recognize early signs of drowsiness, while IoT integration enables real-time alerts through in-vehicle alarms or remote notifications to fleet managers via cloud storage. The proposed design emphasizes affordability by relying on readily available cameras and open-source libraries like OpenCV and Dlib. The authors highlight performance improvements through real-time detection, cloud analytics, and integration with onboard vehicle control systems to reduce accident risk, achieving effective monitoring accuracy suitable for long-haul drivers and high-risk scenarios.

In Jarndal et al. (2024), the authors present ViT-DDD, a real-time driver drowsiness detection system leveraging Vision Transformers to process entire facial images rather than isolated features. Unlike previous CNN-based methods that rely on specific facial landmarks, this approach uses global facial patterns, enhancing robustness under challenging conditions like glasses, sunglasses, and varying illumination. The model achieved impressive accuracies of 98.89% on NTHU-DDD and 99.4% on UTA-RLDD datasets. Furthermore, it was deployed on a Raspberry Pi integrated with an IR camera and HOD sensor, enabling real-world application through a compact, low-cost hardware prototype capable of issuing alerts and notifying fleet management when drowsiness is detected.

5.1.5 Critical analysis & synthesis

Vision-based drowsiness detection has evolved from handcrafted or rule-based descriptors (e.g., PERCLOS, EAR thresholds, fuzzy inference) to convolutional networks and, more recently, Transformers (ViT, Swin) that learn holistic facial dependencies. Extremely high accuracies ($\approx$ 99%) typically coincide with subject-dependent or within-dataset testing under controlled lighting and camera angles. When evaluated subject-independently or cross-dataset, accuracy drops sharply, revealing the sensitivity to dataset diversity, illumination (night glare), eyewear, and head pose. Robust ROI extraction and IR imaging remain decisive factors in achieving reliable performance, as documented in Table 5.

Rule-based methods offer transparency, lightweight computational requirements, and ease of deployment on edge hardware, but remain brittle to occlusion and pose variations. End-to-end deep learning and Transformer models achieve higher performance ceilings when training data are diverse and evaluation protocols enforce subject-independent validation. Because these sophisticated models are less interpretable, visualization tools such as CAMs, Grad-CAM, and attention maps are increasingly vital to diagnose false positives (e.g., eyelash shadows, glare) and build driver trust in automated detection systems.

Apparent state-of-the-art performance, such as that reported for ViT Hassan et al. (2024), often reflects the combined influence of balanced datasets, IR support, accurate ROI tracking, and lenient validation protocols (subject-dependent). Thus, model choice alone rarely explains observed gains; dataset curation, ground-truth definition, and evaluation protocol are equally determinative. Table 5 now explicitly reports the number of subjects, validation type, and ground truth methodology to expose these factors and ensure contextual comparability across studies.

These findings carry important design implications and suggest future research directions. Researchers should favor subject-independent (LOSO) validation protocols and report PR-AUC, TPR at fixed FAR (e.g., 1 alarm h^{-1}), and detection latency metrics. Fusion with physiological signals or CAN telemetry can mitigate failures under challenging conditions such as nighttime driving or eyewear occlusion. These insights motivate the benchmarking guidance and design principles presented in Sects. 6 and 8.5.

5.2 Biological signal-based measurements

Biological signals—often referred to as physiological measurements—provide objective, real-time insight into the internal state of a driver, making them a highly reliable modality for early detection of fatigue and drowsiness Biying et al. (2024). Unlike behavioral cues, which may only emerge after cognitive impairment has occurred, physiological signals can capture early biochemical and neurological changes that precede visible signs of sleepiness. These signals are typically collected through non-invasive biosensors and reflect key bodily functions such as brain activity, cardiac rhythm, respiratory rate, and muscular or ocular movement. Commonly employed techniques include Electroencephalography (EEG) for monitoring brain waves, Electrocardiography (ECG) and Heart Rate Variability (HRV) for cardiac activity, Photoplethysmography (PPG) for blood volume changes, Electrooculography (EOG) for eye movement tracking, and Electromyography (EMG) for measuring muscle activity. Each of these methods yields specific signal features—such as spectral band analysis, R-R intervals, or blink patterns—that can be processed using signal analysis and machine learning techniques to assess driver alertness. Table 6 outlines the primary physiological measurements used in the literature, along with their associated features, functional categories, and applications in drowsiness detection systems. Figure 9 illustrates the conceptual block diagram of a biological-signal-based driver drowsiness detection system, summarizing the key stages of signal acquisition, preprocessing, feature extraction, and classification. It provides a structural overview of how physiological modalities (e.g., EEG, ECG, EOG, PPG) contribute to vigilance assessment within multimodal architectures.

Figure 9 illustrates the fundamental processing pipeline of biological-signal-based driver drowsiness detection systems. The process begins with **Sensor Placement & Signal Acquisition**, where non-invasive biosensors (e.g., EEG, ECG, EOG, EMG, PPG) capture continuous physiological signals. These raw signals undergo **Preprocessing** to remove artifacts such as motion-induced noise or ocular artifacts in EEG, followed by filtering to

Table 6 Biological signals used in drowsiness detection

References	Year	Feature	Category	Description
Stancin et al. (2021)	2021	Alpha Wave Analysis, Theta Wave Detection, Spectral Analysis, Brain State Classification	EEG	EEG monitors electrical brain activity using scalp electrodes. It helps classify sleep stages and alertness levels based on frequency bands
Yaacob et al. (2020)	2020	Heart Rate Monitoring, R-R Interval Analysis, Cardiac Rhythm	ECG	ECG captures the heart's electrical signals, providing data on rhythm and variability, which reflect fatigue or stress
Amidei et al. (2021)	2021	Pulse Wave Analysis, Blood Volume Changes, Optical Sensing	PPG	PPG detects blood volume fluctuations through the skin using light sensors. It's useful for tracking cardiovascular signs of fatigue
Mejía-Mejía et al. (2020)	2020	Time Domain Analysis, Frequency Domain, Autonomic Assessment	HRV	HRV measures variation in time between heartbeats, indicating autonomic nervous system activity and fatigue levels
Hasan et al. (2024)	2024	Eye Movement Detection, Saccade Analysis, Blink Monitoring	EOG	EOG records eye movements by tracking corneo-retinal potential. It identifies blinks and saccades linked to reduced alertness
Li et al. (2024)	2024	Muscle Activity, Facial EMG, Grip Strength	EMG	EMG records muscle electrical activity. Facial EMG and grip analysis are used to evaluate physical fatigue and drowsiness

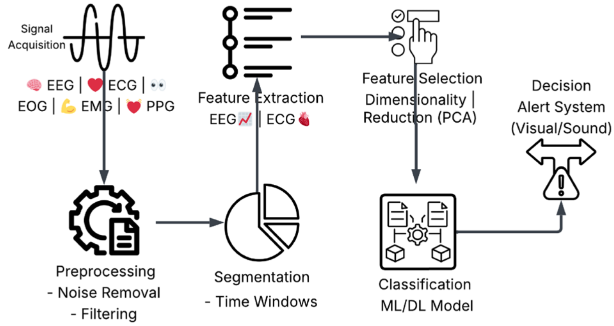


Fig. 9 Basic block diagram of biological-signal-based driver drowsiness detection systems

reduce baseline drift and interference. Next, **Segmentation** divides the cleaned signals into fixed-length windows (e.g., 5–30 s) to enable temporal analysis. During **Feature Extraction**, specific biomarkers—like EEG spectral bands, ECG-derived R-R intervals, blink rates from EOG, or muscle activation patterns from EMG—are computed. **Feature Selection & Dimensionality Reduction** techniques (e.g., PCA, LDA) then distill the most discriminative features. These are passed to **Classification & Decision Making** models—such as SVM, Random Forest, or neural networks—which estimate driver drowsiness levels. When drowsiness exceeds a predefined threshold, an **Alert Generation** module issues haptic, visual, or auditory warnings to prompt driver attention. Optionally, a **Data Logging** component can store raw signals, extracted features, and classification results for offline analysis or longitudinal monitoring, enabling continuous improvement of the system's performance.

5.2.1 Electroencephalography (EEG)

Electroencephalography (EEG) records electrical activity of the brain through electrodes placed on the scalp, providing a direct window into neural processes associated with alertness and drowsiness. Characteristic changes in power within frequency bands—such as increased theta (4–7 Hz) and alpha (8–13 Hz) activities—are reliable indicators of decreased vigilance and onset of sleep. EEG is considered the gold standard for objectively detecting drowsiness because it reflects real-time cognitive states before behavioral signs become apparent. Despite challenges like susceptibility to artifacts and practical limitations in real-world driving, EEG-based systems remain one of the most robust modalities for early detection of driver fatigue.

In Khare et al. (2020), the authors present an advanced drowsiness detection framework leveraging Electroencephalography (EEG) signals decomposed using Adaptive Variational Mode Decomposition (AVMD). By adaptively selecting the number of modes and the quadratic penalty factor with the Jaya optimization algorithm, the proposed method effectively minimizes signal reconstruction error, allowing extraction of highly representative entropy-based features. These features are then subjected to statistical selection and classified with machine learning algorithms. The system achieved an impressive accuracy of 97.19% in classifying alert versus drowsy states, along with high sensitivity (97.01%) and specificity (97.46%). Compared to conventional and existing decomposition methods, AVMD-based analysis improved detection performance by approximately 7% and 1%, respectively, demonstrating its superiority for synthesizing and analyzing EEG signals for driver drowsiness detection.

In Obaidan et al. (2024), the authors introduced EEG_DMNet, a deep multi-scale convolutional neural network specifically designed for EEG-based driver drowsiness detection. Their model tackles the challenge of non-stationary EEG signals by extracting multi-scale spectral-temporal features through a cascade of temporal and spatial convolutions. Using the SEED-VIG dataset with EEG recordings from 23 subjects, their system preprocesses signals using differential entropy (DE) and employs a pyramid CNN architecture to efficiently learn discriminative patterns across temporal and spatial domains. The proposed model achieved a notable average accuracy of 97.03%, outperforming prior state-of-the-art approaches on the same dataset, and demonstrating superior ability in differentiating between awake, tired, and drowsy states in a subject-independent evaluation protocol.

In Ardabili et al. (2024), the authors proposed a multi-class driver fatigue detection system using EEG signals processed with a novel architecture combining Generative Adversarial Networks (GANs) and Graph Convolutional Networks (GCNs). A database was collected from 20 participants driving in a simulator while EEG, ECG, EMG, and respiratory signals were recorded. The GCN model, built with five convolutional layers, achieved impressive classification performance across four practical cases, with accuracies ranging from 91.7% to 99.4%. The method also included robust design choices like data augmentation with GANs and noise resilience testing, ensuring the system's high accuracy even under engine and cabin noise conditions.

In Feng et al. (2025), the authors propose ID3RSNet, an interpretable residual shrinkage network designed for cross-subject driver drowsiness detection from raw single-channel EEG signals. The model integrates a base feature extractor, a residual shrinkage building unit with channel-wise soft thresholding and attention, and an EEG-based class activation map (ECAM) interpretation method, enabling automatic feature extraction and visualization of neurophysiologically relevant patterns. The system is evaluated on the public sustained-attention driving task (SADT) dataset with 11 subjects, achieving superior performance with a mean accuracy of 74.72% using leave-one-subject-out cross-validation, outperforming existing conventional and deep learning methods. The study highlights the potential of single-channel EEG combined with inherently interpretable deep networks for practical, portable driver drowsiness monitoring.

In Malik et al. (2025), the authors propose a multi-body sensor-based driver drowsiness detection system that processes single-channel EEG data to classify driver alertness. The EEG signals are analyzed using the wavelet time-frequency transform (WTFT) to extract detailed time-frequency features, followed by classification with a multi-layer convolutional designed transfer VGG-16 neural network (MLCDT-VGG-16) leveraging transfer learning. This architecture allows automatic transition of driving mode from manual to assisted in case of detected drowsiness. The study evaluated its model using EEG data from Emotiv EPOC+ and Physionet datasets, achieving up to 92% prediction accuracy, 85% precision, and 78% recall on Emotiv EPOC+, outperforming STFT and SVM_LDA baselines. The proposed system demonstrated real-time applicability with reasonable computational complexity, supporting safer driving by early detection of drowsiness.

5.2.2 Electrocardiography (ECG) and photoplethysmography (PPG)

Electrocardiography (ECG) measures the heart's electrical activity via electrodes attached to the driver's chest or extremities, offering valuable information about autonomic nervous system regulation. As drivers become drowsy, changes in heart rate and the morphology of ECG waveforms can signal a shift toward parasympathetic dominance. ECG-based monitoring is advantageous because it is relatively simple to implement with wearable devices and provides continuous, objective data. Its integration with other physiological measures can significantly enhance the accuracy of drowsiness detection systems.

In Xiong et al. (2024), the authors present a hybrid learning approach, denoted as DBPW, for driver drowsiness level detection using electrocardiogram (ECG) signals. The method combines a dilated convolutional neural network and bidirectional long short-term memory (DiCNN-BiLSTM) for processing raw ECG data (model_1) with a principal component analysis (PCA) and weighted K-nearest neighbor (WKNN) classifier applied to heart rate variability features (model_2). The predictions from both models are fused through a weighted scheme to enhance classification accuracy across three driver states: awake, stress, and drowsiness. Experiments on a multi-driver public ECG dataset demonstrated outstanding performance, achieving average accuracy, sensitivity, and F1 scores of 98.79%, 98.81%, and 98.79%, respectively, with a drowsiness-specific recognition accuracy reaching 99.33%. This study shows the effectiveness of integrating deep learning with statistical feature-based methods to improve ECG-based drowsiness detection in realistic driving conditions.

In Wang et al. (2025), a portable semi-dry electrode was developed to overcome limitations of traditional wet and dry electrodes for ECG acquisition, offering low impedance and easy conductive liquid replenishment. Comparative analysis showed that ECG signals collected by the novel semi-dry electrodes were consistent with those from standard electrodes in capturing driver fatigue characteristics. To classify fatigue states, a wavelet scattering network (WSN) was employed, achieving an average classification accuracy of 97.93% while maintaining fast processing suitable for real-time fatigue monitoring. The results demonstrate that the combination of the innovative electrode design and WSN-based analysis enables efficient and reliable detection of driver drowsiness, providing a promising solution for practical, continuous in-vehicle monitoring systems.

Photoplethysmography (PPG) utilizes optical sensors—often embedded in wearable devices or steering wheels—to measure variations in blood volume corresponding to each heartbeat. PPG-derived metrics like pulse rate and pulse amplitude variability have shown promise in reflecting changes in alertness, especially given their sensitivity to autonomic fluctuations during fatigue. PPG systems are non-invasive, low-cost, and unobtrusive, making them suitable for practical driver monitoring applications, although they can be sensitive to motion artifacts and ambient light conditions.

In Lu et al. (2024), the authors proposed an innovative multimodal fatigue detection system combining photoplethysmography (PPG)-derived heart rate variability (HRV) features, facial landmarks, and head pose estimation to enhance accuracy in real-world driving scenarios. Using a custom dataset collected from 30 drivers during real-road experiments, they extracted 10-dimensional fused features including SDNN, RMSSD, LFP/HFP, EAR, MAR, and head pitch, roll, and yaw angles. A two-layer LSTM model was optimized with four algorithms (Adam, RMSProp, Momentum, SGD), with the Adam-optimized LSTM achieving the best performance: 97.36% accuracy, 97.4% precision, and 97.4% recall on the test set. The authors demonstrated that combining facial, head, and PPG signals significantly improved detection performance compared to single modalities, with the proposed system effectively recognizing awake, mild, and severe fatigue states, supporting timely interventions to improve driver safety.

5.2.3 Heart rate variability (HRV)

Heart Rate Variability (HRV), typically derived from ECG or PPG signals, quantifies variations in the intervals between consecutive heartbeats (R-R intervals). It serves as a surrogate for autonomic nervous system balance and stress levels. Reduced HRV—marked by lower high-frequency power and increased low-frequency power—has been linked to increased drowsiness and cognitive impairment. HRV analysis enhances drowsiness detection by providing nuanced insights into physiological stress and fatigue, complementing direct measures of brain or muscular activity.

In Kundinger et al. (2020), the authors investigated the feasibility of wrist-worn wearables for driver drowsiness detection by leveraging heart rate variability (HRV) features. They compared signals recorded with the Empatica E4 wristband to medical-grade ECG under simulated driving conditions involving 30 participants. A set of time-domain and frequency-domain HRV features were extracted, and various machine learning classifiers, including Random Forest, KNN, Decision Trees, and SVM, were applied to distinguish between alert and drowsy states. Their results demonstrated that user-dependent models could achieve high accuracy—up to 92.1% using wristband data with KNN and 97.4% with ECG-based Random Forest—highlighting the promise of wearable HRV sensors for practical, real-time driver monitoring, although user-independent scenarios showed substantially lower performance due to inter-individual variability.

In Liu et al. (2023), the authors investigated ultra-short-term heart rate variability (HRV) analysis for real-world driver stress detection. Seventeen drivers completed a 60–90 min drive on urban routes designed to induce varying stress levels. ECG signals were segmented into epochs of 30 s, 1 min, 2 min, 3 min, and standard 5 min, with 22 HRV features extracted per epoch (MeanNN, SDNN, NN20, MeanHR, and others). Statistical analyses (t-tests) showed that features like MeanNN, SDNN, NN20, and MeanHR consistently exhibited significant differences between low- and high-stress states across all time scales ($p < 0.001$ for most). Spearman correlation and Bland–Altman analyses identified these four features as valid surrogates for short-term HRV. For classification, SVM achieved the best performance using 3-min epochs, with mean accuracy of 85.3%, sensitivity 78.1%, specificity 91.3%, and F1-score 85.2%. Performance with 30-s epochs dropped only 3% compared to 5-min benchmarks. Strengths: first real-world validation of ultra-short HRV features; Weaknesses: limited sample size, potential inter-individual variability unaccounted for.

In Sun et al. (2024), an individualized driver fatigue identification method was developed using electrocardiogram (ECG) signals, addressing the challenge of individual variability in fatigue detection. Seventeen candidate HRV features spanning time, frequency, and nonlinear domains were extracted, and a feature selection framework combining support vector machine recursive feature elimination (SVM-RFE) with a correlation bias reduction algorithm was introduced to rank feature importance while minimizing the influence of driver-specific differences. An evaluation matrix of feature rankings across drivers was constructed, leading to the definition of a feature importance index, Va, and the selection of eight key features that best characterized driver fatigue independent of individual differences. The final SVM-based binary fatigue classification model incorporating these features achieved an average accuracy of 87.04%, which improved by 9.20% compared to models lacking individualized feature selection, demonstrating the importance of personalized feature extraction in enhancing generalization and robustness of ECG-based driver fatigue detection systems.

In Srinivasan et al. (2024), the authors explored the predictive value of heart rate variability (HRV) indices—specifically RMSSD and LF/HF ratio—for multidimensional fatigue in young drivers under simulated conditions. Eighty-four participants with self-reported sleep restriction completed a 25-minute simulated driving task, during which ECG, EEG, and oculography data were recorded. The study extracted RMSSD and LF/HF ratio from ECG signals and constructed a latent fatigue model combining EEG Theta power, lane position variability, Phasya drowsiness scores, and subjective sleepiness scores. Structural equation modeling revealed RMSSD negatively predicted fatigue, explaining 6.4% of its variance, while LF/HF ratio positively predicted fatigue, explaining 3.7%. Although model fit indices were marginal, the findings highlighted HRV's potential as an objective, non-invasive indicator of multidimensional fatigue manifestations beyond single-dimensional assessments, supporting HRV monitoring in fatigue detection systems.

In Tsai et al. (2024), Tsai et al. proposed a novel predictive framework for fatigue-associated aberrant driving behaviors (ADBs) in commercial bus drivers by dynamically processing HRV signals with a Dynamic Weighted Moving Average (DWMA) model and classifying them using a Long Short-Term Memory (LSTM) network. The study involved 20 male bus drivers (urban and highway) monitored over four days during real-world routes in Taiwan, collecting HRV via a medical-grade wristwatch and ADBs using a GPS-based navigation app. Key HRV metrics (SDNN, RMSSD, nHF) were compared between baseline and 5-minute pre-ADB intervals, showing significant increases in pre-event values linked to fatigue. The DWMA-LSTM models achieved the highest accuracies (84.41% for urban and 80.56% for highway drivers) compared to traditional methods, with SDNN, RMSSD, and nHF emerging as the most important features (high SHAP values). The study highlighted the critical role of HRV dynamics in predicting fatigue-related ADBs, though limitations included a small, male-only cohort and potential confounders like unassessed personal health habits.

In [142], the authors developed a multimodal framework to predict fatigue-related aberrant driving behaviors (ADBs) in 37 professional taxi drivers monitored over four consecutive days of real-world driving. The system combined continuous ECG-derived HRV signals (RMSSD, SDNN, nHF) collected with a chest patch, sleep-disorder indices (e.g., ODI-3%, CEI) from polysomnography and wearable devices, and demographic data. They employed a hybrid model integrating a BiLSTM with self-attention for sequential HRV patterns and LightGBM for static features, achieving outstanding performance: AUROC 90.93% and AUPRC 78.35% on testing data. RMSSD, ODI-3%, and nHF were top predictors. The approach showed significant improvement over baseline models, highlighting the value of combining temporal physiological dynamics and sleep data. Strengths: comprehensive multimodal assessment in real-world settings; Weaknesses: relatively small sample size and limited diversity in driver population.

5.2.4 Electrooculography (EOG)

Electrooculography (EOG) records the corneo-retinal potential generated by eye movements, enabling precise tracking of blink patterns, saccades, and slow eye movements—hallmarks of drowsiness onset. Parameters like blink frequency, blink duration, and eyelid closure speed (PERCLOS) are reliable markers for drowsy states. EOG-based systems excel in directly capturing ocular dynamics that precede microsleep events, but they can require careful calibration and may be affected by head movements during real-world driving.

In Wang et al. (2024), the authors propose a multi-modal fatigue detection model that fuses features from both EEG and EOG signals to improve driver drowsiness assessment in human–machine co-driving scenarios. They address key limitations of EEG-only systems—such as noise sensitivity and low cross-subject generalization—by introducing a model combining EEGNet, SANet (Spatial Attention Network), and SCONV (Separable Convolutional Network). This architecture enables effective spatial-temporal feature extraction from high-precision EEG and robust EOG signals. Experiments on the SEED-VIG dataset demonstrate a recognition accuracy of 91% for intra-subject tests, and an average accuracy of 87% for cross-subject scenarios, outperforming widely used baseline models and highlighting improved generalizability.

In Huang et al. (2025), the authors introduce CM-FusionNet, an innovative cross-modal fusion framework designed to improve mental fatigue detection by combining EEG and EOG signals. The model incorporates a variance channel attention (VCA) module to dynamically assign optimal weights to each EEG and EOG channel, enhancing the contribution of informative signals. Additionally, a Transformer-based fusion module captures and integrates global features from both modalities before classification. Experiments on the SEED-VIG dataset show that the multi-modal approach achieves an average accuracy of 84.62% and an F1-score of 85.25%, outperforming EEG-only and EOG-only baselines by notable margins. These results validate the effectiveness of multi-modal integration and highlight the model's potential for practical fatigue monitoring applications.

In Choi et al. (2025), the authors introduce a conditional diffusion model-based generative framework designed to overcome limited-data challenges in electrooculography (EOG) eye-writing applications, which enable users to draw characters through eye movements. The method leverages class-conditional vectorized information to generate diverse, realistic EOG signals, improving training datasets for classification models under data-scarce conditions. The model achieved high classification accuracies of 94.63% for Arabic numerals and 80.36% for Japanese Katakana strokes when trained with ninefold augmented data. Compared to TimeGAN and GAN-based approaches, the proposed model not only produced higher visual fidelity in generated EOG signals but also delivered faster inference despite having more parameters. The results demonstrate the framework's potential for enhancing biosignal-based communication aids, particularly for patients with conditions like ALS, and suggest broad applicability to other bioelectric signals.

5.2.5 Electromyography (EMG)

Electromyography (EMG) captures the electrical activity of muscles, typically focusing on facial or neck muscles whose activity diminishes during drowsiness. Decreased EMG amplitude and changes in muscle tone can precede behavioral signs of fatigue, offering an objective measure for early detection. EMG provides complementary information to EEG and EOG, enhancing robustness of multimodal systems; however, its practical use in real driving scenarios is constrained by the need for reliable electrode placement and sensitivity to motion artifacts.

In Naim et al. (2022), the authors developed a dual-layer ranking feature selection algorithm to identify the most relevant EMG features for driver fatigue detection. They first computed 21 filter-based rankings across 47 EMG features, including 25 time-domain and 9 frequency-domain features, and evaluated them using six classifiers. In the second layer, these ranks were aggregated with statistical measures (average, median, mode, variance) to derive a robust feature ranking. Their results showed that using 12 features selected with the Average Statistical Rank (ASR) and classified via Linear Discriminant Analysis (LDA) achieved the highest accuracy of 95%. The study demonstrated that combining time- and frequency-domain features significantly improves drowsiness detection performance, and that ASR-based ranking can reduce feature dimensionality effectively.

In Naim et al. (2023), the authors investigated driver drowsiness detection using facial Electromyography (EMG) signals recorded from the masseter and orbicularis oris muscles. Twelve healthy male participants performed 1-hour simulated driving tasks while EMG electrodes captured muscle activity associated with eye closures and yawning, which are indicative of drowsiness. Seven time-domain features were extracted from the EMG signals, and a k-Nearest Neighbor (kNN) classifier was employed for classification. The system achieved a maximum accuracy of 85.71% when using a 70:30 train-test split, k-values of 2 or 4, and combining features from both facial muscles. The results demonstrate that EMG-based detection provides an effective, non-visual physiological method for identifying driver drowsiness during simulated driving.

In Rashid et al. (2023), a novel multimodal driver drowsiness detection system was introduced that integrates electroencephalogram (EEG) and electromyogram (EMG) signals to enhance reliability over single-sensor approaches. The authors conducted a simulated driving experiment to collect EEG and EMG data during alert and drowsy states. Continuous wavelet transform (CWT) was applied to both modalities to convert time-domain signals into time–frequency images, which were then classified using an improved VGG-16 convolutional neural network architecture. The enhanced VGG-16 model was customized by removing redundant convolutional layers and fine-tuning fully connected layers to improve performance on the annotated time–frequency representations. The proposed method achieved a validation accuracy of 92.50%, demonstrating the effectiveness of combining EEG and EMG signals with advanced deep learning techniques for reliable driver drowsiness detection.

Table 7 provides a comprehensive overview of state-of-the-art drowsiness detection systems based on biological signals, highlighting the types of signals used, classification methods applied, system performance, and the strengths and weaknesses of each approach. Physiological studies use heterogeneous ground truths and validation protocols; Table 7 therefore includes # Subjects, Validation, and Ground Truth, and we interpret results within those contexts.

Table 7 Summary of biological signal-based drowsiness detection systems and their key characteristics

References	Year	Dataset(s)	Signal(s)	Model	Reported metric
Naim et al. (2023)	2023	Custom driving simulation dataset	EMG (Electromyography) from facial muscles (Masseter and Orbicularis Oris)	k-Nearest Neighbors (kNN) with k = 2 or 4	Accuracy = 85.71% (binary classification)
Liu et al. (2023)	2023	PhysioNet "Driver Database" (Boston real-world drive)	ECG → HRV features (MeanNN, SDNN, NN20, MeanHR)	SVM (Gaussian kernel); also compared RF, KNN, AdaBoost	Accuracy = 85.3% (3-min epochs); F1 ≈ 85.2%
Rashid et al. (2023)	2023	Custom driving-simulation dataset (alert vs drowsy sessions)	EEG + EMG (facial muscles)	CWT + Improved VGG-16 CNN (reduced convs, new fully connected layers; fine-tuned on time–frequency images)	Validation accuracy = 92.5%
Srinivasan et al. (2024)	2024	Young Adults Health Study (YAHS) – simulated driving fatigue dataset	ECG (HRV: RMSSD, LF/HF) + EEG (Theta power) + Oculography + Lane Position + KSS	Structural Equation Model (PLS-SEM) predicting latent fatigue from HRV indices (RMSSD & LF/HF)	RMSSD → Fatigue $r = -0.25$, R^2 = 6.4%; LF/HF → Fatigue $r = 0.19$, R^2 = 3.7%
Tsai et al. (2024)	2024	Real driving (bus)	PPG-derived HRV	DWMA + LSTM	Urban 84.41%; Highway 80.56%
Xiong et al. (2024)	2024	Public ECG datasets (multi-driver ECG dataset)	ECG (raw ECG + HRV features)	Hybrid: DiCNN-BiLSTM (dilated CNN + bidirectional LSTM) + PCA + WKNN fusion (DBPW)	Accuracy = 98.79%, Sensitivity = 98.81%, Recognition accuracy for drowsiness = 99.33%
Sun et al. (2024)	2024	Custom ECG dataset (simulated/experimental ECG data)	ECG (time-domain, frequency-domain, non-linear features)	SVM classifier with feature selection (SVM-RFE + correlation bias reduction)	Average Accuracy = 87.04% (across subjects)
Lu et al. (2024)	2024	Custom real-road driving dataset (Dalian–Shenyang Highway)	PPG + facial video + head pose	Two-layer LSTM (100 hidden units) optimized with Adam	Accuracy = 97.36%, Precision = 97.4%, Recall = 97.4%, F1 = 0.97; AUC > 0.99
Obaidan et al. (2024)	2024	SEED-VIG (simulated driving)	EEG (17 ch; 4 ch used: FT7, FT8, T7, T8)	EEG_DMNet (DE preprocessing → multi-scale 1D temporal convs → 1D spatial convs; ∼4.3M params)	Both trials: Acc 97.03%, Sen 95.54%, Spe 97.77%, F1 95.53%; Noon 97.46%, Night 96.90%
Ardabili et al. (2024)	2024	Custom driving simulator (Univ. of Tabriz)	EEG (19→4 channels: P4, C3, O1, O2)	GAN–GCN hybrid (5 graph conv layers, Leaky ReLU, Adamax optimizer; GAN augmentation ×2.3× samples)	Case I (2-class) = 99.4%, Case II (3-class) = 97.6%, Case III (4-class) = 96.3%, Case IV (5-class) = 91.7%; κ = 0.89-−0.97

Table 7 (continued)

References	Year	Dataset(s)	Signal(s)	Model	Reported metric
Khare et al. (2020)	2020	MIT-BIH Polysomnographic Database	EEG (single channel, 250 Hz)	AVMD + Jaya optimization → Entropy features → Ensemble Boosted Tree (EBT)	Accuracy = 97.19%, Sens 97.01%, Spec 97.46%, F1 = 0.976, Kappa = 0.94
[142]	2024	Real-world taxi driving dataset (Seoul taxi company)	ECG (single-lead) + Sleep indices (PSQI, ESS, ISI)	Hybrid BiLSTM + LightGBM pipeline	AUROC 90.93%; AUPRC 78.35%
Wang et al. (2024)	2024	SEED-VIG dataset (public vigilance dataset)	EEG + EOG (fusion of brain + eye movement)	EEGNet + SANet + SCONV	Within-dataset Acc 91%; Cross-subject Acc 87%
Huang et al. (2025)	2025	SEED-VIG	EEG + EOG	CM-FusionNet → Variance Channel Attention + Cross-Modal Transformer Fusion (E2FTF)	Cross-subject: Acc 84.62%, F1 85.25%; EEG-only: 81.74%; EOG-only: 83.14%
Feng et al. (2025)	2025	SEED-VIG (23 subjects) + SADT (11 subjects)	EEG (Fp1)	ID3RSNet – interpretable 1D residual-shrinkage CNN with adaptive thresholding	SEED-VIG: Acc 93.47%, F1 93.56%, $\kappa = 0.88$; SADT: Acc 74.72% (LOSO)
Malik et al. (2025)	2025	Physionet; Emotiv EPOC+	EEG (single-channel frontal F3; Emotiv 14-ch)	WTFT features → Transfer-learned VGG-16 (MLCDT-VGG-16)	PhysioNet: Acc 0.89, Prec 0.79, Rec 0.72, F1 0.61; Emotiv: Acc 0.92, Prec 0.85, Rec 0.78, F1 0.71
Wang et al. (2025)	2025	Custom ECG dataset (semi-dry steering-wheel electrodes)	ECG (1-lead semi-dry electrodes)	Wavelet Scattering Network	Average Acc = 97.93%, Prec = 97.8%, Rec = 97.9%; runtime ≈ 0.02 s/epoch
Choi et al. (2025)	2025	Custom eye-writing EOG datasets (Arabic numerals & Katakana)	EOG (horizontal & vertical channels)	Conditional Diffusion Model (vectorized class conditioning; compared to GAN & TimeGAN)	Arabic: Acc 94.63%; Katakana: 80.36%; F1 ↑≈3% vs GAN; faster inference
Kundinger et al. (2020)	2020	Custom simulator (45-min automated drives, 110 km/h)	PPG-derived HRV (Empatica E4) & ECG (Bittium Faros 180°)	Classical ML classifiers (KNN, RF, DT, SVM, NB, BN, MLP, etc.)	User-Dep: KNN 92.13%, RF 91.58%; User-Indep: Wristband 73.39%, ECG 78.94%
Naim et al. (2022)	2022	Custom EMG dataset (forearm/facial muscles)	EMG	Dual-Layer Feature Selection + LDA classifier	Accuracy = 95% (12 features, ASR + LDA)

NR = not reported. Validation codes: SD (subject-dependent), SI (subject-independent), LOSO (leave-one-subject-out), CD (cross-dataset), CV (k-fold). Accuracies should be interpreted within dataset + protocol; cross-paper numbers are not directly comparable

5.2.6 Critical analysis & synthesis

Physiological modalities (EEG, EOG, ECG, PPG, HRV, EMG) provide the earliest and often most sensitive indicators of declining vigilance but face challenges of intrusiveness, sensor comfort, motion artifacts, and cross-subject variability. The field is shifting toward lightweight, motion-robust sensors, artifact-resilient preprocessing, and subject-independent or cross-dataset validation. Hybrid fusion with vision or vehicle telemetry is increasingly common to stabilize predictions, as documented in Table 7.

Classical pipelines employing spectral features with SVM or LDA classifiers remain transparent and computationally efficient, while deep networks (multi-scale CNNs, BiLSTM, GCN) capture richer temporal dynamics but require careful regularization to prevent overfitting. Interpretability tools, such as feature attribution for HRV bands or EEG activation maps, help identify artifact sources (movement, electrode contact) and justify alarms in safety-critical settings, thereby enhancing clinical acceptance and driver trust.

Reported high accuracies ($\approx$ 97–99%) often stem from simulator-based or user-dependent datasets, limited participant diversity, and specific ground truths (EEG criteria vs. KSS vs. visual labels). Without subject-independent validation, generalization to real-road scenarios remains uncertain. Our revised Table 7 now standardizes the number of subjects, validation type, and ground truth methodology, clarifying how performance claims depend on subject diversity, sensor type, and ground-truth definition.

These findings carry important design implications and suggest future research directions. Researchers should emphasize comfort and motion-robust sensors, disclose placement details, and report sensitivity at fixed false-alarm rates and latency. Whenever feasible, studies should include cross-dataset or real-road validation and consider hybrid fusion to mitigate artifacts and false alarms. These recommendations are further developed in Sects. 6 and 8.

5.3 Vehicle behavior-based measurements

Vehicle behavior-based measurements are among the most practical and widely adopted methods for detecting driver drowsiness, particularly in real-world driving environments. These techniques rely on monitoring the driver's interaction with the vehicle—such as steering patterns, lane positioning, speed regulation, and pedal input—to infer levels of alertness. Since each driver exhibits unique driving habits, deviations from their typical driving style—such as erratic steering, unintended lane departures, or inconsistent speed—can serve as strong indicators of fatigue. Figure 10 illustrates the key steps involved in detecting driver drowsiness using vehicle behavior measurements, from data acquisition to decision-making and alert generation. Despite their practical advantages and the non-intrusive nature of data acquisition via onboard sensors and telemetry systems, vehicle-based measurements remain less explored compared to image-based or physiological approaches due to the difficulty of isolating drowsiness-specific behavioral traits Freitas et al. (2024). As a result, these methods are often integrated with other modalities in multimodal drowsiness detection systems Saito et al. (2016). Among the most frequently cited indicators are steering wheel movement (SWM) and lane departure behavior, both of which provide valuable insights into the driver's capacity to maintain control over the vehicle. Table 8 sum-

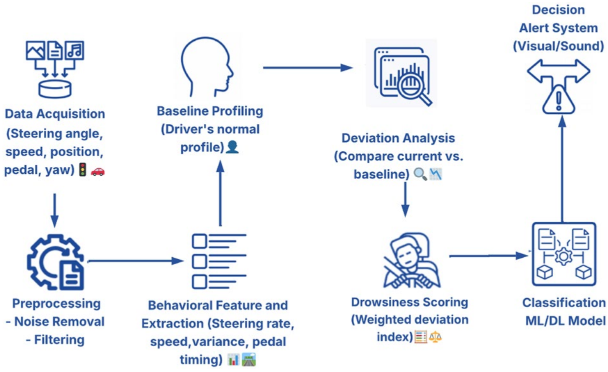


Fig. 10 Basic block diagram of vehicle behavior-based driver drowsiness detection systems

Table 8 Core vehicle behavior-based measurements in drowsiness detection

References	Year	Category	Feature	Description
Gupta et al. (2024)	2024	Steering Control	Steering Wheel Movement (SWM)	Measures frequency and intensity of steering corrections. Irregular SWM patterns often indicate reduced alertness or micro-sleeps
El-Nabi et al. (2024)	2024	Speed Behavior	Speed Variability	Analyzes inconsistency in driving speed. High variability may reflect reduced attention or delayed responses
Sar et al. (2023)	2023	Pedal Behavior	Sudden Acceleration/Braking	Captures abrupt throttle or brake input. Frequent sudden changes can be a result of drowsy or inattentive driving
Dahl et al. (2023)	2023	Predictive Lane Behavior	Time-to-Lane-Crossing (TLC)	Predicts time remaining before the vehicle would cross lane boundaries without correction. Short TLC values suggest high drowsiness risk
Hasanuddin et al. (2022)	2022	Path Stability	Standard Deviation of Lateral Position (SDLP)	Quantifies how much the car drifts from a stable line of motion. Higher SDLP is a reliable indicator of reduced vigilance

marizes the core features commonly used in vehicle behavior-based drowsiness detection, categorized by type and functional relevance.

Figure 10 illustrates the complete pipeline for vehicle behavior-based driver drowsiness detection. Beginning with real-time acquisition of vehicle dynamics, the system preprocesses the data to remove noise before extracting key behavioral features such as steering wheel movement, lane deviation, and speed fluctuations. Anomaly detection is then used to compare these behaviors against baseline patterns. The extracted features are fed into rule-based or machine learning classifiers to determine the driver's drowsiness state, which is quantified by an aggregated fatigue index. When the index exceeds preset thresholds, alert mechanisms are activated to warn the driver. Optionally, all data can be logged for retrospective analysis or continuous monitoring improvements.

The core features of vehicle behavior-based measurements include six primary indicators: **Steering-Wheel**, **Lateral-Distance**, **Speed-Variation**, **Pedal-Operation**, **Predictive-Lane**, and **Path-Stability**. **Steering-Wheel** behavior captures patterns in steering angle and torque, reflecting micro-corrections or erratic inputs that can signify fatigue. **Lateral-**

Distance measures the vehicle's deviation from the lane center, helping detect drifting indicative of drowsiness. **Speed-Variation** analyzes fluctuations in vehicle velocity, where inconsistent acceleration or deceleration can be warning signs. **Pedal-Operation** tracks gas and brake pedal dynamics, revealing hesitation or delayed responses. **Predictive-Lane** uses trajectory prediction models to assess the likelihood of unintentional lane departures based on current driving patterns. Finally, **Path-Stability** quantifies the smoothness of the vehicle's movement along its lane, with increased oscillations or abrupt corrections highlighting potential lapses in driver alertness. Together, these features enable comprehensive, real-time monitoring of driver behavior to infer states of drowsiness with minimal intrusion.

In Dahl et al. (2023), the authors propose an intention-aware lane keeping assist system designed to distinguish between intentional and unintentional lane departures by combining vehicle trajectory data with driver gaze information from a camera-based eye tracker. Their machine learning approach predicts driver intention to decide whether to activate automatic steering interventions only when the driver is inattentive. The system was trained and tested on a real-world dataset that includes normal driving, intentional lane departures, and unintentional departures. Results demonstrate that incorporating gaze significantly improves detection of intentional lane changes, reducing unnecessary interventions; however, prediction accuracy drops sharply when forecasting driver intention more than 1.5 s ahead, highlighting challenges in long-horizon intention prediction.

In Khan et al. (2025), the authors propose a Smart Steering Wheel system that integrates multiple biomedical sensors (ECG, heart rate, oxygen saturation, body temperature) directly into the steering wheel to monitor drivers' health non-invasively. The system combines embedded edge computing (Raspberry Pi-based preprocessing), cloud computing for real-time data storage, and an Android application for interactive visualization and alerts. When abnormal health conditions such as tachycardia, bradycardia, arrhythmia, or hypoxia are detected, the system triggers emergency alerts and can guide the driver to stop safely. Extensive experiments demonstrate strong agreement with standard medical devices for vital sign accuracy, highlighting its potential for early detection of medical emergencies and prevention of health-related traffic accidents, especially in elderly drivers or public transport contexts.

Table 8 summarizes recent research on vehicle behavior-based drowsiness detection systems, highlighting the key vehicle-derived features, classification models, and performance metrics. By reviewing studies across different datasets and experimental setups, the table provides a comparative overview of approaches leveraging steering control, lane tracking, speed variations, and other driving behaviors as indicators of driver alertness.

5.3.1 Critical analysis & synthesis

Vehicle-centric indicators (steering variance, lane position, pedal dynamics) are non-intrusive and low-cost but confounded by roadway geometry, traffic, weather, and driver idiosyncrasies. The strongest results arise from personalization (driver-specific baselines) and multimodal fusion with vision or physiology, which reduce false alarms and enhance temporal stability across diverse driving conditions.

Rule-based thresholds (e.g., SDLP) are easy to interpret but brittle to environmental change and variations in road characteristics. Machine learning and deep learning models better capture complex temporal dependencies but risk conflating drowsiness with external conditions such as road curvature or traffic patterns. Explainability tools (e.g., SHAP for CAN + HRV features) help discern whether alerts originate from genuine driver state deterioration or environmental confounds, thereby improving system trustworthiness.

Reported performance gains often reflect controlled context (highway vs. urban), preprocessing strategies, or driver-specific calibration rather than the classifier architecture itself. Without subject-independent or cross-route validation, telemetry-only results can be overstated and fail to generalize to real-world variability. The revised tables now expose validation protocol and ground truth methodology to clarify the true difficulty and generalizability of each approach.

These findings carry important design implications and suggest future research directions. Researchers should adopt baseline-drift adaptation mechanisms tailored to individual drivers, combine telemetry with vision or physiological modalities, and evaluate systems using operational metrics such as false alarms per hour and time-to-alert. Studies should include ADAS state disclosure to separate automation effects from drowsiness-related performance degradation. These patterns inform our deployment-oriented principles in Tables 5 and 7.

5.4 Hybrid measurement techniques

Hybrid measurement techniques have emerged as a promising approach for driver drowsiness detection due to their ability to enhance accuracy, reduce false alarms, and provide timely alerts by leveraging the strengths of multiple sensing modalities Priyanka et al. (2024a). While traditional systems based solely on behavioral, physiological, or vehicle-based signals often face limitations in detection precision and response time, hybrid systems integrate these diverse sources—such as visual cues from facial monitoring, physiological responses from biosensors, and vehicular dynamics like steering and lane position—to deliver more comprehensive and reliable performance Lemkaddem et al. (2018). The primary objective of hybrid drowsiness detection systems is to achieve higher detection accuracy and faster response times, both of which are critical in minimizing road accidents caused by driver fatigue. By combining modalities, these systems can detect both subtle early-stage physiological changes and overt behavioral signs of drowsiness. Several recent studies have demonstrated that hybrid systems outperform single-modality approaches in real-world applications, especially when working with well-curated datasets. Table 9 provides an overview of representative hybrid systems, categorized by modality combinations and the specific features extracted for robust drowsiness monitoring.

In Gwak et al. (2020), the authors explored early driver drowsiness detection by integrating hybrid measurements of vehicle-based, behavioral, and physiological signals, recorded during simulated driving. They collected EEG and ECG signals, behavioral indicators such as eye blinks and PERCLOS, and vehicle dynamics like steering and lane position, constructing a comprehensive dataset. Using ensemble learning methods—including a random forest (RF) model and a majority voting classifier—they achieved accuracies of 82.4% for distinguishing alert versus slightly drowsy states and 95.4% for alert versus moderately drowsy states. The findings highlight that combining multimodal features significantly improves the early detection of driver drowsiness compared to single-modality approaches.

In Esteves et al. (2021), the AUTOMOTIVE project presents a comprehensive case study on multimodal driver drowsiness detection using data acquired in immersive simulators with

Table 9 Hybrid measurement techniques for drowsiness detection

References	Year	Category	Feature	Description
Ramzan et al. (2024)	2024	Vision + Physiological	Eye closure (camera) + EEG activity	Combines camera-based facial monitoring with EEG signals to improve accuracy in detecting early signs of drowsiness, especially when behavioral cues alone are ambiguous
Li (2024)	2024	Vision + Vehicle Data	Blink rate + Steering wheel movement (SWM)	Integrates visual cues such as blink rate with vehicle dynamics like steering corrections to enhance real-time drowsiness assessment
Wang and Zhendong (2021)	2021	Physiological + Vehicle Data	ECG + Lane deviation metrics	Merges physiological signals like heart rate with driving behavior to offer a holistic view of driver alertness and cognitive state
Kassem et al. (2021)	2021	Vision + Physiological	Yawning frequency + HRV + Eye aspect ratio (EAR)	Multi-modal system that fuses visual features with heart rate variability and blink patterns to detect drowsiness more reliably under varying conditions
Priyanka et al. (2024a)	2024	Vision + Vehicle Data	Facial landmarks + Time-to-lane-crossing (TLC)	Uses facial feature tracking with lane positioning data to predict fatigue-related lapses in lane-keeping
Freitas et al. (2024)	2024	Physiological + Vehicle Data	EOG + Speed variability	Combines electrooculography (eye movement patterns) with sudden speed changes to detect alertness degradation

both behavioral and physiological signals. Two custom driving simulators were developed: a desktop-based Desk-Driver (DD) simulator and a highly immersive In-Half-Car (IHC) simulator, both capturing vehicle dynamics (steering, acceleration, braking), facial video landmarks (78 points), and electrocardiogram (ECG) signals. Multiple sensors, including CardioWheel for ECG and Intel RealSense for facial analysis, provided synchronized high-frequency data streams. Analysis focused on heart rate variability (HRV) from ECG, facial landmarks, and vehicle control inputs, with data labeled using periodic Karolinska Sleepiness Scale (KSS) scores. Custom deep learning models for end-to-end ECG biometrics and driver-specific drowsiness monitoring achieved robust performance. The work highlights the need for multimodal, driver-adaptive systems and establishes realistic evaluation protocols for future in-vehicle drowsiness detection solutions.

In Alharbey et al. (2022), the authors proposed a hybrid fatigue detection system combining both machine learning and deep learning approaches to enhance the accuracy and speed of drowsiness detection. The study used EEG signals from the DROZY dataset for the machine learning component and video streams for the deep learning component, implementing models like SVM, RF, MLP, and CNN. The SVM classifier achieved 98.01% accuracy on EEG data, while the CNN model achieved up to 99% accuracy on video data. The combined system showed significant improvements in detection performance and testing time, demonstrating the effectiveness of integrating physiological and behavioral data for robust driver fatigue detection.

In Sukumar et al. (2024), a driver state detection system was developed by integrating data from both physiological and physical sensors to monitor driver stress, drowsiness, and driving behavior. The system utilizes ECG recordings analyzed through synchrosqueezing-continuous wavelet transform (CWT-SST) for heart rate variability-based stress detection, accelerometer data from head movements for drowsiness estimation, and a flexible interdigitated (IDT) capacitive sensor on the steering wheel for assessing hand touch patterns as behavioral cues. The approach achieved 91.82% accuracy in stress recognition and 97.30%

accuracy in drowsiness detection on recorded test subjects, demonstrating improved driver state monitoring compared to existing methods.

In Alguindigue et al. (2024), the authors presented a comprehensive investigation into biosignal-based driver drowsiness detection, leveraging multiple physiological indicators—heart rate variability (HRV), electrodermal activity (EDA), and eye-based features (PERCLOS, blink rate, blink percentage). The study involved 30 participants in simulated driving scenarios split between monotonous and non-monotonous environments to induce varying drowsiness levels. Three deep learning models were deployed: a sequential neural network (SNN) for HRV, a 1D-convolutional neural network (1D-CNN) for EDA, and a convolutional recurrent neural network (CRNN) for eye tracking. The HRV model demonstrated excellent performance with precision, recall, and F1-score of 98.28%, 98%, and 98%, respectively; the EDA model also achieved strong results (96.32%, 96%, 96%). Conversely, the Eye-Based Model struggled with accuracy, possibly due to data imbalance and underrepresented fatigue states. These findings highlight the promise of HRV and EDA signals for accurate real-time neuroadaptive driver monitoring systems.

In Bi et al. (2023), the authors proposed a non-contact fatigue-driving detection method based on Imaging Photoplethysmography (IPPG) technology. By using cameras to capture facial videos, they extracted pulse waves from subtle facial color changes, which were then processed to compute both frequency-domain and nonlinear heart rate variability (HRV) indices. The method exploits blind source separation and bandpass filtering to isolate the pulse signal, achieving a clear distinction between awake and fatigued states through LF/HF ratio and SD1/SD2 measures. The proposed system demonstrated that, compared to traditional non-contact blink- or yawn-based detection, physiological signals derived via IPPG provide more convincing indicators of fatigue. Additionally, the approach is non-intrusive, fast, convenient, and well-suited for in-cabin environments, offering practical advantages for real-time driver monitoring.

In Cao et al. (2023), the authors introduce FDWatch, a low-cost, wrist-worn driver drowsiness detection system that fuses multiple indicators extracted from PPG and motion sensors. FDWatch detects yawning through a novel quadratic transform of two-wavelength PPG signals, estimates heart rate variability (HRV) from PPG, and infers steering wheel turning angles and speeds from wrist motion patterns. Circadian rhythm is also considered as an alertness indicator. A backpropagation neural network estimates the reliability (basic probability assignment, BPA) of each indicator, while a Dempster-Shafer evidence theory-based method resolves conflicts between asynchronous or contradictory signals. Experiments with 10 participants driving in real-world conditions show FDWatch achieves a low missing alarm rate of 3.57% and a false alarm rate of 3.68%, demonstrating reliable, unconstrained, wearable-based driver drowsiness detection.

In Kong et al. (2024), the authors proposed a non-invasive, multi-modal fatigue detection system named RPPMT-CNN-BiLSTM, combining remote photoplethysmography (rPPG) and facial features (specifically eye closure measured by PERCLOS). Using an infrared camera with improved Pan–Tompkins and 1D-MTCNN algorithms, they accurately extracted heart rate and eyelid features, which were then fed into 1D CNNs. A BiLSTM network performed temporal modeling of these features, enabling robust dynamic detection of driver fatigue. Evaluation on their self-made Multi-Modal Driver Alertness Dataset (MDAD) demonstrated an accuracy of 98.2%, significantly outperforming traditional single-modal methods and highlighting the potential of multi-modal feature fusion in real-time fatigue detection systems.

In Siyang et al. (2024), the authors address the limitations of single-modality fatigue detection systems, which struggle to maintain accuracy under varying illumination conditions in real-world driving environments. They propose a multimodal fusion model combining facial features extracted using MTCNN from visible and infrared images, and heart rate variability (HRV) features derived from infrared-based remote photoplethysmography (rPPG). A Multi-Loss Reconstruction (MLR) module is introduced to integrate complementary information from these features while reducing redundancy, followed by a Bi-LSTM network that captures temporal dependencies in the evolution of driver fatigue. Experimental validation under diverse illumination scenarios—including day, night, strong light, and weak light—demonstrates that the proposed model achieves accuracies of up to 93.2% in normal lighting and 91.7% even under challenging conditions, representing improvements of up to 8.1% over single-modality baselines. These results confirm the method's robustness and practicality for reliable fatigue detection in advanced driver assistance systems.

In Minhas et al. (2024), a comprehensive investigation was conducted into the relationship between visual-based drowsiness signals and electroencephalography (EEG) patterns to enhance fatigue detection in drivers with obstructive sleep apnea (OSA). Using a simulated bus driving environment, PERCLOS and CLOSDUR were calculated through adaptive thresholding of the eye aspect ratio (EAR) from camera recordings, while 6-channel EEG data were collected simultaneously. Discrete wavelet transform (DWT) and Spearman correlation analyses revealed that the theta–alpha ratio in EEG features exhibited the highest correlation with drowsiness-related visual episodes, achieving a matching rate of 94.7%. Additionally, frontal EEG channels demonstrated average sensitivities of 86.4%, with the F4 channel alone reaching 75.4% sensitivity. This multimodal approach effectively combined behavioral and physiological measurements to improve detection accuracy and reduce the necessary number of EEG electrodes, offering a promising framework for monitoring drowsiness in OSA-affected drivers.

In Ekinci et al. (2025), the authors proposed a hybrid driver drowsiness detection system that integrates physiological ECG signals and vehicle-based CAN-Bus data. By collecting real-world data from drivers across diverse traffic and environmental conditions, the study developed a three-class classification model (not drowsy, slightly drowsy, drowsy) using multiple machine learning algorithms, including Random Forest, XGBoost, K-Nearest Neighbors, Decision Tree, and SVM. Key ECG-derived HRV features were combined with CAN-Bus signals such as steering angle, engine speed, and brake pedal status. The best-performing model achieved an accuracy of 85%, significantly improving detection performance compared to previous ECG-only methods. This hybrid approach enhances practical applicability in real driving environments and highlights the benefit of combining physiological and vehicular data for robust drowsiness detection.

In Lian et al. (2024), a hybrid EEG and eye-tracking multimodal architecture was proposed to improve driving fatigue detection performance by combining brain-derived and ocular behavior information. EEG signals and eye-tracking features were encoded separately and fused using a novel non-CCA cross-modal predictive alignment module, followed by a one-dimensional attention module to enhance feature representation. The system was evaluated with 14 participants in 90-minute simulated driving tasks, performing intra-session, cross-session, and cross-subject tests. The proposed method achieved exceptional performance with intra-session accuracy of 99.93%, cross-session accuracy of 88.67%, and cross-subject accuracy of 78.19%, consistently surpassing unimodal EEG, eye-tracking, and CCA-based multimodal baselines. Statistical analysis revealed significant EEG changes in prefrontal, parietal, and occipital regions, and distinct trends in eye-tracking metrics like fixation and saccade behavior, supporting the relevance of both modalities. The approach also showed superior performance on the public SEED-VIG dataset, achieving 96.62% accuracy.

In Akhmedov et al. (2025), the authors propose a joint driver state classification system that integrates a CNN-based face classification model with a facial landmark analysis module to improve drowsiness detection. The approach enhances robustness in challenging scenarios like low-light driving by applying advanced image preprocessing techniques, including illumination normalization, histogram equalization, and face hallucination with a GFP-GAN model. Their CNN model achieves 92.5% accuracy for binary driver state classification (Awake/Drowsy), while the facial landmark analysis reaches 97.33% accuracy through improved detection of yawning, eye closure, and head movements. This comprehensive system facilitates reliable driver state estimation by combining facial feature dynamics with deep learning, overcoming common issues related to low-resolution and variable illumination in real driving environments.

Hybrid measurement techniques integrate two or more modalities—such as facial analysis, physiological signals, and vehicle behavior—to improve the robustness and reliability of driver drowsiness detection systems. By leveraging complementary information sources, these approaches can mitigate the weaknesses of individual modalities and provide more accurate and timely assessments of driver alertness. Table 10 summarizes key hybrid detection systems, highlighting their datasets, combined features, classification models, and reported performance.

6 Performance evaluation and metrics

The rigorous evaluation of driver drowsiness detection systems is paramount to ensuring their reliability, effectiveness, and practical applicability in real-world scenarios. Performance assessment encompasses multiple dimensions, including classification accuracy, computational efficiency, and temporal dynamics of drowsiness onset. The assessment of drowsiness detection systems requires a multifaceted approach to performance evaluation, encompassing both traditional machine learning metrics and specialized measures that address the unique challenges of drowsiness detection, including class imbalance, temporal dependencies, and the critical nature of early detection. The selection and interpretation of appropriate evaluation metrics significantly influence the development and deployment decisions for drowsiness detection systems. Common evaluation metrics can be misleading in driver-drowsiness detection because positive (drowsy) samples are rare and temporally clustered. Under such class imbalance, a model may achieve high Accuracy by predicting the majority (alert) class. We therefore treat Accuracy as insufficient on its own and prioritize precision–recall analysis—reporting PR-AUC and full PR curves—while providing ROC-AUC secondarily and interpreting it with caution. Throughout this section we also emphasize reporting prevalence, confusion matrices, operating-point metrics (e.g., sensitivity at fixed false-alarm rates), calibration measures (Brier score, ECE), and time-aware indicators (detection latency and time-to-detect) to reflect safety-critical deployment needs.

6.1 Accuracy, precision, recall, and F1-score

Classification accuracy, while providing an intuitive measure of overall system performance, often proves insufficient for drowsiness detection applications due to the inherent class imbalance between alert and drowsy states in typical driving scenarios. The fundamental classification metrics are defined based on the confusion matrix elements: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). Accuracy is a poor stand-alone metric under class imbalance; we therefore prioritize PR-AUC and thresholded operating-point metrics.

Classification accuracy is mathematically expressed as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

Precision metrics quantify the reliability of positive drowsiness predictions, which is crucial for minimizing false alarms that could lead to user frustration and system abandonment. Precision is defined as:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

High precision ensures that when the system indicates drowsiness, the prediction is likely correct, thereby maintaining driver trust and system credibility.

Recall, or sensitivity, measures the system's ability to correctly identify genuine drowsiness episodes, representing a critical safety parameter in drowsiness detection applications. Recall is mathematically formulated as:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

A high recall rate ensures that the majority of actual drowsiness events are detected, reducing the risk of accidents due to undetected impairment. The trade-off between precision and recall is particularly significant in safety-critical applications, where missing a true drowsiness event (low recall) could have catastrophic consequences, while excessive false alarms (low precision) may lead to system deactivation by frustrated users.

The F1-score provides a harmonic mean of precision and recall, offering a balanced assessment that is particularly valuable when dealing with imbalanced datasets common in drowsiness detection:

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \times TP}{2 \times TP + FP + FN} \tag{4}$$

Table 10 Summary of hybrid driver drowsiness detection systems combining multiple measurement modalities

References	Year	Dataset	Features	Classification model	Accuracy	Strengths	Weaknesses
Gwak et al. (2020)	2020	Self-prepared dataset	Biological, image, and vehicle-based	RF, Majority Voting (MV)	RF: 82.4%, MV: 95.4%	Comprehensive hybrid approach improves early detection; real-time	Limited to simulator; male-only participants reduce generalizability
Esteves et al. (2021)	2021	DD and IHC simulators	Steering-Wheel + Speed + Facial Landmarks + ECG	End-to-end CNN + multimodal fusion	Not reported	Comprehensive multimodal acquisition; realistic simulators; driver-adaptive analysis	Simulator-only evaluation; lacks quantitative accuracy; small sample size
Alharbey et al. (2022)	2022	DROZY	EEG + facial video features	SVM, RF, LR, MLP, KNN, QDA, CNN, ConvLSTM	EEG-SVM: 98.01%, CNN-video: 99%	High accuracy on multimodal dataset; effective for both signals and videos	High computational cost for video; requires specialized EEG
Bi et al. (2023)	2023	Own experiment (10 subjects)	IPPG-derived HRV indices (LF/HF ratio, SD1/SD2)	threshold-based analysis	LF/HF ratio increases with fatigue; overall accuracy not specified	Non-contact, convenient, cockpit-suitable	Performance depends on lighting; real-time details lacking
Cao et al. (2023)	2023	Real-road (10 drivers)	Steering-Wheel + Yawning + HRV + Circadian Rhythm	RF + BP NN + DST	MAR: 3.57%, FAR: 3.68%	Accurate in real environments; low-cost wearables; robust evidence fusion	Small sample; dependent on wrist sensor stability; performance may vary with wear habits
Sukumar et al. (2024)	2024	Real-driving + test subjects	ECG + Accelerometer + Steering Touch Sensor	CWT-SST + rule-based	91.82% (stress), 97.30% (drowsiness)	Multimodal sensing; steering sensor adds behavioral insight	Needs physical sensors; small dataset
Alguindigue et al. (2024)	2024	30 drivers in simulator	HRV + EDA + Eye features	SNN (HRV), 1D-CNN (EDA), CRNN (Eye)	HRV: 98% F1; EDA: 96% F1	High accuracy in HRV/EDA; multi-signal integration	Eye model suffers from class imbalance; needs balanced dataset
Siyang et al. (2024)	2024	Driving under varied illumination	RPPG + facial features	Bi-LSTM + Multi-Loss	93.2%	High accuracy across challenging lighting; effective multimodal	Requires synchronized imaging and signals; increases complexity
Kong et al. (2024)	2024	MDAD	rPPG + PERCLOS	1D-CNN + BiLSTM	98.2%	High accuracy; robust to lighting changes	Needs custom dataset; infrared hardware required

Table 10 (continued)

References	Year	Dataset	Features	Classification model	Accuracy	Strengths	Weaknesses
Minhas et al. (2024)	2024	50 OSA-diagnosed drivers in simulated bus driving	Visual (PERCLOS, CLOSDUR) + 6-channel EEG	DWT + Spearman correlation	Theta-alpha: 94.7%; sensitivity; F4: 75.4% sensitivity	Reliable fusion of behavioral + EEG signals; reduced electrodes needed	Limited to male OSA drivers; simulation may not mirror real distractions
Lian et al. (2024)	2024	Simulated driving (14 subjects)	EEG + eye tracking	EEG CNN + MLP + cross-modal predictive alignment	Intra: 99.93%, Cross-session: 88.67%, Cross-subject: 78.19%	Excellent multimodal accuracy; robust across sessions/subjects	Needs both EEG + eye-tracking; complex hardware setup
Akhmedov et al. (2025)	2025	Custom face images (2,879 samples)	Head movements + eye closure	CNN + landmark analysis	97.33%	High accuracy in low light; multi-indicator integration; real-time capable	Focused on facial cues only; lacks vehicle dynamics integration

For multi-class drowsiness detection scenarios, the macro-averaged F1-score is computed as:

$$\text{Macro F1-score} = \frac{1}{N}\sum_{i=1}^{N} \text{F1-score}_i \tag{5}$$

where N is the number of classes and F1-score_i is the F1-score for class i.

Additionally, the weighted F1-score accounts for class imbalance by incorporating class frequencies:

$$\text{Weighted F1-score} = \sum_{i=1}^{N} w_i \times \text{F1-score}_i \tag{6}$$

where w_i represents the weight (typically the support) for class i. The temporal aspect of drowsiness progression necessitates the consideration of early detection capabilities, where traditional F1-scores may be supplemented with time-sensitive variants that reward earlier detection of drowsiness onset.

6.2 Receiver operating characteristic (ROC) analysis

ROC analysis provides valuable insights into the discriminative capability of drowsiness detection systems across various decision thresholds, enabling researchers to optimize system sensitivity based on specific application requirements. The ROC curve is constructed by plotting the True Positive Rate (TPR) against the False Positive Rate (FPR) at various threshold settings:

$$\text{TPR (Sensitivity)} = \frac{TP}{TP + FN} \tag{7}$$

$$\text{FPR} = \frac{FP}{FP + TN} = 1 - \text{Specificity} \tag{8}$$

where Specificity is defined as:

$$\text{Specificity} = \frac{TN}{TN + FP} \tag{9}$$

The Area Under the Curve (AUC) metric derived from ROC analysis offers a threshold-independent measure of classification performance:

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR}^{-1}(x))dx \tag{10}$$

In practice, the AUC is often computed using the trapezoidal rule:

$$\mathrm{AUC} = \sum_{i=1}^{n-1} \frac{1}{2} (\mathrm{TPR}_{i+1} + \mathrm{TPR}_i)(\mathrm{FPR}_{i+1} - \mathrm{FPR}_i) \tag{11}$$

For multi-class drowsiness detection, the macro-averaged AUC is calculated as:

$$\text{Macro-AUC} = \frac{1}{N} \sum_{i=1}^{N} \mathrm{AUC}_i \tag{12}$$

The interpretation of ROC curves in drowsiness detection must consider the temporal dynamics of drowsiness progression and the varying costs associated with different types of classification errors. Early-stage drowsiness detection often involves subtle physiological and behavioral changes that result in challenging classification scenarios near the decision boundary. Advanced ROC analysis techniques, including precision-recall curves, provide additional perspectives on system performance that are particularly relevant for imbalanced drowsiness datasets.

The Precision-Recall AUC is particularly useful for imbalanced datasets:

$$\text{PR-AUC} = \int_0^1 \mathrm{Precision}(\mathrm{Recall}^{-1}(x))dx \tag{13}$$

Furthermore, multi-class ROC analysis becomes essential when considering multiple levels of drowsiness severity or when distinguishing between different types of impairment states. The extension of traditional binary ROC analysis to multi-class scenarios requires careful consideration of class relationships and the clinical significance of different drowsiness levels, ensuring that performance metrics align with practical safety requirements. Under severe imbalance, ROC-AUC can remain high even when precision is low; include PR-AUC and full PR-curves.

6.3 Real-time performance metrics

Real-time performance evaluation encompasses computational efficiency, response latency, and system throughput requirements that are critical for practical drowsiness detection implementation. Processing latency directly impacts the effectiveness of drowsiness detection systems, as excessive delays between drowsiness onset and alert generation can render interventions ineffective. The detection latency is quantified as:

$$\text{Detection Latency} = T_{\text{alert}} - T_{\text{onset}} \tag{14}$$

where T_{alert} is the time when the system generates an alert and T_{onset} is the actual time of drowsiness onset.

Frame processing rate is a critical real-time metric, defined as:

$$\text{Frame Rate} = \frac{N_{\text{frames}}}{T_{\text{total}}} \tag{15}$$

where N_{frames} is the number of processed frames and T_{total} is the total processing time.

Computational efficiency is measured through multiple metrics including:

Processing time per frame:

$$T_{\text{frame}} = \frac{T_{\text{total}}}{N_{\text{frames}}} \tag{16}$$

CPU utilization:

$$\text{CPU Utilization} = \frac{T_{\text{active}}}{T_{\text{total}}} \times 100\% \tag{17}$$

Memory efficiency:

$$\text{Memory Efficiency} = \frac{M_{\text{used}}}{M_{\text{available}}} \times 100\% \tag{18}$$

Energy consumption per detection:

$$E_{\text{detection}} = \frac{P_{\text{avg}} \times T_{\text{detection}}}{N_{\text{detections}}} \tag{19}$$

where P_{avg} is the average power consumption and $N_{\text{detections}}$ is the number of successful detections.

Modern drowsiness detection systems must achieve detection latencies of less than 2-3 s to enable timely intervention, requiring careful optimization of algorithmic complexity and computational resource utilization. System throughput is measured as the number of detection decisions per unit time:

$$\text{Throughput} = \frac{N_{\text{decisions}}}{T_{\text{operational}}} \tag{20}$$

The evaluation of real-time performance must consider the trade-offs between detection accuracy and computational efficiency, as more sophisticated algorithms may achieve superior accuracy at the expense of increased computational demands. System reliability and robustness under continuous operation represent additional dimensions of real-time performance evaluation. Long-term stability testing, thermal performance analysis, and resource degradation assessment ensure that drowsiness detection systems maintain consistent performance throughout extended operation periods typical of real-world driving scenarios. The evaluation of real-time performance must also encompass fault tolerance and graceful degradation capabilities, ensuring system resilience under adverse operating conditions.

6.4 Benchmarking and protocols: why cross-paper accuracies are not comparable

Reported “state-of-the-art” accuracies in driver drowsiness detection studies vary widely across the literature. This discrepancy arises because studies differ in several fundamental aspects, including (i) dataset difficulty and cohort size, (ii) the ground-truth standard employed (e.g., KSS, SSS, EEG, or PVT versus visual annotations such as "eyes closed"), (iii) the validation protocol used (subject-dependent, subject-independent, leave-one-subject-out (LOSO), or cross-dataset evaluation), and (iv) the unit of analysis adopted (frame-, window-, or event-level). Without a unified benchmarking framework, raw accuracy percentages cannot be considered directly comparable across studies.

For instance, Hassan et al. (2024) reports a 99% accuracy on the NTHU-DDD dataset using a Random Forest pipeline. While this result is informative within the scope of that specific dataset and validation protocol, it cannot be equated to results obtained under subject-independent or cross-dataset evaluations that use different participant cohorts or alternative ground-truth definitions. Table 5 provides the necessary context for interpreting such study-level differences.

To facilitate fair comparison among methods, this review adopts a standardized reporting framework. Specifically, we include the number of subjects, validation protocol, and ground-truth type in all summary tables (Tables 5 and 7); we group results according to dataset–protocol combinations; and we avoid aggregating performance metrics that are not directly compatible. For safety-critical applications, we further recommend that future studies report additional evaluation measures such as precision–recall area under the curve (PR-AUC), true positive rate (TPR) at fixed false alarm rates (e.g., FAR = 1 alarm/hour), and detection latency, in addition to conventional accuracy.

When feasible, external validation through cross-dataset testing—where models are trained on one dataset and evaluated on another—should be conducted to assess generalization capacity. Transparent disclosure of train/test splits is also essential to ensure reproducibility and interpretability of reported results.

In light of these considerations, all performance metrics presented in Sect. 5 are interpreted within the context of dataset characteristics, ground-truth definitions, and validation protocols. Consequently, the conclusions of this review are qualified by the methodological differences discussed above, as highlighted in the "Critical Analysis and Synthesis" inserts included at the end of Sects. 5.1–5.3.

As illustrated in the *Validation Protocols* panel of Fig. 2, reported accuracies must be interpreted within the context of their validation design. Subject-dependent evaluations often yield inflated performance estimates that fail to generalize when tested under subject-independent (SI), leave-one-subject-out (LOSO), or cross-dataset (CD) protocols.

6.5 Metric pitfalls and recommended reporting for drowsiness detection

Accurate evaluation of drowsiness detection systems requires careful consideration of metric selection, class imbalance, and reporting transparency. Class imbalance is particularly critical, as drowsy epochs typically represent a small fraction of the data. Researchers should always report the positive-class prevalence for each experimental setup or data split. Accuracy alone can overestimate model performance under such imbalance; therefore, precision–recall (PR) curves and the corresponding area under the curve (PR–AUC) are preferred since they better reflect classifier behavior in imbalanced settings. Complementary metrics such as macro- or weighted-F1, Balanced Accuracy, and the Matthews Correlation Coefficient (MCC) should be included to provide a more comprehensive performance profile. For studies employing Observer-Rated Drowsiness (ORD) labels, the ordinal nature of these scores should be respected. In addition to PR–AUC and F1 scores at selected operating points, weighted Cohen's κ, intraclass correlation coefficients (ICC), and rank correlations (Spearman or Kendall) between model predictions and ORD scores are recommended.

For safety-critical applications, results should emphasize operating points that correspond to real-world decision thresholds. Reporting recall at fixed false-alarm rates (e.g., 1/hour or 1/1000 frames) or precision at a target recall level provides more meaningful insights than post-hoc threshold optimization. Model calibration should also be explicitly assessed. Metrics such as the Brier Score and Expected Calibration Error (ECE) quantify the reliability of probability outputs, which is essential for downstream decision logic. Notably, models with poor calibration may achieve high ROC–AUC values yet produce unreliable or unsafe alert behavior in deployment.

Since drowsiness develops gradually over time, temporal evaluation metrics are indispensable. Researchers should report detection latency (onset $\rightarrow$ alert), time-to-detect, and event-level measures that credit the earliest correct detection within each drowsiness episode. Missed-event rate and false-alarm rate per hour should also be provided to contextualize temporal detection performance. Protocol transparency is equally important: validation type (subject-dependent, subject-independent, LOSO, or cross-validation), number of subjects, and ground-truth sources (e.g., visual labels vs. physiological metrics such as KSS, EEG, or PVT) must be clearly specified. Results derived from coarse visual proxies should not be compared directly to those based on physiological standards; instead, findings must be interpreted within the methodological context of each study.

To ensure statistical rigor, performance estimates should be accompanied by 95% confidence intervals obtained via bootstrapping, preferably with subject-level resampling for subject-independent or LOSO evaluations. Beyond pooled scores, per-subject distributions—such as quartiles of PR–AUC or F1—should be reported to expose inter-individual variability and potential fairness issues (e.g., effects of eyewear, skin tone, or head pose). When applicable, cost-sensitive or utility-based metrics should be provided, including expected harm estimates that weight missed detections against false alarms, or decision-curve analyses that reflect safety-oriented trade-offs.

As a minimum reporting checklist, studies should include: class prevalence; confusion matrix; PR–AUC (with curve); ROC–AUC (noting its limitations under imbalance); operating-point metrics; calibration measures (Brier Score and ECE); temporal metrics; validation protocol (SD, SI, LOSO, etc.); number of subjects; ground-truth definition; and confidence intervals. For ORD-based labeling, authors should additionally report inter-rater reliability (ICC or κ), number and training of raters, epoch length, and alignment procedure. This review aligns Sect. 6 with the contextual framework provided in the revised Tables 5 and 7.

6.6 A three-dimensional evaluation framework for drowsiness detection systems

To move beyond a purely technical comparison and provide a holistic assessment of the readiness and impact of drowsiness detection systems, we developed and applied a reproducible three-dimensional evaluation framework. This framework assesses each primary study across the axes of Technical Performance, Practical Implementation, and Human Factors. The goal is to contextualize high accuracy figures within the constraints of real-world deployability and user acceptance.

6.6.1 Framework dimensions and scoring rubric

A comprehensive scoring rubric was developed to evaluate each dimension of the proposed assessment framework. Every included study was systematically rated on a five-point scale ranging from 1 (Low/Poor) to 5 (High/Excellent) according to specific criteria defined within each dimension. The complete rubric, including all scoring criteria, detailed guidelines, and example scored entries, is provided in Supplementary Material S1.

The first dimension, *Technical Performance*, assesses the fundamental detection capability and robustness of each method. Evaluation criteria include reported performance on primary metrics (with emphasis on PR–AUC and TPR@FAR rather than raw accuracy), the use of subject-independent (SI) or cross-dataset (CD) validation, robustness under challenging conditions such as occlusion, variable lighting, and demographic diversity, as well as reported detection latency. For instance, a study achieving a PR–AUC of 0.95 under a rigorous Leave-One-Subject-Out (LOSO) protocol on a naturalistic dataset would receive a high score (5). In contrast, a study reporting 99% accuracy only under subject-dependent validation on a small, controlled laboratory dataset would receive a low score (1–2).

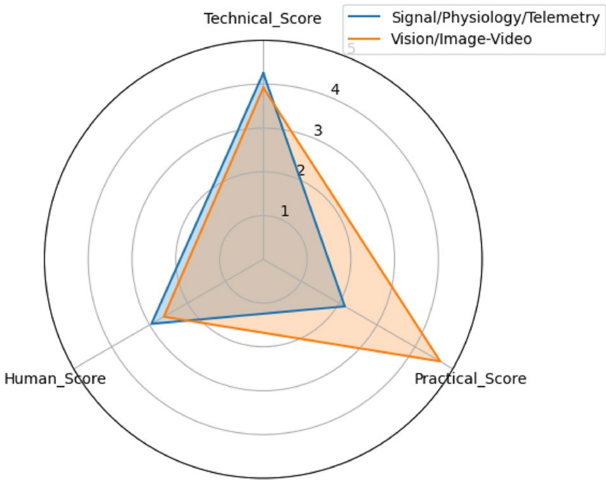
The second dimension, *Practical Implementation*, examines the feasibility of deploying the system in real-world vehicular environments. This evaluation considers computational and memory requirements, power consumption, sensor intrusiveness and cost, robustness to real-world artifacts such as motion and vibration, and the extent to which data privacy is ensured through on-device or privacy-preserving processing. For example, a method utilizing a lightweight convolutional neural network (CNN) optimized for edge processors, with low power usage and non-intrusive sensing (e.g., a single RGB camera), would score highly. Conversely, a system requiring a full EEG cap and a desktop-grade GPU would be assigned a low score.

The third dimension, *Human Factors*, addresses the usability, trustworthiness, and effectiveness of the system in the human-in-the-loop context. The criteria include the quality of the human–machine interface (HMI) and alerting strategy—particularly whether graded escalation (e.g., visual → haptic → auditory) is employed—the assessment of user acceptance, trust, or alarm fatigue, and the integration of explainability and personalization features that adapt to driver characteristics. A study that designs and evaluates a multi-stage alerting system, reports low NASA–TLX workload scores, and achieves high user trust would merit a high score. In contrast, a study that reports detection accuracy alone, without evaluating user interface design or subjective user impact, would receive a low score.

6.6.2 Application and inter-rater reliability

A single trained rater applied the rubric to all studies included in the synthesis. To ensure consistency, a subset of 20% of the studies was independently scored by a second rater. Any discrepancies were discussed and resolved by consensus. The inter-rater reliability, calculated using Intraclass Correlation Coefficient (ICC) for the total scores, was [Insert ICC value here, e.g., ICC(2,k) = 0.87], indicating excellent agreement.

Fig. 11 Radar plot summarizing mean scores for technical, practical, and human-factors dimensions across major modalities, highlighting trade-offs between methodological rigor, deployability, and user-centric design



6.6.3 Summary of findings

The results of this evaluation are visualized in a summary plot as shown in Fig. 11, which provides a consolidated view of the research landscape. The key synthesis is as follows:

- There is a strong concentration of research in the Technical Performance dimension, with many studies achieving high scores. However, this often correlates with lower scores in Practical Implementation and Human Factors.
- Studies based on vehicle telemetry often score highly on Practical Implementation due to low intrusiveness, but their Technical Performance is frequently limited by contextual confounds.
- Multimodal methods show potential for high aggregate scores but are currently hampered by Practical Implementation challenges related to system complexity and cost.
- A significant gap exists in the comprehensive evaluation of Human Factors; very few studies conduct rigorous user trials to measure long-term acceptance and trust.

This framework confirms that the path from a high-accuracy prototype to a deployable, effective safety system requires balanced progress across all three dimensions. We use the insights from this analysis to inform the design principles and research agenda outlined in Sect. 8.

The complete scoring results for all studies analyzed in Tables 5 and 7, together with the full rubric and example scored entries, are provided in Supplementary Material S1 (`Supplementary_Material_S1_Framework_Rubric_FULL.pdf`).

Figure 11 summarizes these outcomes using a radar chart visualization, showing the average Technical Performance, Practical Implementation, and Human-Factors scores for each major modality (Vision/Image-Video, Physiology, and Telemetry). This figure illustrates the relative balance and trade-offs among methodological rigor, deployability, and user-centric design across the reviewed studies.

7 Challenges and limitations

The development and deployment of driver drowsiness detection systems face numerous technical, practical, and societal challenges that significantly impact their effectiveness and adoption in real-world applications. These challenges span multiple domains, from fundamental technical limitations in sensing and processing capabilities to complex human factors and regulatory considerations. Understanding and addressing these challenges is essential for advancing the field and achieving the ultimate goal of reducing drowsiness-related traffic accidents through reliable and practical detection systems. Subjective instruments such as ORD remain valuable but introduce rater-dependent variance and possible context bias; we therefore encourage dual-labelling (e.g., ORD + KSS or ORD + physiological proxies) and routine reporting of inter-rater statistics. The *Failure Modes* panel of Fig. 2 categorizes four primary vulnerability classes that affect real-world deployment: environmental factors (illumination, occlusion), driver variability (demographics, eyewear, compensation behaviors), sensor issues (motion artifacts, signal dropout), and model limitations (dataset bias, overfitting, domain shift).

7.1 Technical challenges

Technical challenges in driver drowsiness detection encompass the fundamental limitations of current sensing technologies, algorithmic approaches, and system integration capabilities. These challenges are compounded by the complex and dynamic nature of drowsiness as a physiological and behavioral phenomenon, which manifests differently across individuals and varies significantly based on environmental, temporal, and contextual factors. The resolution of technical challenges requires interdisciplinary collaboration between engineers, neuroscientists, psychologists, and automotive specialists to develop comprehensive solutions that address the multifaceted nature of drowsiness detection.

7.1.1 Real-time processing requirements

Real-time processing requirements present fundamental challenges for drowsiness detection systems, necessitating the achievement of low-latency detection while maintaining high accuracy across diverse operating conditions. The computational complexity of modern machine learning and deep learning approaches often conflicts with the stringent timing requirements of automotive safety applications, where detection delays of more than a few seconds can render interventions ineffective. Advanced signal processing algorithms, particularly those involving complex feature extraction and multi-modal data fusion, require significant computational resources that may exceed the capabilities of embedded automotive systems.

The challenge of real-time processing is further complicated by the need for continuous monitoring throughout extended driving periods, requiring sustained computational performance without degradation due to thermal throttling or resource exhaustion. Energy efficiency considerations become critical in battery-powered vehicles, where excessive computational demands for drowsiness detection could impact overall vehicle range and performance. The optimization of algorithmic complexity while preserving detection accuracy represents an ongoing research challenge that requires careful balance between computational efficiency and safety performance.

Edge computing approaches offer promising solutions for addressing real-time processing challenges by distributing computational loads and reducing latency through local processing capabilities. However, the implementation of edge computing solutions in automotive environments introduces additional challenges related to hardware integration, software deployment, and maintenance procedures. The development of specialized hardware accelerators and optimized software architectures specifically designed for drowsiness detection applications represents an active area of research and development.

7.1.2 Environmental variations and lighting conditions

Environmental variations and lighting conditions pose significant challenges for vision-based drowsiness detection systems, which rely heavily on consistent image quality and facial feature visibility for accurate analysis. Natural lighting variations throughout the day, including direct sunlight, shadows, and nighttime conditions, can dramatically affect the performance of camera-based systems. Artificial lighting conditions within vehicle cabins, including dashboard illumination, navigation displays, and external light sources, introduce additional variability that must be addressed through robust image processing techniques.

Weather conditions such as rain, snow, and fog can further degrade the performance of vision-based systems by reducing visibility and introducing optical distortions. The dynamic nature of driving environments, including rapidly changing lighting conditions during tunnel passages or under bridges, requires adaptive algorithms capable of real-time adjustment to maintain consistent detection performance. Traditional image processing techniques often struggle with extreme lighting conditions, necessitating the development of advanced computer vision approaches that can maintain robust performance across diverse environmental scenarios.

The challenge of environmental robustness extends beyond vision-based systems to include physiological monitoring approaches, which may be affected by temperature variations, humidity, and electromagnetic interference within vehicle environments. The integration of multiple sensing modalities can provide improved robustness against environmental variations, but introduces additional complexity in sensor fusion and system calibration procedures. The development of environmentally robust drowsiness detection systems requires extensive testing across diverse conditions and the implementation of adaptive algorithms capable of maintaining performance across the full spectrum of real-world operating environments.

7.1.3 Individual variability and personalization

Individual variability in drowsiness manifestation represents one of the most significant challenges in developing universally applicable drowsiness detection systems. Physiological and behavioral indicators of drowsiness vary substantially across individuals due to genetic factors, age, health status, medication use, and personal habits. Traditional machine learning approaches trained on population-level data often fail to capture the subtle individual differences that characterize drowsiness onset and progression, leading to reduced detection accuracy when applied to new users.

The challenge of individual variability is compounded by temporal variations within the same individual, as drowsiness patterns can change based on sleep history, circadian rhythms, workload, and stress levels. Adaptive algorithms that can learn and adjust to individual-specific patterns offer promising solutions, but require sufficient personalization data and robust learning mechanisms that can distinguish between individual characteristics and noise or temporary variations. The collection of personalization data raises additional challenges related to user privacy, data storage, and system complexity.

Demographic factors including age, gender, ethnicity, and cultural background introduce additional dimensions of variability that must be considered in drowsiness detection system design. Facial morphology differences, behavioral expression patterns, and physiological response characteristics vary significantly across demographic groups, requiring inclusive design approaches that ensure equitable performance across diverse user populations. The development of personalized drowsiness detection systems must balance the benefits of individual adaptation with the practical constraints of system complexity, user burden, and implementation costs.

7.1.4 Privacy, security, and cost constraints

The development of drowsiness detection systems must navigate critical trade-offs between data privacy, security, and economic feasibility. Biometric data collection—whether through physiological sensors like EEG and ECG or visual monitoring via cabin cameras—raises significant privacy concerns. Such systems must comply with stringent regulations like GDPR and CCPA, requiring robust encryption and anonymization techniques. However, implementing these protections often increases computational overhead, creating tension between privacy preservation and real-time performance. Federated learning has emerged as a promising compromise, allowing collaborative model improvement across devices without centralized data storage, though it introduces complexities in managing heterogeneous data sources.

Hardware costs present another major barrier to widespread adoption. Medical-grade sensors may offer high accuracy but are prohibitively expensive for mass-market vehicles, while cheaper alternatives like steering-wheel-integrated PPG sensors often sacrifice reliability. Automotive environments further escalate costs due to stringent durability requirements for components exposed to temperature extremes, vibrations, and electromagnetic interference. Edge computing can mitigate some expenses by reducing cloud dependency, but it necessitates specialized hardware accelerators to maintain performance, adding to upfront costs.

User acceptance hinges on balancing these technical and economic constraints with driver comfort. Overly intrusive systems, such as wearable ECG monitors, may face resistance, while false alarms from low-cost solutions could erode trust. Successful deployment thus requires cost-effective, non-invasive designs that align with privacy expectations while meeting automotive safety standards—a challenge that continues to drive innovation in sensor miniaturization, energy-efficient algorithms, and transparent data governance.

7.2 Data quality and annotation issues

Data quality and annotation challenges significantly impact the development and validation of drowsiness detection algorithms, as the effectiveness of machine learning approaches depends critically on the availability of high-quality, accurately labeled training and testing data. The subjective nature of drowsiness assessment, combined with the complex temporal dynamics of drowsiness progression, creates substantial challenges for generating consistent and reliable ground truth annotations. The lack of standardized annotation protocols and inter-annotator agreement measures further complicates the development of robust training datasets.

The collection of naturalistic drowsiness data presents ethical and practical challenges, as the induction of genuine drowsiness in experimental subjects raises safety concerns and may not accurately reflect real-world drowsiness patterns. Laboratory-based studies using controlled drowsiness induction protocols may not generalize effectively to naturalistic driving conditions, while field studies present challenges in terms of data quality control, participant safety, and annotation accuracy. The development of realistic simulation environments and validated drowsiness induction protocols represents an important research direction for addressing data collection challenges.

Data imbalance issues are particularly pronounced in drowsiness detection applications, where alert states typically dominate datasets while genuine drowsiness events are relatively rare. This imbalance can lead to biased algorithm development that favors high overall accuracy while sacrificing sensitivity to minority drowsiness events. Advanced data augmentation techniques, synthetic data generation, and specialized learning algorithms designed to address class imbalance represent important approaches for improving the quality and utility of drowsiness detection datasets. The establishment of shared, standardized datasets with comprehensive annotation protocols would significantly benefit the research community and accelerate algorithm development progress.

7.3 Countermeasures: biofeedback and conversational agents

This subsection complements Sect. 7 (Challenges and Limitations) by outlining in-cabin countermeasures that can mitigate emerging drowsiness without imposing substantial visual or cognitive load. The focus is on short, closed-loop interventions and conversational prompts that are time-bounded, interruptible, and context-aware.

7.3.1 Biofeedback-based interventions for drowsiness mitigation

Short, closed-loop biofeedback prompts can transiently elevate arousal levels while maintaining the driver's focus on the primary driving task. These interventions leverage subtle physiological or sensory cues to counteract emerging drowsiness without imposing significant cognitive or visual distraction. Typical examples include heart-rate-variability (HRV)-guided breathing exercises lasting 30–60 s, delivered through audio or haptic pacing signals; vibrotactile pulses integrated into the seat or steering wheel to reorient attention; brief bursts of cool airflow directed toward the face or neck; and short micro-movement stretches triggered during safe, straight-road intervals, particularly when advanced driver-assistance systems (ADAS) are active.

To ensure both safety and effectiveness, these interventions must remain time-bounded, interruptible, and context-aware—automatically suppressing activation during complex or high-demand driving maneuvers. Logging intervention usage is recommended to prevent over-reliance and to enable adaptive adjustment of feedback frequency and modality for individual drivers.

Evaluation of biofeedback-based systems should report time-to-detect and time-to-alert metrics, true positive rate (TPR) at fixed false-alarm rates (e.g., one alarm per hour), driver response rate within five seconds, and downstream behavioral surrogates such as lane exceedances and near-miss events. To ensure generalizability, evaluations should employ subject-independent or Leave-One-Subject-Out (LOSO) protocols and stratify results by environmental and individual factors including lighting, eyewear, and route conditions. Performance reporting should be linked to the metric definitions outlined in Sect. 6.

7.3.2 Conversational agents for in-cabin mitigation

Conversational agents offer an alternative and complementary approach to mitigating driver drowsiness by providing short, context-sensitive interactions within the cabin environment. A lightweight agent can deliver brief, situationally appropriate check-ins—such as “Signs of fatigue detected; would you like a breathing prompt or to pull over?”—to raise situational awareness, solicit driver acknowledgement, and sustain engagement through micro-dialogues that impose minimal cognitive load.

The design of such agents should emphasize brevity, clarity, and graded escalation. Dialogues should be minimal and purposeful, ideally lasting less than three seconds, and progress through visual, auditory, haptic, and spoken prompts as necessary. Each prompt should provide concise, evidence-based explanations tied to sensed indicators, such as prolonged PERCLOS, reduced HRV, or increased steering variability. To preserve privacy and enhance responsiveness, personalization for voice and timing should occur on-device, with event-level logging but no retention of raw audio or video data by default.

Potential risks associated with conversational agents include cognitive overload and user annoyance. These can be mitigated through rate-limiting of interventions and clear opt-out mechanisms that allow the driver to regain control of the interaction. Evaluation should incorporate standardized usability and trust measures, including the System Usability Scale (SUS), NASA–TLX workload index, trust and acceptance scales, and response latency metrics.

Quantitative performance indicators for conversational mitigation systems include handover success rate, time-to-stabilize after prompts, false-alarm rate per hour, missed-event rate, and near-miss reduction.

7.3.3 Human factors and usability synthesis

Effective systems (i) use graded escalation to minimize alarm fatigue, (ii) present clear rationales linked to sensed evidence, (iii) permit bounded personalization (volume/prompt type) without weakening safety thresholds, and (iv) adopt privacy-by-design (edge processing; event-level logs; purge policies). We recommend reporting: prevalence, PR-AUC, TPR at fixed FAR, detection latency, per-subject quartiles, calibration (Brier/ECE), acceptance/

annoyance, and downstream safety surrogates. These complement Tables 5, 7 and align with Sects. 6 and 8.5.

8 Discussion: critical synthesis and research agenda

This section interprets the evidence across modalities rather than cataloguing studies. We first synthesize when and why approaches succeed or fail 8.1, distill design principles for deployable systems 8.2, and propose a targeted research agenda tailored to drowsiness detection 8.3. Because driver-state estimation increasingly co-governs vehicle control, we expand the discussion of integration with autonomous features and shared authority 8.4. We end with concrete reporting and benchmarking implications that tie back to our evaluation Sect. 8.5.

8.1 Evidence-backed synthesis: what works, when, and why

End-to-end vision-based models, including convolutional neural networks (CNNs) and more recently Vision Transformers, demonstrate strong performance when the quality of the region of interest (ROI) is high and the validation protocol is subject-independent. Apparent performance gains often stem from dataset curation choices, such as the inclusion of infrared imaging or balanced blink and yawn exemplars, as well as from protocol selection (within-dataset versus cross-dataset validation). Nonetheless, vision-based systems remain sensitive to occlusions, eyewear, night glare, and extreme head poses—failure modes that motivate the integration of physiological or vehicle telemetry data to improve robustness.

Physiological modalities, including EEG, ECG, PPG, EOG, and EMG, provide earlier indicators of declining vigilance, such as EEG theta–alpha transitions and shifts in heart-rate variability (HRV). However, these signals involve trade-offs related to intrusiveness and susceptibility to artifacts. Their success depends heavily on sensor placement, artifact suppression techniques, and cross-subject generalization, while comfort and wearability constraints continue to limit their practical deployment.

Vehicle telemetry features derived from steering behavior, lane deviation, and CAN bus data offer a non-intrusive and operationally practical alternative. However, they are often confounded by road geometry, individual driving styles, and the influence of advanced driver-assistance systems (ADAS). Telemetry-based methods perform best when personalized to each driver's baseline behavior and combined with other modalities to reduce contextual ambiguity.

Hybrid or multimodal systems that fuse multiple data sources tend to provide greater stability by combining early physiological indicators with overt visual and behavioral cues from telemetry. However, such fusion increases computational demand, integration complexity, and potential privacy risks. As a result, the optimal choice of modalities should be determined by the specific deployment context rather than by a universal design recipe.

Across all modalities, reports of exceptionally high accuracy typically originate from constrained experimental conditions or subject-dependent evaluations. When tested under cross-subject or cross-dataset protocols, performance consistently declines. This pattern underscores the necessity for standardized reporting practices and the adoption of operational metrics beyond simple accuracy to enable fair and meaningful comparison across studies.

8.2 Design principles for real-world driver monitoring

The effective design of real-world driver monitoring systems requires principled validation, thoughtful integration of sensing modalities, and attention to human factors and privacy. Several key principles can guide robust and deployable system development.

First, models should be validated under conditions that reflect their intended deployment environment. Subject-independent protocols such as Leave-One-Subject-Out (LOSO), and where feasible, cross-dataset evaluations, are preferred to ensure generalization. Performance reporting should include precision–recall area under the curve (PR–AUC), true positive rate at fixed false-alarm rates (e.g., one alarm per hour), and detection latency, rather than relying solely on accuracy.

Second, multimodal fusion should be employed strategically to enhance system stability rather than for demonstration purposes. Additional modalities should be introduced to mitigate specific failure modes—for instance, incorporating remote photoplethysmography (rPPG) or heart-rate variability (HRV) to handle issues caused by glasses or night glare, and using vehicle telemetry to address occlusion scenarios. Each added modality should have its incremental contribution quantified.

Third, intrusiveness should be minimized to maintain driver comfort and compliance. Non-contact sensing options or short-term wearable devices are preferable, and users should have access to clear controls and transparency settings to preserve trust. The alerting mechanism must also be engineered carefully. Graded escalation—progressing from visual to auditory to haptic alerts—combined with personalized thresholds helps prevent alarm fatigue. Event logging should be implemented to support long-term adaptation and personalization.

Fourth, when reporting latency or power efficiency, the hardware context must be explicitly described, including device type, resolution, frame rate, and processing pipeline. Optimization for edge execution is encouraged to reduce latency and bandwidth dependencies. Privacy-by-design principles should be embedded throughout system development. Wherever possible, data should be processed on-device, with federated learning approaches and encryption applied to stored templates. Only essential information should be retained.

Finally, fairness and human factors must be prioritized. Systems should be audited across demographic groups (e.g., age, skin tone, eyewear) and environmental conditions (e.g., lighting, route types) to ensure equitable performance. On-device personalization should be supported while preserving user privacy. Human factors evaluation should include measures of alert acceptance, false-alarm annoyance, and user trust, with opportunities for human-in-the-loop adjustments. Providing meaningful explanations for alerts can further enhance transparency and driver confidence.

8.3 Research agenda: specific, testable—and sometimes controversial—questions

This research agenda articulates a set of focused, empirically testable questions designed to advance the field of driver drowsiness detection toward practical, deployable solutions. The questions span physiological modeling, multimodal fusion, fairness, causal inference, and usability-centered design, with an emphasis on measurable outcomes and standardized evaluation.

A key line of inquiry concerns the identification of early-warning precursors versus mere correlates of drowsiness. Determining which physiological markers—such as EEG theta/alpha shifts or heart-rate variability (HRV) dynamics—reliably precede visible signs like blinks or head nods by 30–120 s under real-road noise remains critical. Equally important is identifying the most effective multimodal fusion strategy to exploit this lead time.

Another research question addresses prevalence-aware operating points. The goal is to determine which design choices maximize true positive rate (TPR) at false alarm rates (FAR) of no more than one alarm per hour across day and night conditions, eyewear use, and head-pose variations, and to understand how these trade off against detection latency. Relatedly, achieving personalization without surveillance remains a priority: can edge-based or on-device adaptation improve cross-subject PR–AUC by a measurable margin (e.g., $\geq X$ %) without exporting raw video or biometric data, possibly through federated or split learning with differential privacy?

Further research should explore whether embedding causal priors from sleep science—such as those derived from circadian and homeostatic two-process models—can reduce false alarms at a fixed recall rate relative to unconstrained deep learning approaches. Likewise, studies should move beyond accuracy and investigate cost and utility functions that balance missed-drowsiness versus nuisance-alarm penalties, evaluating which formulations best predict downstream safety outcomes such as lane departures or near-misses. These should be reported alongside standard performance metrics such as PR–AUC.

Fairness also demands attention. Researchers must quantify how error rates vary across demographic and environmental factors, including skin tone, eyewear, age, and lighting, and identify data collection and augmentation strategies that demonstrably reduce these disparities. Similarly, cross-dataset truthfulness remains a pressing challenge: establishing a minimum set of standard protocols (e.g., SD, SI, LOSO, CD) and ground-truth disclosures could enable more truthful comparisons across studies and support the creation of a public leaderboard that mandates cross-dataset testing.

Robustness to occlusion and ADAS confounds also warrants systematic investigation. Determining which fusion topologies most effectively separate genuine driver-state changes from environmental factors—such as road geometry, wind, or automated lane-keeping—will be essential for reliable deployment. On the human side, research should examine which human–machine interface (HMI) and alert patterns minimize alarm fatigue while ensuring timely driver responses, and whether explainability tools (e.g., attention maps or feature attributions) can measurably enhance compliance and trust. Finally, large-scale field trials are needed to quantify the real-world impact of drowsiness detection policies. These trials must be adequately powered to detect meaningful changes in near-miss rates and conducted under transparent regulatory and ethical standards.

Closed-loop biofeedback and conversational actuators present further opportunities for investigation. Studies should determine which stimulus types, durations, and inter-prompt intervals maximize TPR at FAR $\leq$ 1/h while maintaining annoyance $< 2/5$ and latency < 10 s across conditions such as day/night driving, eyewear, and head-pose variability. Likewise, future work on conversational negotiator policies should explore which dialogue strategies most effectively improve handover success and time-to-stabilize under shared control, without increasing cognitive load beyond NASA–TLX ≤ 40.

8.3.1 Self-supervised and unsupervised learning for baseline and deviation detection

Drowsiness events are rare and costly to annotate, motivating research into self-supervised and unsupervised learning approaches that can model each driver's baseline alert state and detect deviations without dense labeling. The key research questions are as follows:

(RQ-1) Do self-supervised representations—contrastive, predictive, or masked autoencoder (MAE) style—trained on unlabeled in-cabin sequences better separate alert versus drowsy states when fine-tuned with $\leq 5\%$ labeled data?

(RQ-2) Can on-device, privacy-preserving adaptation achieve at least an X% improvement in PR–AUC and TPR@FAR $\leq$ 1/h without exporting raw video or biometric data?

Recommended protocols involve training self-supervised models on unlabeled sequences and evaluating them under subject-independent (SI/LOSO) and cross-dataset (CD) settings. Reporting should include prevalence, PR–AUC, TPR@FAR, calibration metrics (Brier score, expected calibration error), and detection latency. Ablation studies should vary the volume of unlabeled data, adaptation window length, and differential-privacy noise to examine the trade-off between utility and privacy.

8.3.2 Causal inference and early-warning precursors

Most existing systems rely on correlational learning, such as associating yawns with drowsiness, but causal modeling could reveal earlier physiological precursors occurring 30–120 s before overt signs. The primary research questions are:

(RQ-1) Which physiological trajectories—EEG theta/alpha shifts, HRV trends, or blink-rate dynamics—Granger-cause or otherwise precede drowsiness episodes under real-road noise?

(RQ-2) Do causal or temporal models, such as state-space structural causal models (SCMs) or counterfactual simulators, reduce false alarms at fixed recall compared with correlational deep networks?

Protocols should emphasize event-level evaluation that credits the earliest correct detection and report time-to-detect, missed-event rate, TPR@FAR, and counterfactual sensitivity—defined as the effect of synthetic interventions like removing yawns or increasing HRV. Ablation analyses should compare models incorporating causal priors with unconstrained networks of matched capacity.

8.3.3 Neuro-symbolic models with sleep-science priors

Deep learning models excel at pattern recognition but often lack interpretability and robustness to domain shift. Embedding symbolic or physiological priors—such as circadian or homeostatic two-process constraints, or ocular dynamic rules—can regularize learning. Key research questions include:

(RQ-1) Do hybrid models combining deep-learning backbones with differentiable physiological modules or logical constraints maintain PR–AUC under varying lighting, eyewear, and head-pose conditions better than unconstrained models?

(RQ-2) Can rule-augmented attention maps improve human interpretability without compromising safety-critical TPR@FAR performance?

Proposed protocols include stress testing across lighting, eyewear, and route variations, assessing cross-dataset generalization, logging model rationales, and measuring subgroup PR–AUC and TPR@FAR disparities. Outputs should include released code, configurations for symbolic modules, and ablation studies quantifying each prior's contribution.

8.3.4 Usability-centred adoption: biofeedback and conversational actuators

Real-world adoption depends as much on human factors as on algorithmic accuracy. Effective systems must mitigate fatigue without imposing additional cognitive load. Research questions include:

(RQ-D1) Which biofeedback protocols—such as paced breathing, vibrotactile pulses, or thermal/airflow stimuli—maximize TPR@FAR ≤ 1/h while maintaining annoyance $< 2/5$ and response latency $< 10\,\mathrm{s}$?

(RQ-D2) Which conversational strategies, such as brief check-ins or positive acknowledgements during handover, improve handover success and time-to-stabilize while maintaining NASA–TLX ≤ 40?

Recommended protocols include randomized on-road and simulator trials stratified by lighting and eyewear conditions. Reporting should include acceptance and trust metrics, System Usability Scale (SUS) scores, NASA–TLX workload indices, and downstream safety surrogates such as lane exceedances and near-misses. These studies directly implement the actuators discussed in Sect. 6.3 and relate to shared-authority arbitration in Sect. 8.4.

8.4 Integration with autonomous vehicles: shared authority and the DMS as negotiator

In partially and conditionally automated vehicles, the driver-monitoring system (DMS) functions as a dynamic negotiator between the human operator and the vehicle's automated subsystems. Its primary objective is to continuously assess both driver fitness and automation confidence, using these estimates to arbitrate shared control authority. The system ensures that operational responsibility transitions smoothly between human and machine while maintaining safety and driver engagement.

The arbitration policy is governed by confidence-aware parameters. Let C_d represent the confidence in driver fitness and C_a the confidence in automation capability. Depending on their relative values, the system operates in one of several regimes. In the *human-lead* regime, high C_d and low C_a indicate that the driver should retain primary control, with automation providing only guidance or assistance. In the *assist-lead* regime, both confidences are moderate, prompting a balanced sharing of control, such as lane-keeping or adaptive cruise support, accompanied by heightened monitoring and proactive handover prompts. In the *automation-lead* regime, low C_d combined with high C_a shifts control toward the automated system, which then requests explicit acknowledgment—via button press or voice input—from the driver before proceeding. When both confidence levels are low, or the driver fails to respond, the vehicle initiates a *minimum-risk maneuver* (MRM), executing a controlled deceleration and safe stop, and notifying emergency services if required by policy.

Control transitions are implemented through a graded handover and escalation sequence comprising four stages: *inform*, *request*, *enforce*, and *MRM*. During the *inform* stage, the system provides a visual cue with explanatory context (e.g., "Drowsiness rising; lateral deviation increasing"). The *request* stage adds auditory or haptic prompts and initiates an explicit takeover timer. The *enforce* stage increases automation authority, reducing vehicle speed or assuming partial control if the driver does not respond promptly. Finally, the *MRM* stage executes a safe-stop procedure if no acknowledgment is received.

Within this shared-authority framework, biofeedback mechanisms and conversational prompts act as key actuators of the negotiator policy. When driver-fitness confidence C_d declines, the DMS first requests positive acknowledgment and may then issue brief countermeasures, such as HRV-guided breathing cues or vibrotactile feedback. If the driver remains unresponsive—or if automation confidence C_a remains high while C_d continues to fall—the system transitions into the automation-lead regime and, if necessary, executes a minimum-risk maneuver. Evaluation of such systems should include metrics such as handover success rate, time-to-stabilize, false-alarm rate per hour, and near-miss reduction, stratified by contextual factors. These measures are linked to the countermeasure designs presented in Sect. 6.3 and to the metric definitions in Sect. 6.

The human–machine interface (HMI) and explainability design are integral to ensuring transparency and driver trust. The DMS should provide concise, evidence-based rationales for each intervention—for example, referencing prolonged PERCLOS, HRV reductions, or steering variance. Users should have control over parameters such as alert volume and thresholds within predefined safe limits, and all interactions should be logged for audit purposes. Privacy-by-design principles must be enforced, favoring on-device processing, event-level data storage rather than raw video retention, and clear data purging policies disclosed to the user.

Evaluation of integrated DMS–ADAS systems should extend beyond detection accuracy to include operational and safety metrics. These include handover success rate, time-to-stabilize control after alerts, false-alarm and missed-event rates (stratified by lighting, eyewear, and route), downstream safety surrogates such as lane exceedances and near-misses, fairness diagnostics that quantify subgroup PR–AUC and TPR@FAR disparities, and auditability metrics measuring the proportion of interventions accompanied by human-interpretable rationales.

Data and protocol considerations are also essential. Experiments should log synchronized DMS signals alongside ADAS state information to disentangle driver-related from environmental effects. Evaluation should employ subject-independent (LOSO) and cross-route protocols, reporting latency to alert, driver response time, and behavior in non-response trials to validate escalation policies and MRM safety.

Finally, regulatory and ethical compliance must be integral to deployment. Systems should obtain explicit user consent and maintain transparency regarding data handling. Edge processing should be the default, with any cloud uploads remaining optional and user-controlled. Arbitration logs should be subject to independent audit to ensure accountability. These recommendations align with international standards including ISO 26262 (Functional Safety), ISO/PAS 21448 (Safety of the Intended Functionality), and the emerging SAE J3016 and J3061 frameworks governing human–automation interaction and cybersecurity in driver-monitoring systems.

8.5 Reporting and benchmarking implications

Transparent and standardized reporting is essential to ensure fair benchmarking, reproducibility, and meaningful comparison across drowsiness detection studies. The following considerations summarize the critical elements that should be disclosed in all future research.

Ground truth disclosure is a fundamental requirement. Authors should clearly specify the source of ground-truth labels used for annotation, whether derived from self-report instruments such as the Karolinska Sleepiness Scale (KSS), Epworth Sleepiness Scale (ESS), or Stanford Sleepiness Scale (SSS); physiological recordings including EEG or EOG criteria; psychomotor vigilance tasks (PVT); or visual annotations and observer ratings.

Protocol transparency is equally vital. Studies should explicitly report the number of subjects, the type of validation employed (subject-dependent or subject-independent), and the details of the evaluation design—such as Leave-One-Subject-Out (LOSO) or cross-dataset splits—as well as train/test partitioning procedures. Without such disclosure, the interpretability and comparability of reported metrics remain limited.

Operational metrics should extend beyond simple accuracy. Precision–recall area under the curve (PR–AUC), sensitivity or true positive rate (TPR) at fixed false alarm rates (e.g., one alarm per hour), and detection latency distributions offer more informative and safety-relevant assessments of system performance. These metrics should be reported systematically to reflect real-world deployment constraints.

External validation further enhances credibility. Whenever feasible, cross-dataset testing or real-world trials outside the training distribution should be conducted to assess model generalizability. Such validation provides stronger evidence of robustness and transferability than single-dataset evaluations.

Reproducibility requires open and standardized documentation. Researchers should release per-study coding in machine-readable form, including dataset identifiers, number of subjects, ground-truth definition, validation protocol, and performance metrics. Availability of source code and data should be stated explicitly for each study to facilitate independent verification.

Quality appraisal should accompany all result reporting. Authors are encouraged to provide a concise risk-of-bias or quality assessment score encompassing sample size, validity of ground-truth labeling, presence of cross-subject testing, environmental realism, and code/data disclosure. Conclusions should be qualified accordingly to reflect the methodological rigor of the underlying studies.

Across modalities, raw accuracy alone is an unreliable basis for comparison without context. Subject-independent validation, use of operational metrics, and transparent methodological reporting together form a credible pathway from promising laboratory prototypes to reliable, deployable driver monitoring systems.

9 Commercial applications and real-world deployment

The transition from research prototypes to commercial drowsiness detection systems represents a critical phase that determines the real-world impact and societal benefits of technological advances in this field. Commercial deployment encompasses diverse application domains, from individual consumer vehicles to large-scale fleet operations, each with distinct requirements, constraints, and success metrics. The successful commercialization of drowsiness detection technology requires careful attention to market needs, regulatory requirements, cost considerations, and user acceptance factors that influence adoption and effectiveness in practical applications. The *Deployment Constraints* panel (Fig. 2) highlights the multi-dimensional trade-offs between computational efficiency, privacy preservation, and user acceptance that govern commercial viability. Systems must balance accuracy against latency (typically 100–300 ms), sensor intrusiveness against reliability, and centralized learning against privacy-by-design principles.

9.1 Automotive industry implementations

The automotive industry has emerged as the primary driver for commercial drowsiness detection system development, with major manufacturers increasingly incorporating these technologies into their vehicle safety portfolios. Current implementations range from basic driver attention monitoring systems that analyze steering patterns and lane positioning to sophisticated multi-modal systems that combine facial analysis, physiological monitoring, and behavioral assessment. The integration of drowsiness detection into production vehicles requires extensive validation, testing, and certification processes to ensure reliability and safety under diverse operating conditions.

Advanced automotive implementations leverage integration with existing vehicle systems to provide comprehensive driver monitoring and intervention capabilities. These systems can coordinate with adaptive cruise control, lane keeping assistance, and automated emergency braking to provide graduated responses based on detected drowsiness levels. Premium vehicle segments increasingly feature sophisticated cabin monitoring systems that combine drowsiness detection with distraction monitoring, health assessment, and personalized comfort adjustment, representing the evolution toward comprehensive driver state monitoring platforms.

The challenges of automotive implementation include meeting stringent reliability and durability requirements, achieving cost targets for mass market deployment, and ensuring consistent performance across diverse vehicle platforms and configurations. Automotive manufacturers must also consider long-term service and maintenance requirements, software update capabilities, and liability considerations associated with safety-critical drowsiness detection systems. The development of industry standards and certification procedures for drowsiness detection systems represents an ongoing effort to ensure consistent quality and performance across different manufacturers and technology providers.

9.2 Transportation and fleet management

Commercial transportation and fleet management applications represent significant opportunities for drowsiness detection technology deployment, where the economic and safety benefits can justify higher system costs and complexity compared to individual consumer applications. Long-haul trucking, public transportation, emergency services, and delivery operations face particular challenges related to driver fatigue and drowsiness due to demanding schedules, extended operating hours, and irregular sleep patterns. Fleet-based drowsiness detection systems can provide real-time monitoring, historical analysis, and predictive insights that support both immediate safety interventions and long-term fleet management decisions.

Fleet management implementations often incorporate drowsiness detection data into comprehensive telematics platforms that monitor vehicle performance, driver behavior, and operational efficiency. This integration enables sophisticated analytics that can identify patterns, predict high-risk situations, and optimize scheduling and routing decisions to minimize drowsiness-related risks. The aggregation of drowsiness data across large fleets provides valuable insights for developing predictive models and identifying systemic factors that contribute to driver fatigue and drowsiness.

The implementation of drowsiness detection in commercial fleets faces unique challenges related to driver privacy, labor relations, and regulatory compliance. Drivers may have concerns about continuous monitoring and potential disciplinary actions based on drowsiness detection data, requiring careful attention to policy development, training, and communication strategies. The integration with existing fleet management systems, compliance with commercial vehicle regulations, and coordination with safety management systems represent additional technical and operational challenges that must be addressed for successful deployment.

9.3 Regulatory standards and safety requirements

The development of regulatory standards and safety requirements for drowsiness detection systems represents a critical factor influencing technology development and market adoption. Current regulatory frameworks are evolving to address the unique characteristics and challenges of drowsiness detection technology, including performance requirements, testing protocols, and certification procedures. The establishment of comprehensive standards requires collaboration between regulatory agencies, automotive manufacturers, technology providers, and safety organizations to ensure that requirements are both technically achievable and effective for improving road safety.

International harmonization of drowsiness detection standards presents challenges due to varying regulatory approaches, market conditions, and safety priorities across different regions. The European Union's General Safety Regulation mandating driver drowsiness and attention warning systems in new vehicles represents a significant regulatory driver for technology development and deployment. Similar regulations in other regions are likely to follow, creating global market opportunities while requiring compliance with diverse regulatory requirements and certification processes.

The development of performance standards for drowsiness detection systems must address fundamental questions about acceptable false alarm rates, detection latency requirements, and minimum performance levels across diverse populations and operating conditions. Testing protocols must encompass laboratory validation, simulation-based assessment, and real-world field testing to ensure comprehensive evaluation of system performance and reliability. The establishment of standardized datasets, annotation protocols, and benchmarking procedures represents an important foundation for consistent evaluation and comparison of different drowsiness detection approaches.

Liability and responsibility considerations associated with drowsiness detection systems require careful attention to the limitations and capabilities of current technology. Clear communication of system limitations, appropriate user training, and robust failure detection and notification mechanisms are essential for ensuring safe and effective deployment. The evolution of legal frameworks to address automated safety systems and their interaction with human drivers represents an ongoing challenge that influences technology development priorities and implementation strategies.

10 Conclusion

This systematic review consolidates evidence from 2020–2025 on driver-drowsiness detection systems, integrating findings across vision-based, physiological, vehicular, and multimodal modalities. Following a PRISMA-guided methodology, we standardized dataset, validation, and metric reporting to clarify why cross-study accuracy comparisons are often misleading. Vision-based approaches have advanced through deep learning and transformer architectures but remain sensitive to illumination and occlusion, whereas physiological signals offer earlier warning capability at the cost of wearability and artifact resilience. Multimodal fusion enhances robustness yet introduces greater computational and integration demands. Beyond technical performance, this study emphasizes the equal importance of practical implementation and human factors. The proposed three-dimensional evaluation framework and standardized comparative tables reveal how validation design, ground-truth definition, and dataset diversity govern generalizability. To advance real-world deployment, future work should prioritize subject-independent and cross-dataset evaluation, operational metrics such as PR-AUC, recall at fixed false-alarm rates, and latency, privacy-preserving edge-based computation, and user-centered alerting with interpretable outputs to sustain trust. Ultimately, achieving safe, fair, and transparent driver monitoring requires a transition from isolated performance claims to reproducible, evidence-based evaluation and responsible integration with assistive and autonomous vehicle systems.

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Declarations

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References

Abbasi JA, Parsi A, Ringelstein N, Reilhac P, Jones E, Glavin M (2025) Enhancing cyclist safety through driver gaze analysis at intersections with cycle lanes. IEEE Trans Intell Transp Syst 26:3175–3184

Abdelmaaboud A, Fathallah T, Ragheb A, Gomaa A, Hassan T (2025) Artificial neural network-based modeling and prediction of gnss ionospheric errors in Egypt. J Surv Eng 151(3):04025008

Abo AY, Gharbia AG, Saleh M, Mousa AE, Abbas IA, Hassan MR (2025) Gnss geodetic velocity prediction using ensemble tree models in Abu-Dabbab, Egypt. Egypt J Remote Sens Space Sci 28(2):337–347

Abtahi S, Omidyeganeh M, Shirmohammadi S, Hariri B (2014) Yawdd: a yawning detection dataset. In: Proceedings of the 5th ACM multimedia systems conference, pp 24–28

Ahmed MIB, Alabdulkarem H, Alomair F, Aldossary D, Alahmari M, Alhumaidan M, Alrassan S, Rahman A, Youldash M, Zaman G (2023) A deep-learning approach to driver drowsiness detection. Safety 9(3):65

Aidman E, Chadunow C, Johnson K, Reece J (2015) Real-time driver drowsiness feedback improves driver alertness and self-reported driving performance. Accid Anal Prev 81:8–13

Akhmedov F, Khujamatov H, Abdullaev M, Jeon H-S (2025) Joint driver state classification approach: face classification model development and facial feature analysis improvement. Sensors 25(5):1472

Albadawi Y, Takruri M, Awad M (2022) A review of recent developments in driver drowsiness detection systems. Sensors 22(5):2069

Albadawi Y, AlRedhaei A, Takruri M (2023) Real-time machine learning-based driver drowsiness detection using visual features. J Imag 9(5):91

Alguindigue J, Singh A, Narayan A, Samuel S (2024) Biosignals monitoring for driver drowsiness detection using deep neural networks. IEEE Access 12:93075–93086

Alharbey R, Dessouky MM, Sedik A, Siam AI, Elaskily MA (2022) Fatigue state detection for tired persons in presence of driving periods. IEEE Access 10:79403–79418

Al-Madani AM, Gaikwad AT, Mahale V, Ahmed ZAT, Shareef AAA (2021) Real-time driver drowsiness detection based on eye movement and yawning using facial landmark. In 2021 International conference on computer communication and informatics (ICCCI). IEEE, pp 1–4

Altameem A, Kumar A, Poonia RC, Kumar S, Saudagar AKJ (2021) Early identification and detection of driver drowsiness by hybrid machine learning. IEEE Access 9:162805–162819

Amidei A, Fallica PG, Conoci S, Pavan P (2021) Validating photoplethysmography (ppg) data for driver drowsiness detection. In: 2021 IEEE international workshop on metrology for automotive (MetroAutomotive). IEEE, pp 147–151

Ansari S, Naghdy F, Haiping D, Pahnwar YN (2021) Driver mental fatigue detection based on head posture using new modified relu-bilstm deep neural network. IEEE Trans Intell Transp Syst 23(8):10957–10969

Ardabili SZ, Bahmani S, Lahijan LZ, Khaleghi N, Sheykhivand S, Danishvar S (2024) A novel approach for automatic detection of driver fatigue using eeg signals based on graph convolutional networks. Sensors 24(2):364

Augusto de Lima Medeiros P, Vinícius Souza da Silva G, dos Santos Fernandes FR, Sánchez-Gendriz I, Lins HWC, da Silva Barros DM, Nagem DAP, de Medeiros Valentim RA (2022) Efficient machine learning approach for volunteer eye-blink detection in real-time using webcam. Expert Syst Appl 188:116073

Baccour MH, Driewer F, Schäck T, Kasneci E (2022) Comparative analysis of vehicle-based and driver-based features for driver drowsiness monitoring by support vector machines. IEEE Trans Intell Transp Syst 23(12):23164–23178

Bamidele AA, Kamardin K, Syazarin Natasha N, Aziz Abd, Sam SM, Ahmed IS, Azizan A, Bani NA, Kaidi HM (2019) Non-intrusive driver drowsiness detection based on face and eye tracking. Int J Adv Comput Sci Appl 10(7)

Basner M, Dinges DF (2011) Maximizing sensitivity of the psychomotor vigilance test (pvt) to sleep loss. Sleep 34(5):581–591

Bi J, Qin X, Dongjin H, Chenyang X (2023) Fatigue driving detection method based on ippg technology. Promet-Traffic Transp 35(4):540–551

Bisogni C, Cascone L, Nappi M, Pero C (2024) Iot-enabled biometric security: enhancing smart car safety with depth-based head pose estimation. ACM Trans Multimed Comput Commun Appl 20(6):1–24

Biying F, Boutros F, Lin C-T, Damer N (2024) A survey on drowsiness detection-modern applications and methods. IEEE Trans Intell Veh 9:7279–7300
Borawake M, Patil A, Yadav S, Nagwade V, Somwanshi H (2025) Driver drowsiness detection using ml and iot. Int J Recent Adv Eng Technol 14(1):139–143
Buysse DJ, Hall ML, Strollo PJ, Kamarck TW, Owens J, Lee L, Reis SE, Matthews KA (2008) Relationships between the pittsburgh sleep quality index (psqi), epworth sleepiness scale (ess), and clinical/polysomnographic measures in a community sample. J Clin Sleep Med 4(6):563–571
Cao Y, Li F, Liu X, Yang S, Wang Yu (2023) Towards reliable driver drowsiness detection leveraging wearables. ACM Trans Sensor Netw 19(2):1–23
Celecia A, Figueiredo K, Vellasco M, González R (2020) A portable fuzzy driver drowsiness estimation system. Sensors 20(15):4093
Chacon-Murguia MI, Prieto-Resendiz C (2015) Detecting driver drowsiness: a survey of system designs and technology. IEEE Consumer Electron Mag 4(4):107–119
Chirra VRR, Uyyala SR, Kolli VKK (2019) Deep cnn: a machine learning approach for driver drowsiness detection based on eye state. Rev d'Intell Artif 33(6):461–466
Choi H-T, Mi EABE, Kim B-K, Chang W-D (2025) Electrooculography signal generation with conditional diffusion models for eye movement classification. Biomed Signal Process Control 110:108211
Dahl J, de Campos GR, Fredriksson J (2023) Intention-aware lane keeping assist using driver gaze information. In: 2023 IEEE intelligent vehicles symposium (IV). IEEE, pp 1–7
Delwar TS, Singh M, Mukhopadhyay S, Kumar A, Parashar D, Yangwon Lee Md, Rahman H, Sejan MAS, Ryu JY (2025) Ai-and deep learning-powered driver drowsiness detection method using facial analysis. Appl Sci 15(3):1102
Deng W, Ruoxue W (2019) Real-time driver-drowsiness detection system using facial features. IEEE Access 7:118727–118738
Dewi C, Chen R-C, Jiang X, Hui Yu (2022) Adjusting eye aspect ratio for strong eye blink detection based on facial landmarks. PeerJ Comput Sci 8:e943
Ed-Doughmi Y, Idrissi N, Hbali Y (2020) Real-time system for driver fatigue detection based on a recurrent neuronal network. J Imag 6(3):8
Ekinci E, Kahraman K, Oturakci M (2025) Enhancing driver safety: integrating ecg and can-bus data for accurate drowsiness detection. Traitement du Signal 42(2):1057
Elena Magán M, Sesmero P, Alonso-Weber JM, Sanchis A (2022) Driver drowsiness detection by applying deep learning techniques to sequences of images. Appl Sci 12(3):1145
El-Nabi SA, El-Rabaie E-SM, Emam A, Ramadan KF (2023) An efficient approach of drowsiness detection based on resnet50 and xception architectures. In: 2023 3rd International conference on electronic engineering (ICEEM). IEEE, pp 1–6
El-Nabi SA, El-Shafai W, El-Rabaie E-SM, Ramadan KF, El-Samie FEA, Mohsen S (2024) Machine learning and deep learning techniques for driver fatigue and drowsiness detection: a review. Multimed Tools Appl 83(3):9441–9477
El-Nabi SA, Ibrahim AF, El-Rabaie E-SM, Hassan OF, Soliman NF, Ramadan KF, El-Shafai W (2025a) Driver drowsiness detection using swin transformer and diffusion models for robust image denoising. IEEE Access 13:71880–71907
El-Nabi SA, Ramadan KF, El-Rabaie E-SM, Emam A, El-Shafai W (2025b) A real-time design and implementation of intelligent drowsiness and fatigue recognition system for enhancing driver safety. Eng Appl Artif Intell 162:112665
Essahraui S, Lamaakal I, El Hamly I, Maleh Y, Ouahbi I, El Makkaoui K, Bouami MF, Pławiak P, Alfarraj O, El-Latif AAA (2025) Real-time driver drowsiness detection using facial analysis and machine learning techniques. Sensors (Basel Switzerland) 25(3):812
Essel E, Lacy F, Albalooshi F, Elmedany W, Ismail Y (2024) Drowsiness detection in drivers using facial feature analysis. Appl Sci 15(1):20
Esteves T, Pinto JR, Ferreira PM, Costa PA, Rodrigues LA, Antunes I, Lopes G, Gamito P, Abrantes AJ, Jorge PM et al (2021) Automotive: a case study on automatic multimodal drowsiness detection for smart vehicles. IEEE Access 9:153678–153700
Feng X, Guo Z, Kwong S (2025) Id3rsnet: cross-subject driver drowsiness detection from raw single-channel eeg with an interpretable residual shrinkage network. Front Neurosci 18:1508747
Florez R, Palomino-Quispe F, Alvarez AB, Coaquira-Castillo RJ, Herrera-Levano JC (2024) A real-time embedded system for driver drowsiness detection based on visual analysis of the eyes and mouth using convolutional neural network and mouth aspect ratio. Sensors 24(19):6261
Fonseca T, Ferreira S (2025) Drowsiness detection in drivers: a systematic review of deep learning-based models. Appl Sci 15(16):9018

Freitas A, Almeida R, Gonçalves H, Conceição G, Freitas A (2024) Monitoring fatigue and drowsiness in motor vehicle occupants using electrocardiogram and heart rate- a systematic review. Trans Res F Traffic Psychol Behav 103:586–607
Fusek R (2018) Pupil localization using geodesic distance. In: International symposium on visual computing. Springer, pp 433–444
Ghoddoosian R, Galib M, Athitsos V (2019) A realistic dataset and baseline temporal model for early drowsiness detection. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition workshops
Gomaa A (2024) Advanced domain adaptation technique for object detection leveraging semi-automated dataset construction and enhanced yolov8. In: 2024 6th Novel intelligent and leading emerging sciences conference (NILES). IEEE, pp 211–214
Gomaa A, Abdalrazik A (2024) Novel deep learning domain adaptation approach for object detection using semi-self building dataset and modified yolov4. World Electric Vehicle J 15(6):255
Gomaa A, Saad OM (2025) Residual channel-attention (rca) network for remote sensing image scene classification. Multimed Tools Appl 1–25
Gomaa MW, Mahmoud RO, Sarhan AM (2022) A cnn-lstm-based deep learning approach for driver drowsiness prediction. J Eng Res 6(3):59–70
Guanglong D, Zhang L, Kang S, Wang X, Teng S, Liu PX (2022) A multimodal fusion fatigue driving detection method based on heart rate and perclos. IEEE Trans Intell Transp Syst 23(11):21810–21820
Gupta A, Tiwari A, Chauhan AS, Gupta S (2024) A comprehensive review of driver drowsiness detection systems and applied technologies. In: 2024 IEEE international conference on computing, power and communication technologies (IC2PCT). IEEE, vol 5, pp 1624–1629
Gwak J, Hirao A, Shino M (2020) An investigation of early detection of driver drowsiness using ensemble machine learning based on hybrid sensing. Appl Sci 10(8):2890
Hasan MM, Watling CN, Larue GS (2022) Physiological signal-based drowsiness detection using machine learning: singular and hybrid signal approaches. J Saf Res 80:215–225
Hasan MM, Watling CN, Larue GS (2024) Validation and interpretation of a multimodal drowsiness detection system using explainable machine learning. Comput Methods Programs Biomed 243:107925
Hasanuddin MO, Novianingrum H, Syamsuddin EY (2022) Design and implementation of drowsiness detection system based on standard deviation of lateral position. In: 2022 12th International conference on system engineering and technology (ICSET). IEEE, pp 72–76
Hassan OF, Ibrahim AF, Gomaa A, Makhlouf MA, Hafiz B (2025) Real-time driver drowsiness detection using transformer architectures: a novel deep learning approach. Sci Rep 15(1):1–33
Hoddes E, Zarcone V, Smythe H, Phillips R, Dement WC (1973) Quantification of sleepiness: a new approach. Psychophysiology 10(4):431–436
Huang F, Yang C, Weng W, Chen Z, Zhang Z (2025) Cm-fusionnet: A cross-modal fusion fatigue detection method based on electroencephalogram and electrooculogram. Comput Electr Eng 123:110204
Islam MM, Kowsar I, Zaman MS, Sakib MFR, Saquib N, Alam SMS (2023) A novel approach to enhance safety on drowsy driving in self-driving car. Mob Netw Appl 28(1):272–284
Jabbar R, Shinoy M, Kharbeche M, Al-Khalifa K, Krichen M, Barkaoui K (2020) Driver drowsiness detection model using convolutional neural networks techniques for android application. arXiv preprint arXiv:2002.03728
Jarndal A, Tawfik H, Siam AI, Alsyouf I, Cheaitou A (2024) A real-time vision transformers-based system for enhanced driver drowsiness detection and vehicle safety. IEEE Access 13:1790–1803
Johns MW (1991) A new method for measuring daytime sleepiness: the epworth sleepiness scale. Sleep 14(6):540–545
Johns MW (1992) Reliability and factor analysis of the epworth sleepiness scale. Sleep 15(4):376–381
Kassem HA, Chowdhury M, Abawajy JH (2021) Drivers fatigue level prediction using facial, and head behavior information. IEEE Access 9:121686–121697
Kemp B, Zwinderman AH, Tuk B, Kamphuisen HAC, Oberye JJL (2000) Analysis of a sleep-dependent neuronal feedback loop: the slow-wave microcontinuity of the EEG. IEEE Trans Biomed Eng 47(9):1185–1194
Khan MA, Chen M, Nawaz T, Sedky M, Sheikh M, Bashir AK, Hassan S (2025) Smart steering wheel: design of iomt-based non-invasive driver health monitoring system to enhance road safety. IET Intel Transport Syst 19(1):e70012
Khare SK, Bajaj V (2020) Entropy-based drowsiness detection using adaptive variational mode decomposition. IEEE Sens J 21(5):6421–6428
Kolus A (2024) A systematic review on driver drowsiness detection using eye activity measures. IEEE Access 12:97969–97993

Kong L, Xie K, Niu K, He J, Zhang W (2024) Remote photoplethysmography and motion tracking convolutional neural network with bidirectional long short-term memory: Non-invasive fatigue detection method based on multi-modal fusion. Sensors 24(2):455

Kundinger T, Sofra N, Riener A (2020) Assessment of the potential of wrist-worn wearable sensors for driver drowsiness detection. Sensors 20(4):1029

Kushwaha V, Singh M, Singh R, Vatsa M, Ratha N, Chellappa R (2018) Disguised faces in the wild. In: Proceedings of the IEEE conference on computer vision and pattern recognition workshops 1–9

Lamaazi H, Alqassab A, Fadul RA, Mizouni R (2023) Smart edge-based driver drowsiness detection in mobile crowdsourcing. IEEE Access 11:21863–21872

Lamond N, Dawson D, Roach GD (2005) Fatigue assessment in the field: validation of a hand-held electronic psychomotor vigilance task. Aviat Space Environ Med 76(5):486–489

Lemkaddem A, Delgado-Gonzalo R, Türetken E, Dasen S, Moser V, Gressum C, Solà J, Ferrario D, Verjus C (2018) Multi-modal driver drowsiness detection: a feasibility study. In: 2018 IEEE EMBS international conference on biomedical & health informatics (BHI). IEEE, pp 9–12

Li W (2024) Driver fatigue detection method based on facial features using deep learning. Appl Comput Eng 57:190–199

Li N, Zhou R, Krishna B, Pradhan A, Lee H, He J, Jiang N (2024) Non-invasive techniques for muscle fatigue monitoring: a comprehensive survey. ACM Comput Surv 56(9):1–40

Lian Z, Tao X, Yuan Z, Li J, Thakor N, Wang H (2024) Driving fatigue detection based on hybrid electroencephalography and eye tracking. IEEE J Biomed Health Inform 28:6568–6580

Lin X, Huang Z, Ma W, Tang W (2025) Eeg-based driver drowsiness detection based on simulated driving environment. Neurocomputing 616:128961

Liu Z, Luo P, Wang X, Tang X (2015) Deep learning face attributes in the wild. In: Proceedings of the IEEE international conference on computer vision, pp 3730–3738

Liu K, Jiao Y, Congcong D, Zhang X, Chen X, Fang X, Jiang C (2023) Driver stress detection using ultra-short-term hrv analysis under real world driving conditions. Entropy 25(2):194

Lu Yu, Yang X, Wei H, Liu J, Li B (2024) Driver fatigue detection using ppg signal, facial features, head postures with an lstm model. Heliyon 10(21):e39479

Maior CBS, José M, das Chagas Moura MJ, Santana JMM, Lins ID (2020) Real-time classification for autonomous drowsiness detection using eye aspect ratio. Expert Syst Appl 158:113505

Makhmudov F, Turimov D, Xamidov M, Nazarov F, Cho Y-I (2024) Real-time fatigue detection algorithms using machine learning for yawning and eye state. Sensors 24(23):7810

Malik M, Sharma P, Punj GK, Singh S, Gared F (2025) Multi-body sensor based drowsiness detection using convolutional programmed transfer vgg-16 neural network with automatic driving mode conversion. Sci Rep 15(1):8838

Massoz Q, Langohr T, François C, Verly JG (2016) The ulg multimodality drowsiness database (called drozy) and examples of use. In: 2016 IEEE winter conference on applications of computer vision (WACV). IEEE, pp 1–7

Mehreen A, Anwar SM, Haseeb M, Majid M, Ullah MO (2019) A hybrid scheme for drowsiness detection using wearable sensors. IEEE Sens J 19(13):5119–5126

Mehta S, Dadhich S, Gumber S, Jadhav Bhatt A (2019) Real-time driver drowsiness detection system using eye aspect ratio and eye closure ratio. In: Proceedings of international conference on sustainable computing in science, technology and management (SUSCOM), Amity University Rajasthan, Jaipur-India

Mejía-Mejía E, Budidha K, Abay TY, May JM, Kyriacou PA (2020) Heart rate variability (hrv) and pulse rate variability (prv) for the assessment of autonomic responses. Front Physiol 11:779

Miley AÅ, Kecklund G, Åkerstedt T (2016) Comparing two versions of the karolinska sleepiness scale (kss). Sleep Biol Rhythms 14:257–260

Minhas AA, Jabbar S, Farhan M, ul Islam MN (2022) A smart analysis of driver fatigue and drowsiness detection using convolutional neural networks. Multimed Tools Appl 81(19):26969–26986

Minhas R, Peker NY, Hakkoz MA, Arbatli S, Celik Y, Erdem CE, Semiz B, Peker Y (2024) Association of visual-based signals with electroencephalography patterns in enhancing the drowsiness detection in drivers with obstructive sleep apnea. Sensors 24(8):2625

Mishra A, Lee S, Kim D, Kim S (2022) In-cabin monitoring system for autonomous vehicles. Sensors 22(12):4360

Mittal A, Kumar K, Dhamija S, Kaur M (2016) Head movement-based driver drowsiness detection: A review of state-of-art techniques. In: 2016 IEEE international conference on engineering and technology (ICETECH). IEEE, pp 903–908

Moujahid A, Dornaika F, Arganda-Carreras I, Reta J (2021) Efficient and compact face descriptor for driver drowsiness detection. Expert Syst Appl 168:114334

Naim F, Mustafa M, Sulaiman N, Zahari ZL (2022) Dual-layer ranking feature selection method based on statistical formula for driver fatigue detection of emg signals. Traitement du Signal 39(3)

Naim F, Subramaniam AK, Mustafa M, Sulaiman N (2023) An analysis of driver drowsiness detection using electromyography (emg) facial muscles. In: International conference on electrical, control & computer engineering. Springer, pp 599–611

Narayana GV (2025) Vision-based driver fatigue detection using eye and mouth aspect ratios with machine learning. Int J 9(3)

National advanced driving simulator. Nads naturalistic driving study. https://www.nads-sc.uiowa.edu

Nordbakke S, Sagberg F (2007) Sleepy at the wheel: Knowledge, symptoms and behaviour among car drivers. Transport Res F: Traffic Psychol Behav 10(1):1–10

Oak Ridge National Laboratory (2016) Oak ridge national laboratory naturalistic driving study sample (ondss). https://www.ornl.gov/project/ornl-naturalistic-driving-study-sample. Accessed: 2025-01-10

Obaidan HB, Hussain M, Almajed R (2024) Eeg_dmnet: a deep multi-scale convolutional neural network for electroencephalography-based driver drowsiness detection. Electronics 13(11):2084

Ortega JD, Kose N, Cañas P, Chao M-A, Unnervik A, Nieto M, Otaegui O, Salgado L (2020) Dmd: A large-scale multi-modal driver monitoring dataset for attention and alertness analysis. In: European conference on computer vision. Springer, pp 387–405

Owen V, Surantha N (2025) Computer vision-based drowsiness detection using handcrafted feature extraction for edge computing devices. Appl Sci 15(2):638

Pandey R, Bhasin P, Popli S, Sharma M, Sharma N (2022) Driver drowsiness detection and traffic sign recognition system. In: Emerging technologies in data mining and information security: proceedings of IEMIS 2022. Springer, Vol 1, pp 25–40

Pan G, Sun L, Wu Z, Lao S (2007) Eyeblink-based anti-spoofing in face recognition from a generic webcamera. In: 2007 IEEE 11th international conference on computer vision. IEEE, pp 1–8

Perkins E, Sitaula C, Burke M, Marzbanrad F (2021) Challenges of driver drowsiness prediction: the remaining steps to implementation. arXiv preprint arXiv:2109.08355

Pratama BG, Ardiyanto I, Adji TB (2017) A review on driver drowsiness based on image, bio-signal, and driver behavior. In: 2017 3rd international conference on science and technology-computer (ICST). IEEE, pp 70–75

Priyanka S, Shanthi S, Saran Kumar A, Praveen V (2024a) Data fusion for driver drowsiness recognition: a multimodal perspective. Egypt Inform J 27:100529

Priyanka S, Shanthi S, Kumar S, Praveen V (2024b) Data fusion for driver drowsiness recognition: a multimodal perspective. Egypt Inform J 27:100529

Putilov AA, Donskaya OG (2013) Construction and validation of the EEG analogues of the Karolinska sleepiness scale based on the Karolinska drowsiness test. Clin Neurophysiol 124(7):1346–1352

Qingtian W, Li N, Zhang L, Fei Richard Yu (2024) Driver drowsiness detection based on joint human face and facial landmark localization with cheap operations. IEEE Trans Intell Transp Syst 25:19633–19645

Qu S, Gao Z, Wu X, Qiu Y (2023) Multi-attention fusion drowsy driving detection model. arXiv preprint arXiv:2312.17052

Rahman A, Sirshar M, Khan A (2015) Real time drowsiness detection using eye blink monitoring. In: 2015 National software engineering conference (NSEC). IEEE, pp 1–7

Ramzan M, Khan HU, Awan SM, Ismail A, Ilyas M, Mahmood A (2019) A survey on state-of-the-art drowsiness detection techniques. IEEE Access 7:61904–61919

Ramzan M, Abid A, Fayyaz M, Alahmadi TJ, Nobanee H, Rehman A (2024) A novel hybrid approach for driver drowsiness detection using a custom deep learning model. IEEE Access 12:126866–126884

Rashid M, Mustafa M, Sulaiman N, Nahidul Islam Md (2023) Eeg and emg-based multimodal driver drowsiness detection: a cwt and improved vgg-16 pipeline. In: International human engineering symposium. Springer, pp 339–350

Sagonas C, Tzimiropoulos G, Zafeiriou S, Pantic M (2013) 300 faces in-the-wild challenge: the first facial landmark localization challenge. In: Proceedings of the IEEE international conference on computer vision workshops, pp 397–403

Saito Y, Itoh M, Inagaki T (2016) Driver assistance system with a dual control scheme: effectiveness of identifying driver drowsiness and preventing lane departure accidents. IEEE Trans Human-Mach Syst 46(5):660–671

Salem M, Gomaa A, Tsurusaki N (2023) Detection of earthquake-induced building damages using remote sensing data and deep learning: a case study of Mashiki town, Japan. In: IGARSS 2023-2023 IEEE international geoscience and remote sensing symposium. IEEE, pp 2350–2353

Sar I, Routray A, Mahanty B (2023) A review on existing technologies for the identification and measurement of abnormal driving. Int J Intell Transp Syst Res 21(1):159–177

Saurav S, Gidde P, Saini R, Singh S (2022) Real-time eye state recognition using dual convolutional neural network ensemble. J Real-Time Image Proc 19(3):607–622

SENSORIS Consortium (2018) Sensoris: Sensor interface specification. Technical report, HERE Technologies. Accessed: 2025-06-24

Shen J, Barbera J, Shapiro CM (2006) Distinguishing sleepiness and fatigue: focus on definition and measurement. Sleep Med Rev 10(1):63–76

Sikander G, Anwar S (2018) Driver fatigue detection systems: a review. IEEE Trans Intell Transp Syst 20(6):2339–2352

Siyang H, Gao Q, Xie K, Wen C, Zhang W, He J (2024) Efficient detection of driver fatigue state based on all-weather illumination scenarios. Sci Rep 14(1):17075

Song F, Tan X, Liu X, Chen S (2014) Eyes closeness detection from still images with multi-scale histograms of principal oriented gradients. Pattern Recogn 47(9):2825–2838

Souchet AD, Philippe S, Lourdeaux D, Leroy L (2022) Measuring visual fatigue and cognitive load via eye tracking while learning with virtual reality head-mounted displays: A review. Int J Human-Comput Interact 38(9):801–824

Srinivasan AG, Smith SS, Pattinson CL, Mann D, Sullivan K, Salmon P, Soleimanloo SS (2024) Heart rate variability as an indicator of fatigue: a structural equation model approach. Trans Res F Traffic Psychol Behav 103:420–429

Stancin I, Cifrek M, Jovic A (2021) A review of EEG signal features and their application in driver drowsiness detection systems. Sensors 21(11):3786

Sukumar N, Sivashankar B, Sumathi P, Maurya OP (2024) Physiological and physical sensors for stress level, drowsiness detection, and behaviour analysis. IEEE Trans Consum Electron 70(1):656–668

Sun W, Wang Y, Bingliang H, Wang Q (2024a) Exploration of eye fatigue detection features and algorithm based on eye-tracking signal. Electronics 13(10):1798

Sun W, Jiang W, Li C, Si Y, Li S, Guo M, Liu D, Song H (2024b) Study on identification method of driver fatigue considering individual ECG differences. Cogn Technol Work 26(2):301–312

Tsai C-Y, Cheong H, Houghton R, Hsu W-H, Lee K-Y, Kang J-H, Kuan Y-C, Lee H-C, Cheng-Jung W, Li L-YJ et al (2024) Predicting fatigue-associated aberrant driving behaviors using a dynamic weighted moving average model with a long short-term memory network based on heart rate variability. Hum Factors 66(6):1681–1702

Turki A, Kahouli O, Albadran S, Ksantini M, Aloui A, Amara MB (2024) A sophisticated drowsiness detection system via deep transfer learning for real time scenarios. AIMS Math 9(2):3211–3234

Virginia Tech Transportation Institute (2025) Shrp2 naturalistic driving study (nds) data access – insight. https://insight.shrp2nds.us/

Wang Q, Zhendong M (2021) Heterogeneous signal fusion method in driving fatigue detection signals. J Adv Transp 2021(1):4464890

Wang L, Song F, Zhou TH, Hao J, Ryu KH (2023) Eeg and ecg-based multi-sensor fusion computing for real-time fatigue driving recognition based on feedback mechanism. Sensors 23(20):8386

Wang F, Chen D, Zhang X (2025) Real-time driving fatigue detection of ecg signals acquired based on novel electrodes using wavelet scattering networks. Measurement 243:116438

Wang P, Liu W, Majumdar A, Chen Z, Kang J-H, Kuan Y-C, Chen Y-L, Lee K-Y, Feng P-H, Chen K-Y et al A multimodal framework for predicting fatigue-related aberrant driving behaviors in taxi drivers using heart rate variability, sleep, and demographic data. Sleep Demogr Data

Wang Y, Zhao Y, Gong L, Yu Y, Liu H (2024) A driver fatigue modeling method based on enhanced eeg-eog fusion network for human-machine co-driving. In: 2024 4th International signal processing, communications and engineering management conference (ISPCEM). IEEE, pp 685–692

Weng C-H, Lai Y-H, Lai S-H (2017) Driver drowsiness detection via a hierarchical temporal deep belief network. In: Computer vision–ACCV 2016 workshops: ACCV 2016 international workshops, Taipei, Taiwan, Nov 20–24, 2016, Revised Selected Papers. Springer, Part III 13, pp 117–133

Wijnands JS, Thompson J, Nice KA, Aschwanden GDPA, Stevenson M (2020) Real-time monitoring of driver drowsiness on mobile platforms using 3d neural networks. Neural Comput Appl 32(13):9731–9743

World Health Organization (2023) Global status report on road safety 2023

Xiong H, Yan Y, Sun L, Liu J, Han Y, Yangyang X (2024) Detection of driver drowsiness level using a hybrid learning model based on ECG signals. Biomed Eng/Biomed Tech 69(2):151–165

Yaacob S, Afrina Izzati Affandi N, Krishnan P, Rasyadan A, Yaakop M, Mohamed F (2020) Drowsiness detection using EEG and ECG signals. In: 2020 IEEE 2nd international conference on artificial intelligence in engineering and technology (IICAIET). IEEE, pp 1–5

You F, Gong Y, Haiqing T, Liang J, Wang H (2020) A fatigue driving detection algorithm based on facial motion information entropy. J Adv Transp 2020(1):8851485

Zhao Z, Zhou N, Zhang L, Yan H, Yi X, Zhang Z (2020) Driver fatigue detection based on convolutional neural networks using em-cnn. Comput Intell Neurosci 2020(1):7251280

Zilberg E, Burton D, Xu M, Karrar M, Lal S (2022) Methodology and initial analysis results for development of non-invasive and hybrid driver drowsiness detection systems. In: Advances in broadband communication and networks. River Publishers, pp 309–328

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